# IGCSE Physics Workbook

2025 · Paper 2 (Multiple Choice) + Paper 4 (Theory) mixed by topic

**Name:** _____

**Class:** _____

Cambridge IGCSE Physics 0625 — 2025 (Feb/Mar, May/Jun, Oct/Nov)

练习册

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# 1 MOTION, FORCES AND ENERGY

90 questions · Paper 2 & Paper 4

# Vocabulary · Topic 1 MOTION, FORCES AND ENERGY

## 词汇表（中英对照）

| | | | |
|---|---|---|---|
| **speed** | 速度 | **moment** | 力矩 |
| **velocity** | 速度（矢量） | **pivot** | 支点 |
| **acceleration** | 加速度 | **centre of gravity** | 重心 |
| **distance** | 距离 | **equilibrium** | 平衡 |
| **displacement** | 位移 | **momentum** | 动量 |
| **scalar** | 标量 | **impulse** | 冲量 |
| **vector** | 矢量 | **conservation of momentum** | 动量守恒 |
| **mass** | 质量 | **energy** | 能量 |
| **weight** | 重量 | **kinetic energy** | 动能 |
| **gravitational field strength** | 重力场强度 | **gravitational potential energy** | 重力势能 |
| **density** | 密度 | **elastic potential energy** | 弹性势能 |
| **force** | 力 | **work done** | 做功 |
| **resultant force** | 合力 | **power** | 功率 |
| **friction** | 摩擦力 | **efficiency** | 效率 |
| **air resistance** | 空气阻力 | **pressure** | 压强 |
| **drag** | 阻力 | **renewable** | 可再生 |
| **spring constant** | 弹簧常数 | **non-renewable** | 不可再生 |
| **extension** | 伸长量 | | |

# 1.1 Physical quantities and measurement techniques

6 MCQ + 0 theory questions · Paper 2 & Paper 4

· Describe the use of rulers and measuring cylinders to find a length or a volume
· Describe how to measure a variety of time intervals using clocks and digital timers
· Determine an average value for a small distance and for a short interval of time by measuring multiples (including
· Understand that a scalar quantity has magnitude (size) only and that a vector quantity has magnitude and direction
· Know that the following quantities are scalars: distance, speed, time, mass, energy and temperature
· Know that the following quantities are vectors: force, weight, velocity, acceleration, momentum, electric field str
· Determine, by calculation or graphically, the resultant of two vectors at right angles, limited to forces or veloci

Date: _____ Score: _____

**1** A student determines the diameter of a metal ball.

Which apparatus is needed to get an accurate measurement?

**A** a ruler and a protractor

**B** a ruler only

**C** a protractor and a set square

**D** a ruler and two rectangular blocks

0625/21/O/N/25 Q1

**1** Which instrument is most suitable to determine the volume of a small irregularly shaped stone?

**A** 30 cm ruler

**B** digital timer

**C** measuring cylinder

**D** tape measure

0625/22/M/J/25 Q1

**1** Which quantities are both vectors?

**A** acceleration and force

**B** acceleration and pressure

**C** density and force

**D** density and pressure

0625/22/O/N/25 Q1

Date: _____ Score: _____

**1** Which quantity can be measured using a ruler?

**A** the distance travelled by a toy train in one second

**B** the temperature of a toy train

**C** the time taken for a toy train to travel one metre

**D** the volume of a toy train

0625/23/M/J/25 Q1

**1** When a pendulum passes a marker, a timer displays 1 minute 30 seconds.

The pendulum passes the marker 12 more times from the same direction.

On the final pass, the timer displays a time of 2 minutes 18 seconds.

What is the time period of oscillation of the pendulum?

**A** 4.0 s **B** 4.4 s **C** 7.5 s **D** 12 s

0625/23/O/N/25 Q1

**4** Which measuring instrument is used to compare masses?

**A** balance

**B** protractor

**C** stop-watch

**D** voltmeter

0625/23/O/N/25 Q4

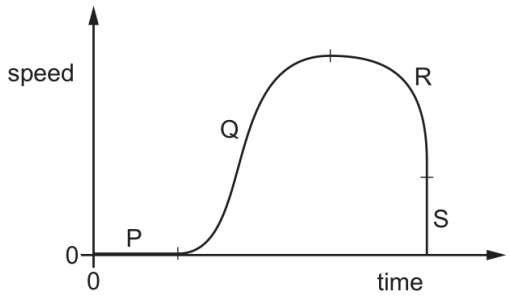
# 1.2 Motion

12 MCQ + 6 theory questions · Paper 2 & Paper 4

· Define speed as distance travelled per unit time; recall and use the equation v = s/t
· Define velocity as speed in a given direction
· Recall and use the equation average speed = total distance ÷ total time
· Sketch, plot and interpret distance–time and speed–time graphs
· Determine, qualitatively, from given data or the shape of a distance–time graph or speed–time graph when an object
· Calculate speed from the gradient of a straight- line section of a distance–time graph
· Calculate the area under a speed–time graph to determine the distance travelled for motion with constant speed or c

Date: _____ Score: _____

**1** The speed–time graph for a racing car is divided into four sections, P, Q, R and S. The car starts the race but soon crashes into a wall of tyres.




Which sections of the graph show that there is an acceleration that is changing?

**A** P, Q, R and S

**B** Q, R and S only

**C** Q and R only

**D** R and S only

0625/21/M/J/25 Q1

**2** An athlete runs at a speed of 8 m / s for 10 s, and then at a speed of 6 m / s for 12 s.

Which calculation gives the average speed of the athlete in m / s?

**A** $\frac{8+6}{2}$

**B** $\frac{(8\times10)+(6\times12)}{22}$

**C** $\frac{(8\div10)+(6\div12)}{22}$

**D** $\frac{(10\div8)+(12\div6)}{22}$

0625/21/O/N/25 Q2

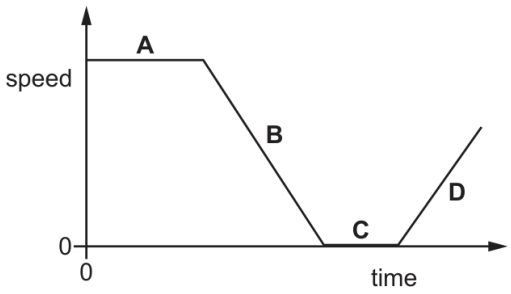
**2** Which row defines speed and velocity?

| | speed | velocity |
|---|---|---|
| **A** | distance travelled in a given direction | speed per unit time |
| **B** | distance travelled per unit time | speed in a given direction |
| **C** | distance travelled in a given direction | speed in a given direction |
| **D** | distance travelled per unit time | speed per unit time |

0625/22/F/M/25 Q2

**3** The graph shows the journey of a motorcyclist.

Which section of the graph shows the time when the motorcyclist is stationary at some traffic lights?




0625/22/F/M/25 Q3

Date: _____ Score: _____

**2** A steel ball is dropped from the top floor of a building. Air resistance can be ignored.

Which statement describes the motion of the ball?

**A** The ball falls with constant acceleration.

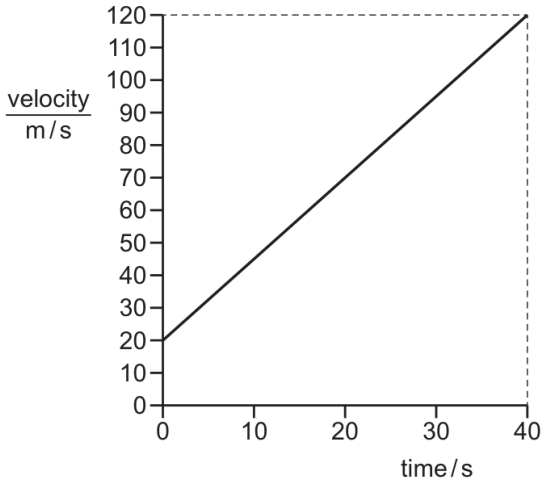
**B** The ball falls with constant speed.

**C** The ball falls with decreasing speed.

**D** The ball falls with increasing acceleration.

0625/22/M/J/25 Q2

**3** The diagram shows a velocity–time graph for an object which is accelerating.




What is the acceleration of the object?

**A** $0.40\,m/s^2$ **B** $2.5\,m/s^2$ **C** $3.0\,m/s^2$ **D** $100\,m/s^2$

0625/22/M/J/25 Q3

Date: _____ Score: _____

**2** How is the deceleration of an object calculated from a speed–time graph?

**A** It is the area under the graph.

**B** It is the gradient of the graph.

**C** It is the difference between two readings on the horizontal ($x$) axis.

**D** It is the difference between two readings on the vertical ($y$) axis.

0625/22/O/N/25 Q2

**2** A car is driven from one town to another town along a road that is **not** straight.

The driver of the car divides the total distance travelled by the total time taken.

Which quantity does the driver calculate?

**A** the acceleration of the car

**B** the average speed of the car

**C** the kinetic energy of the car

**D** the velocity of the car

0625/23/M/J/25 Q2

**3** Which quantity can be determined from the area under a speed–time graph?

**A** acceleration

**B** distance

**C** speed

**D** velocity

0625/23/M/J/25 Q3

Date: _____ Score: _____

**9** A stone is dropped from a bridge which is 22 m above a river.

What is the speed of the stone when it hits the water?

**A** 15 m/s **B** 21 m/s **C** 220 m/s **D** 430 m/s

0625/23/M/J/25 Q9

**2** A car driver applies the brakes to bring a car to rest with a constant deceleration of 4.0 m/s$^2$ in a time of 8.0 s.

What is the average speed of the car during this time?

**A** 4.0 m/s **B** 8.0 m/s **C** 16 m/s **D** 32 m/s

0625/23/O/N/25 Q2

**3** An object is falling in a uniform gravitational field.

After a certain time, it reaches terminal velocity.

Which row is correct?

| | velocity of object | acceleration of object |
|---|---|---|
| **A** | increases then decreases | positive then negative |
| **B** | increases then decreases | positive then zero |
| **C** | increases then constant | positive then negative |
| **D** | increases then constant | positive then zero |

0625/23/O/N/25 Q3

**1** **(a)** Circle the vector quantities in the list.

**acceleration** **mass** **speed** **time** **velocity**

[1]

**(b)** Fig. 1.1 shows the speed–time graph for a train travelling from station A to station B.

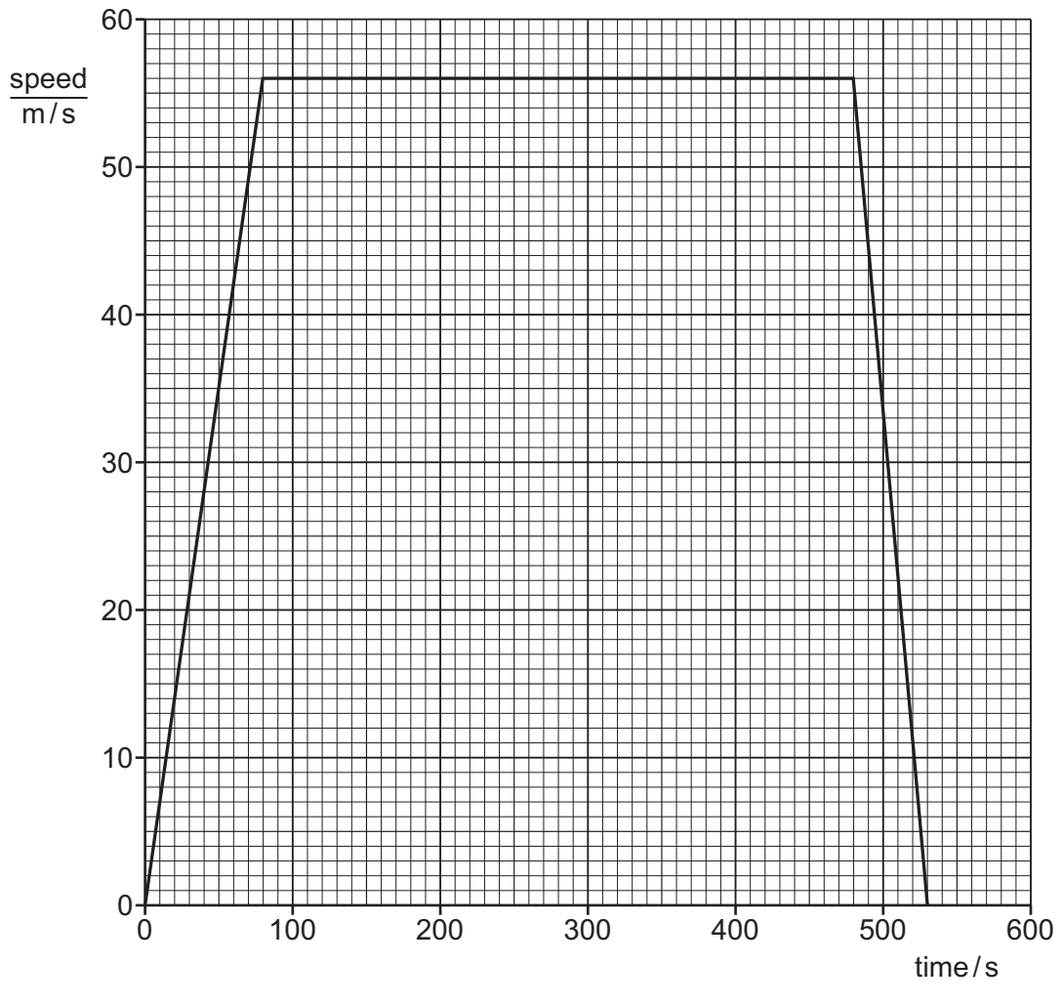



**Fig. 1.1**

**(i)** State the maximum speed of the train.

maximum speed = ..... [1]

**(ii)** Describe the motion of the train between station A and station B.

.....

.....

..... [2]

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**(iii)** Calculate the distance between station A and station B.

distance = ..... [3]

**(iv)** On a different day, the train takes 650 s to travel between station A and station B.

Suggest **one** change to the motion of the train that leads to this longer journey time.

.....

..... [1]

[Total: 8]

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**1** A train travels with a constant velocity of 56 m / s on a horizontal track. The mass of the train is 440 000 kg.

**(a)** State the difference between the velocity of the train and its speed.

.....

..... [1]

**(b)** Calculate the kinetic energy stored in the moving train.

kinetic energy = ..... [2]

**(c)** **(i)** The train has a uniform deceleration of 1.2 m / $s^2$.

Calculate the constant braking force which brings the train to rest.

force = ..... [2]

**(ii)** Calculate the distance travelled by the train as it comes to rest.

distance = ..... [3]

[Total: 8]

* 0000800000008 *



**4** A train has a maximum speed of 200 km / h. It accelerates from rest with constant acceleration of 0.70 m / $s^2$.

**(a) (i)** Define acceleration.

.....

..... [1]

**(ii)** Show that the maximum speed of the train is approximately 56 m / s.

[2]

**(iii)** Calculate the time taken for the train to reach its maximum speed.

time = ..... [2]

**(b) (i)** The train has a total mass of 440 000 kg. Calculate the force which causes the acceleration of the train.

force = ..... [2]

**(ii)** The train travels into a headwind. The force of this headwind opposes the motion of the train. State and explain the effect of this force on the motion of the train.

statement .....

explanation .....

.....

[1]

[Total: 8]

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**1** Fig. 1.1 is a speed–time graph for an ice skater.

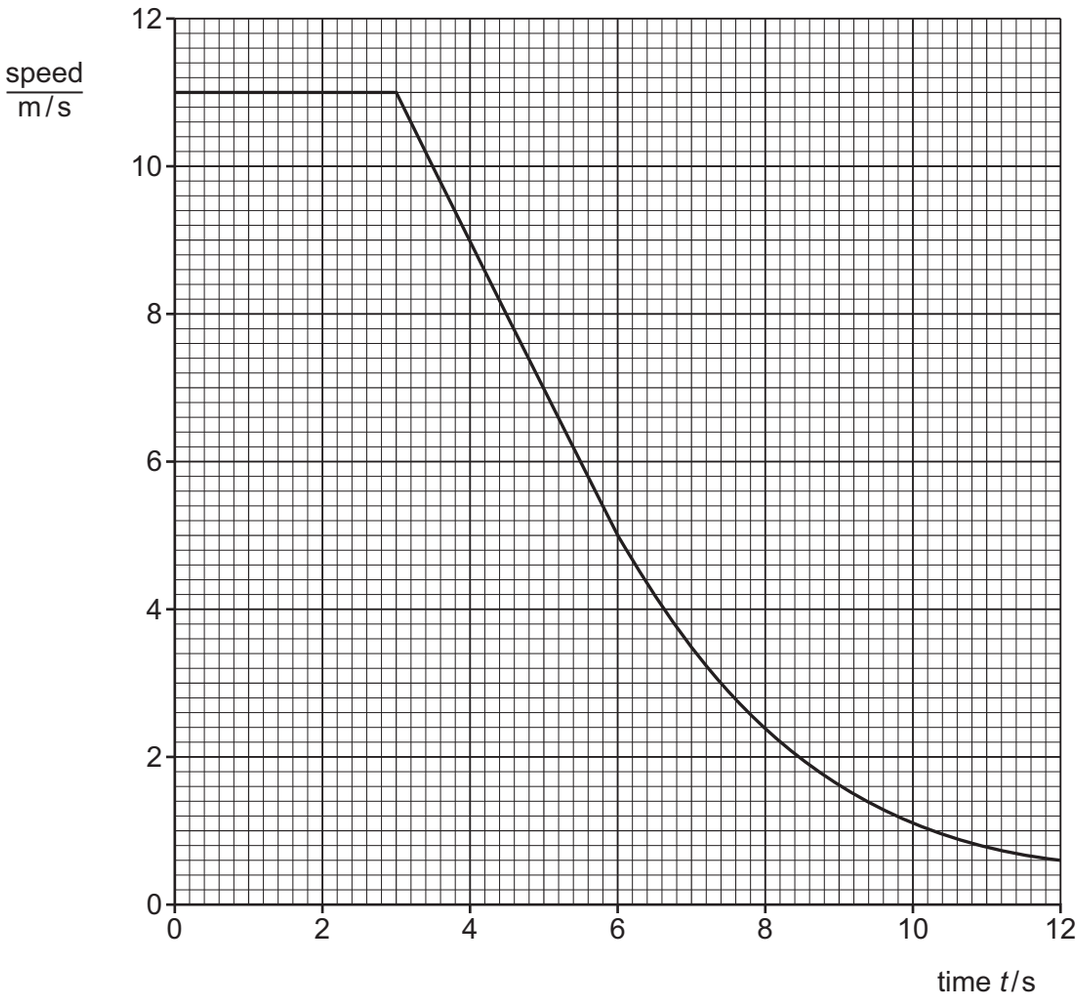


**Fig. 1.1**

**(a)** Describe the motion of the skater between $t = 0$ and $t = 3.0$ s.

..... [1]

**(b)** Calculate the distance travelled by the skater between $t = 0$ and $t = 3.0$ s.

distance = ..... [2]

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**(c) (i)** State what is meant by deceleration.

..... [1]

**(ii)** Draw a tangent to the graph at $t = 9.0\,s$.

Use the tangent to calculate the deceleration of the skater at $t = 9.0\,s$.

deceleration = ..... [3]

[Total: 7]

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**1** A car travels at a speed of 20 m / s. The driver applies the brakes when he sees a red traffic light.

Fig. 1.1 shows the speed–time graph for the car.

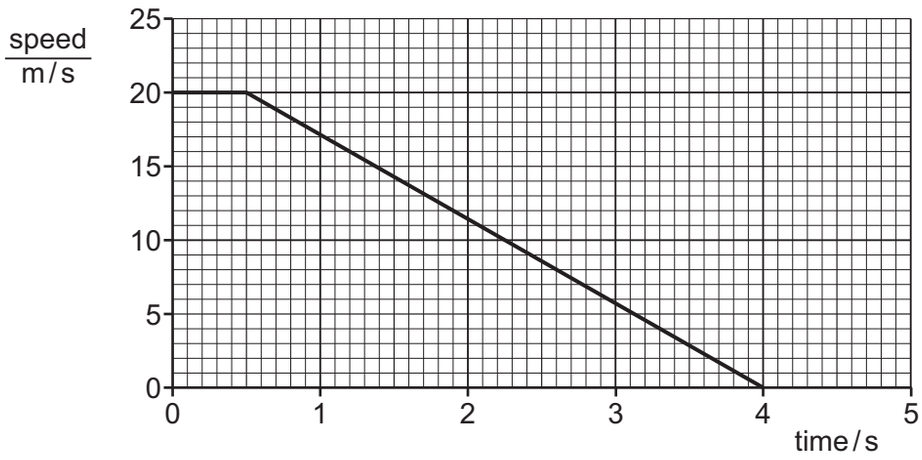


**Fig. 1.1**

**(a)** Determine the speed of the car at time = 2.0 s.

speed of the car = ..... [1]

**(b)** Calculate the distance travelled by the car between time = 0 and time = 4.0 s.

distance travelled = ..... [3]

**(c)** Calculate the deceleration of the car between time = 0.5 s and time = 4.0 s.

deceleration = ..... [3]

[Total: 7]

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**1** **(a)** A hard ball is dropped vertically downwards.

Fig. 1.1 shows the speed–time graph for the ball from when it is dropped until it hits the ground.

Air resistance is ignored.

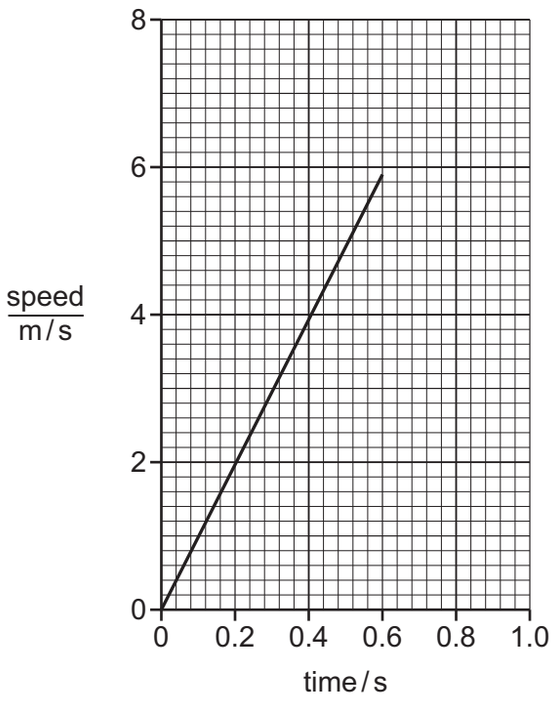


**Fig. 1.1**

**(i)** Describe the motion of the ball in Fig. 1.1.

.....

..... [1]

**(ii)** Show that the distance travelled by the ball is 1.8 m.

[2]

**(iii)** The mass of the ball is 0.54 kg. Calculate the gravitational potential energy stored in the ball immediately before it is dropped.

gravitational potential energy = ..... [2]

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**(iv)** Determine the kinetic energy of the ball when it has fallen half way to the ground.

kinetic energy = ..... [1]

**(b)** The same ball is dropped from an aeroplane. Air resistance acts on the ball.

Explain how the vertical motion of the ball changes, from when it is dropped to just before it hits the ground.

Use the idea of forces in your answer.

.....

.....

.....

.....

..... [3]

**(c)** A soft ball is dropped by a student so that it bounces on the ground.

Fig. 1.2 shows the height–time graph for the ball. The height is measured from the ground.

The ball bounces four times.

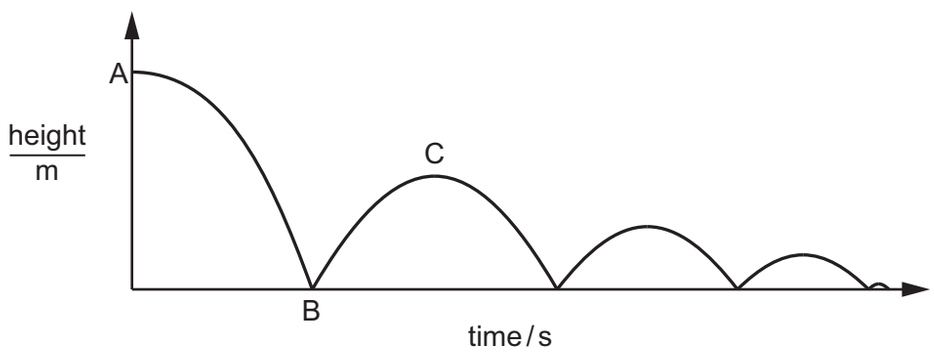



**Fig. 1.2**

Explain why the change in height of the ball between B and C is less than the change in height between A and B.

.....

.....

.....

..... [2]

[Total: 11]

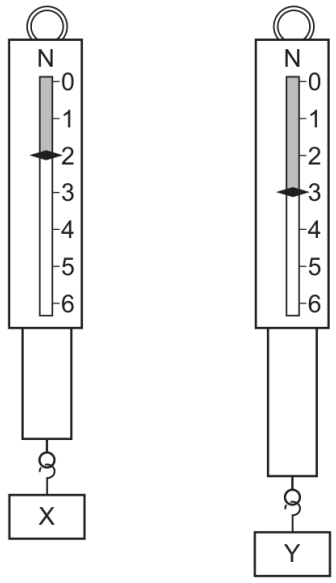
# 1.3 Mass and weight

8 MCQ + 1 theory questions · Paper 2 & Paper 4

· State that mass is a measure of the quantity of matter in an object at rest relative to the observer
· State that weight is a gravitational force on an object that has mass
· Define gravitational field strength as force per unit mass; recall and use the equation g = W/m and know that this
· Know that weights (and masses) may be compared using a balance
· Describe, and use the concept of, weight as the effect of a gravitational field on a mass

Date: _____ Score: _____

**3** The diagrams show two metal blocks, X and Y, suspended from identical force meters. X and Y have identical dimensions.




Which statement about X and Y is correct?

**A** They have different volumes and different weights.

**B** They have different volumes and equal weights.

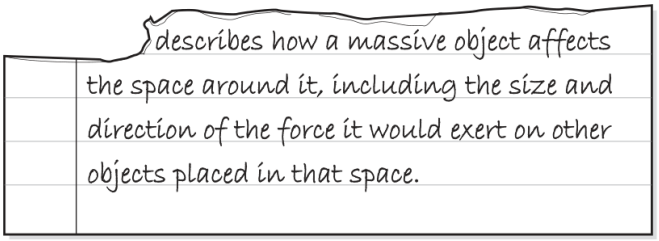
**C** They have equal volumes and equal weights.

**D** They have equal volumes and different weights.

0625/21/M/J/25 Q3

Date: _____ Score: _____

**3** A piece of paper torn out of an exercise book is shown.




What is being described?

**A** gravitational charge

**B** gravitational field

**C** gravitational mass

**D** gravitational potential energy

0625/21/O/N/25 Q3

**4** The acceleration of free fall on the Moon has one sixth of its value on the Earth.

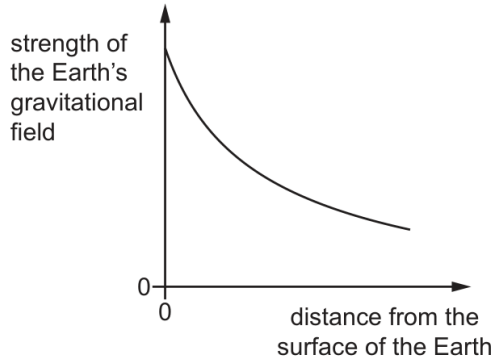
On the Earth, an astronaut's mass is measured as 60 kg.

Which row gives the mass and weight of the astronaut on the Earth and on the Moon?

| | Earth | | Moon | |
|---|---|---|---|---|
| | mass / kg | weight / N | mass / kg | weight / N |
| **A** | 60 | 6.1 | 10 | 1.0 |
| **B** | 60 | 6.1 | 60 | 6.1 |
| **C** | 60 | 590 | 10 | 16 |
| **D** | 60 | 590 | 60 | 98 |

0625/22/F/M/25 Q4

**39** The graph shows how the strength of the Earth's gravitational field varies as the distance from the Earth's surface increases.




Which row describes the effect that this has on the mass and on the weight of an object as it moves further away from the Earth's surface?

| | mass of object | weight of object |
|---|---|---|
| **A** | decreases | decreases |
| **B** | decreases | unchanged |
| **C** | unchanged | decreases |
| **D** | unchanged | unchanged |

0625/22/F/M/25 Q39

**4** Diagram 1 shows a piece of flexible material that contains many pockets of air. Diagram 2 shows the same piece of flexible material after it has been compressed so that its volume decreases.

diagram 1
(before compression)

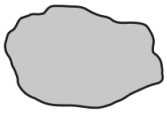
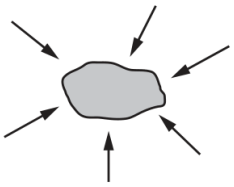


diagram 2
(after compression)



What happens to the mass and to the weight of the flexible material when it is compressed?

| | mass | weight |
|---|---|---|
| **A** | increases | increases |
| **B** | increases | no change |
| **C** | no change | increases |
| **D** | no change | no change |

0625/22/M/J/25 Q4

Date: _____ Score: _____

**3** A space probe that has a mass of 600 kg is on the planet Mars.

The acceleration of free fall on Mars is $4.0\,m/s^2$.

What is the weight of the space probe on Mars?

**A** 60 N **B** 150 N **C** 2400 N **D** 6000 N

0625/22/O/N/25 Q3

**38** A space probe measures the gravitational field strength at four different distances from a planet.

The point furthest from the planet is measured first and the point nearest the planet is measured last.

In which order were the measurements taken?

**A** 6.3 N/kg, 10 N/kg, 14 N/kg, 20 N/kg

**B** 10 N/kg, 14 N/kg, 6.3 N/kg, 20 N/kg

**C** 20 N/kg, 14 N/kg, 6.3 N/kg, 10 N/kg

**D** 20 N/kg, 14 N/kg, 10 N/kg, 6.3 N/kg

0625/22/O/N/25 Q38

**4** Students are asked for a statement about the weight of an object.

Three of the statements are listed.

1 The weight is the quantity of matter in the object.

2 The weight is the effect of a gravitational field on the object.

3 The weight is the force of attraction between the object and the Earth.

Which statements describe the weight of the object?

**A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

0625/23/M/J/25 Q4

**2** Table 2.1 contains information about the planet Mars.

**Table 2.1**

| | |
|---|---|
| mass | $6.4 \times 10^{23}$ kg |
| gravitational field strength at surface | 3.7 N / kg |
| average density | 3900 kg / $m^3$ |

**(a)** Define gravitational field strength.

.....

..... [1]

**(b)** **(i)** An object has a weight of 42 N at the surface of the Earth.

Calculate the weight of the object at the surface of Mars.

weight = ..... [2]

**(ii)** Calculate the volume of Mars.

volume = ..... [2]

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**(c)** Fig. 2.1 shows a space buggy that is tested on Earth. The buggy is travelling at a constant speed in a straight line. The driving force on the buggy is 30 N.

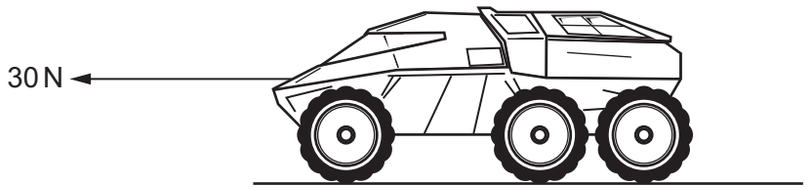



**Fig. 2.1**

**(i)** Draw and label **one** arrow on Fig. 2.1 to show the size and direction of the resistive forces on the buggy. [2]

**(ii)** Air resistance on Mars is less than air resistance on Earth.
The same driving force, 30 N, is exerted on the buggy on Mars.

1. State the effect this has on the resultant force on the buggy on Mars.

.....

2. State the relationship between resistive forces, driving force and resultant force.

.....
[1]

[Total: 8]

# 1.4 Density

6 MCQ + 0 theory questions · Paper 2 & Paper 4

· Define density as mass per unit volume; recall and use the equation ρ = m/V
· Describe how to determine the density of a liquid, of a regularly shaped solid and of an irregularly shaped solid w
· Determine whether an object floats based on density data
· Determine whether one liquid will float on another liquid based on density data given that the liquids do not mix

Date: _____ Score: _____

**4** A girl has a gold necklace.

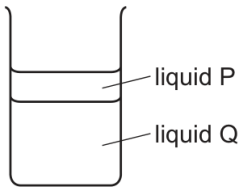
Which apparatus does she need to find the density of the gold?

| | balance | measuring cylinder | thermometer | ruler |
|---|---|---|---|---|
| **A** | ✓ | ✓ | ✓ | ✓ |
| **B** | ✓ | ✓ | ✓ | ✗ |
| **C** | ✓ | ✓ | ✗ | ✗ |
| **D** | ✓ | ✗ | ✗ | ✗ |

key
✓ = needed
✗ = **not** needed

0625/21/M/J/25 Q4

**4** The diagram shows one liquid floating on top of another liquid.




Which statement explains why liquid P floats on top of liquid Q?

**A** The mass of liquid Q is greater than the mass of liquid P.

**B** The mass of 1 $cm^3$ of liquid Q is greater than the mass of 1 $cm^3$ of liquid P.

**C** The volume of liquid Q is greater than the volume of liquid P.

**D** The volume of 1 g of liquid Q is greater than the volume of 1 g of liquid P.

0625/21/O/N/25 Q4

Date: _____ Score: _____

**5** A student determines the density of copper.

She has a rectangular block of copper.

Which other apparatus does she need?

| | balance | ruler | stop-watch | thermometer |
|---|---|---|---|---|
| **A** | ✓ | ✓ | ✓ | ✓ |
| **B** | ✓ | ✓ | ✓ | ✗ |
| **C** | ✓ | ✓ | ✗ | ✗ |
| **D** | ✓ | ✗ | ✗ | ✗ |

key

✓ = needed

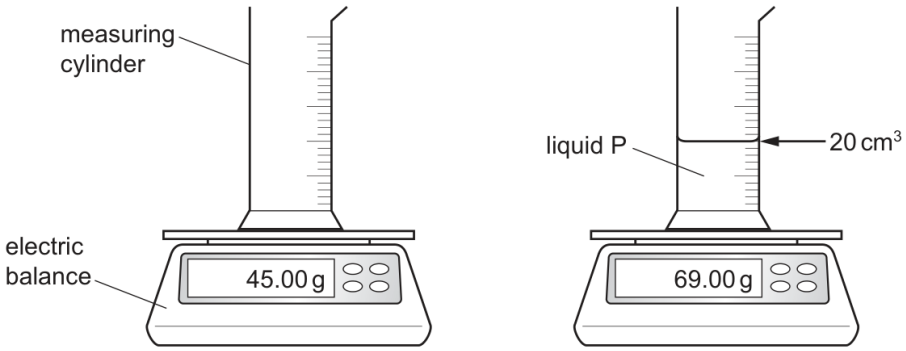
✗ = **not** needed

0625/22/M/J/25 Q5

Date: _____ Score: _____

**4** Students are asked to predict the order in which three liquids, P, Q and R, will float on each other. They are told that the liquids will **not** mix.

The diagrams show how the density of liquid P is determined.



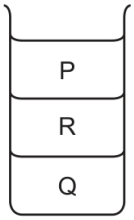

The table shows the densities of Q and R.

| liquid | density / g/cm$^3$ |
|---|---|
| Q | 0.75 |
| R | 1.1 |

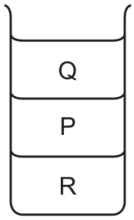
The liquids are poured gently into a tall beaker and allowed to settle.

Which diagram shows the liquids settled in the correct order?

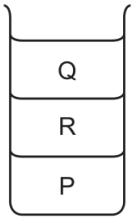
**A**



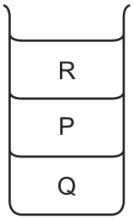

**B**




**C**




**D**




0625/22/O/N/25 Q4

Date: _____ Score: _____

**5** Samples of three different liquids have equal masses.

The volumes of the samples are 190 $cm^3$, 200 $cm^3$ and 235 $cm^3$.

The liquids are poured into a large measuring cylinder and the liquids form three separate layers because they do **not** mix.

What are the readings on the measuring cylinder at the two boundaries between the samples?

**A** 190 $cm^3$ and 390 $cm^3$

**B** 190 $cm^3$ and 425 $cm^3$

**C** 235 $cm^3$ and 425 $cm^3$

**D** 235 $cm^3$ and 435 $cm^3$

0625/23/M/J/25 Q5

**5** Three different liquids are poured into a tall beaker. The liquids do **not** mix.

The liquid densities are shown in the table.

| liquid | density / g/$cm^3$ |
|---|---|
| P | 1.8 |
| Q | 2.4 |
| R | 1.2 |

What is the correct order of the liquids from the top of the beaker to the bottom?

**A** P → Q → R **B** Q → P → R **C** Q → R → P **D** R → P → Q

0625/23/O/N/25 Q5

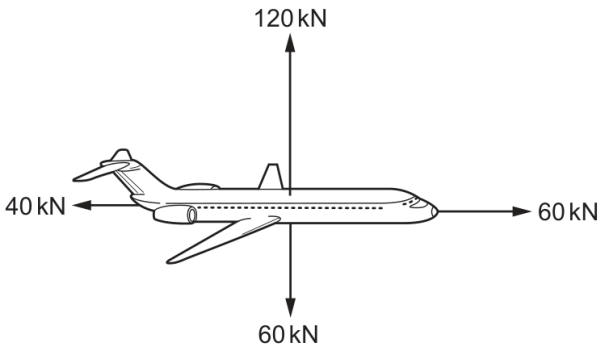
# 1.5 Forces

14 MCQ + 4 theory questions · Paper 2 & Paper 4

· Know that forces may produce changes in the size and shape of an object
· Sketch, plot and interpret load–extension graphs for an elastic solid and describe the associated experimental proc
· Determine the resultant of two or more forces acting along the same straight line
· Know that an object either remains at rest or continues in a straight line at constant speed unless acted on by a r
· State that a resultant force may change the velocity of an object by changing its direction of motion or its speed
· Define the spring constant as force per unit extension; recall and use the equation k = F/x 10 Define and use the t
· Describe solid friction as the force between two surfaces that may impede motion and produce heating

Date: _____ Score: _____

**2** The diagram shows the four forces acting on an aircraft in flight.




Which arrow shows the direction of the resultant of the four forces?

**A** **B** **C** **D**

0625/21/M/J/25 Q2

**5** A spring has a length of 25 cm when a load of 2.0 N is suspended from it.

The spring has a length of 35 cm when a 7.0 N load is used.

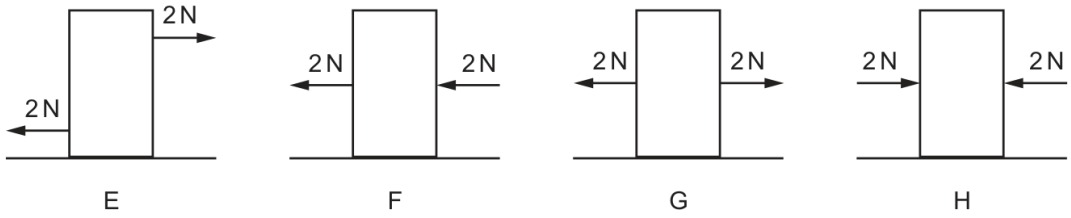
What is the spring constant of the spring?

**A** 0.20 N / cm **B** 0.50 N / cm **C** 2.0 N / cm **D** 5.0 N / cm

0625/21/M/J/25 Q5

Date: _____ Score: _____

**6** The diagrams E–H show a block of wood on a frictionless surface. In each diagram, the block has two forces acting on its sides.




Which diagrams show the block in equilibrium?

**A** E, F, G and H

**B** E, G and H only

**C** E and F only

**D** G and H only

0625/21/M/J/25 Q6

**8** A force acting on a moving ball causes its motion to change. This force stays constant.

What makes the force produce a greater change in the motion of the ball?

**A** decreasing the total mass of the ball

**B** increasing the temperature of the ball

**C** using a ball with a hollow centre but the same mass

**D** using a different material for the ball so that it has a lower density but the same mass

0625/21/M/J/25 Q8

Date: _____ Score: _____

**5** A spring has a spring constant $k$.

What is a possible unit for $k$?

**A** metre per newton

**B** newton metre

**C** newton per metre

**D** newton per metre squared

0625/21/O/N/25 Q5

**6** Which statement about the moment of a force is **not** correct?

**A** If an object is balanced about a pivot, the resultant moment on the object must be zero.

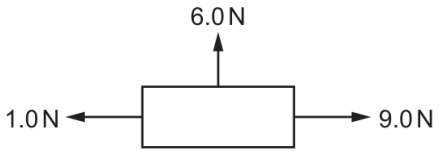
**B** The moment of a force is a measure of its turning effect.

**C** The moment of a force about a point is equal to: force × perpendicular distance from the point.

**D** The moment of a force about a point increases when the perpendicular distance of the force from the point decreases.

0625/21/O/N/25 Q6

**1** The diagram shows three forces acting on an object.



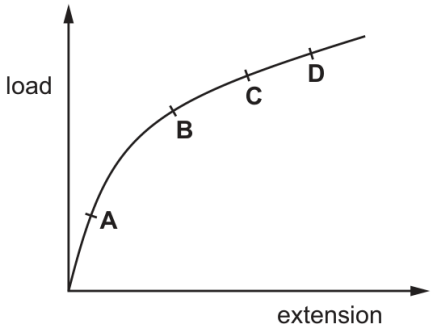

What is the magnitude of the resultant force acting on the object?

**A** 2.0 N **B** 10 N **C** 14 N **D** 16 N

0625/22/F/M/25 Q1

**5** The load–extension graph for a material is shown.

Which point indicates the limit of proportionality?




0625/22/F/M/25 Q5

**6** A 0.15 kg mass is suspended from a vertical spring.

The mass is pulled downwards until the spring has extended by 12 cm from its unstretched length.

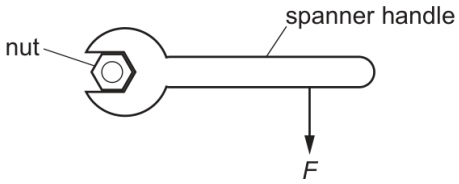
The spring constant of the spring is 30 N / m.

What is the magnitude of the initial acceleration of the spring when it is released?

**A** $14\,m/s^2$ **B** $24\,m/s^2$ **C** $28\,m/s^2$ **D** $34\,m/s^2$

0625/22/M/J/25 Q6

**5** A force *F* is applied to a spanner, as shown.




Which action increases the moment of *F* about the centre of the nut?

**A** applying the force *F* to the right-hand end of the spanner handle

**B** applying the force *F* parallel to the spanner handle

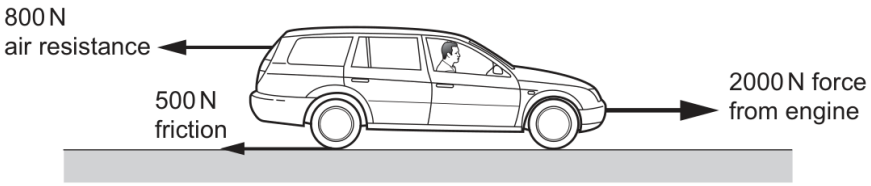
**C** spraying oil on the nut

**D** using a shorter spanner

0625/22/O/N/25 Q5

**6** A car moves along a level road.

The diagram shows all of the horizontal forces acting on the car.




Which statement is correct?

**A** The car is slowing down.

**B** The car is speeding up.

**C** The car is moving at a constant speed.

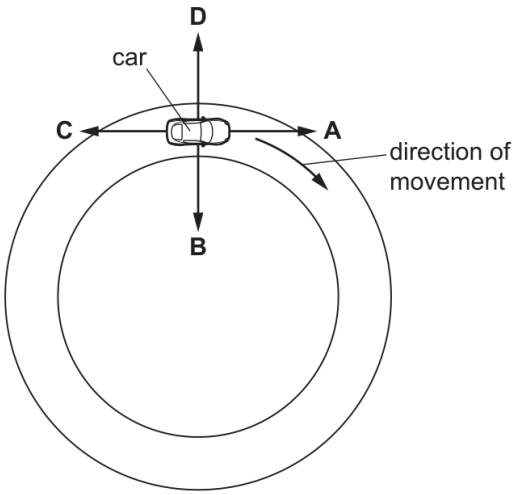
**D** The car is moving backwards.

0625/23/M/J/25 Q6

Date: _____ Score: _____

**7** A car is travelling around a circular track at a constant speed, as shown.

In which direction is the resultant force on the car?




0625/23/M/J/25 Q7

**6** A car is moving in a straight line on a level road. Its engine provides a forward force on the car. A second force of equal size acts on the car in the opposite direction.

Which statement describes what happens?

**A** The car changes direction.

**B** The car moves at a constant speed.

**C** The car slows down.

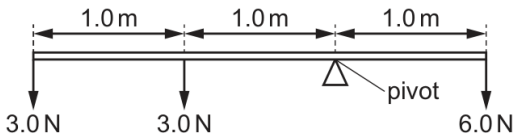
**D** The car speeds up.

0625/23/O/N/25 Q6

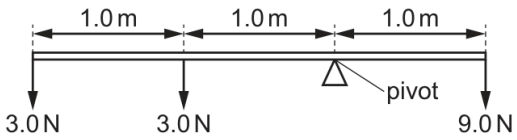
Date: _____ Score: _____

**7** Which beam is balanced? (Ignore the weight of the beam.)

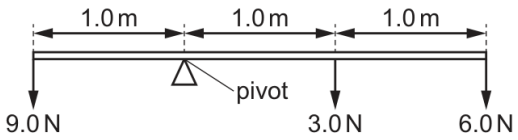
**A**



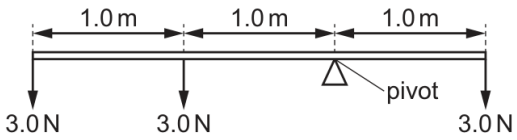

**B**




**C**




**D**




0625/23/O/N/25 Q7

**2** **(a)** A resultant force is applied to an object moving with a velocity *v* in a straight line.

**(i)** State **two** different changes to the motion that the resultant force may cause.

1 .....

2 .....
[2]

**(ii)** State **one** other way that forces may change a stationary object.

..... [1]

**(b)** Describe how a uniform metre ruler, a pivot and a selection of masses can be used to demonstrate that there is no resultant moment on an object in equilibrium.

You may include a labelled diagram in your answer.

.....

.....

.....

.....

.....

..... [4]

[Total: 7]

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**1** Fig. 1.1 shows a force–extension graph for a spring.

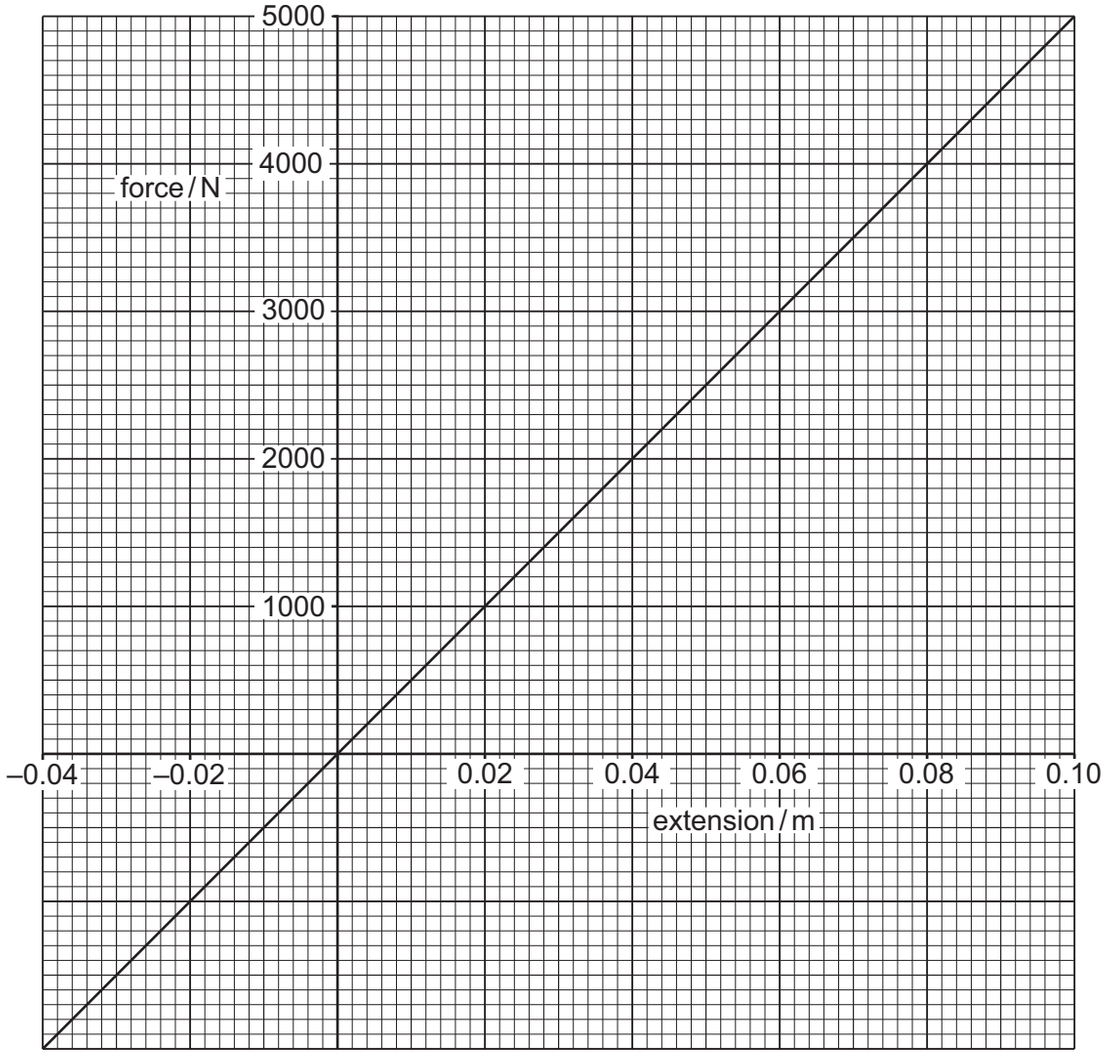


**Fig. 1.1**

**(a)** Calculate the spring constant *k* of the spring.

*k* = ..... [2]

**(b)** A student states that the spring has **not** reached the limit of proportionality when a force of 4500 N is applied to it.

State how the graph shows that this statement is true.

.....

..... [1]

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**(c)** Springs can be compressed by forces. The spring described by Fig. 1.1 is compressed by a force $F$ and has an extension of –0.025 m.

Determine $F$.

$F$ = ..... [2]

**(d)** State whether force is a scalar quantity or a vector quantity. Explain your answer.

.....

..... [1]

[Total: 6]

**2** Fig. 2.1 shows a balanced, uniform metre ruler made of wood.

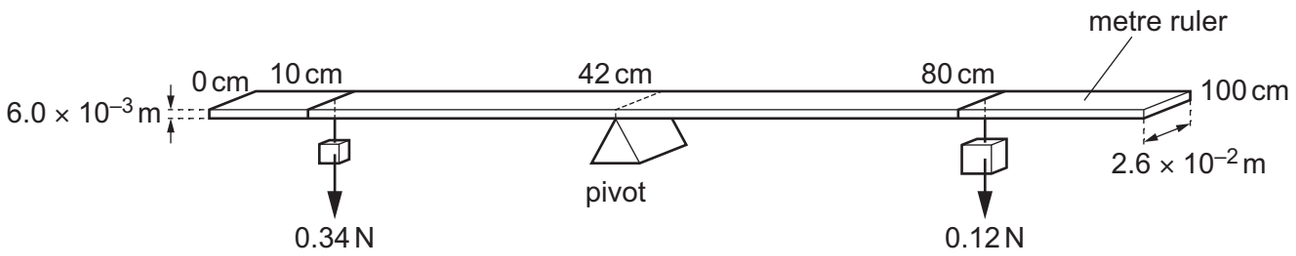


**Fig. 2.1**

The width of the metre ruler is $2.6 \times 10^{-2}$ m and the thickness of the ruler is $6.0 \times 10^{-3}$ m.

**(a)** Define the 'moment' of a force in words.

.....

..... [1]

**(b)** On Fig. 2.1, mark the position of the centre of gravity of the metre ruler with a point labelled X. Label the distance of X from the 0 cm end of the ruler. [1]

**(c)** **(i)** Show that the mass of the metre ruler is 0.081 kg.

[3]

**(ii)** Calculate the density of the wood of the metre ruler.

density = ..... [2]

[Total: 7]

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**2** **(a)** A truck moves with a constant acceleration. The speed of the truck increases from 13 m / s to 22 m / s. The time taken for this increase in speed is 4.5 s. The mass of the truck is 3700 kg.

**(i)** Calculate the resultant force acting on the truck.

resultant force = ..... [3]

**(ii)** The engine of the truck provides the forward force on the truck. State **two** forces acting on the truck in the opposite direction to the forward force.

.....

.....

..... [2]

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**(b)** Fig. 2.1 shows a car as it travels round a circular racing track.

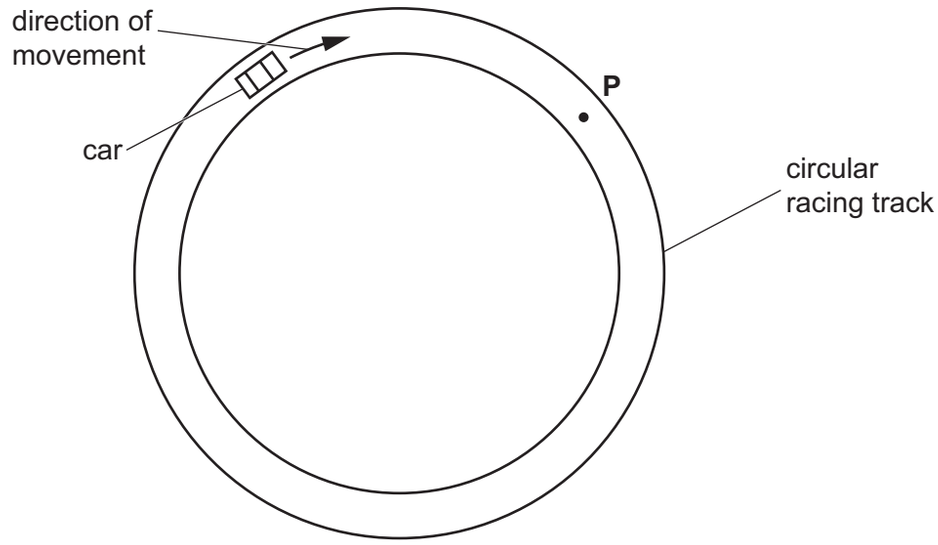



**Fig. 2.1**

**(i)** The car travels at constant speed.

On Fig. 2.1, draw an arrow to show the direction of the force acting on the car. [1]

**(ii)** A different car travels around the same track and slides off the track at point P.

State **two** possible reasons that cause the car to slide off the track.

.....

.....

.....

..... [2]

[Total: 8]

# 1.6 Momentum

8 MCQ + 5 theory questions · Paper 2 & Paper 4

· Define momentum as mass × velocity; recall and use the equation p = mv
· Define impulse as force × time for which force acts; recall and use the equation impulse = $F\Delta t = \Delta(mv)$
· Apply the principle of the conservation of momentum to solve simple problems in one dimension
· Define resultant force as the change in momentum per unit time; recall and use the equation $F = \Delta p/\Delta t$

Date: _____ Score: _____

**7** Which statement is **not** correct?

**A** the momentum of a system is conserved in all interactions

**B** impulse = change of momentum

**C** force acting = change of momentum × time

**D** momentum = mass × velocity

0625/21/M/J/25 Q7

**7** The momentum of a car changes from 10 000 kg m / s to 15 000 kg m / s in a time of 8.0 s.

What is the resultant force acting on the car?

**A** 630 N **B** 3100 N **C** 40 000 N **D** 200 000 N

0625/21/O/N/25 Q7

**6** An object of mass 4.0 kg is moving with a velocity of 3.0 m / s in a straight line.

What is the momentum of the object?

**A** 0.75 kg m / s **B** 1.3 kg m / s **C** 12 kg m / s **D** 24 kg m / s

0625/22/F/M/25 Q6

**8** A tennis ball with momentum 2.3 kg m / s is struck by a tennis racket.

The racket causes the direction of motion of the ball to reverse.

The racket exerts an average force of 1500 N during the impact with the ball. The racket and ball are in contact with each other for a time of 0.0024 s.

What is the magnitude of the momentum of the ball as it leaves the tennis racket?

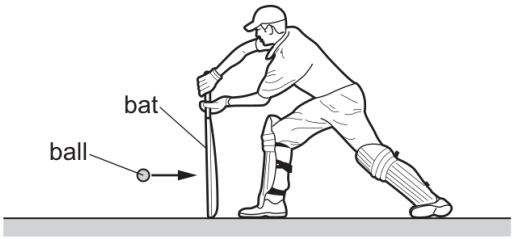
**A** 1.3 kg m / s **B** 3.6 kg m / s **C** 5.9 kg m / s **D** 8.6 kg m / s

0625/22/F/M/25 Q8

**7** A cricket ball of mass 0.15 kg is moving at a speed of 30 m / s.

The diagram shows a batsman holding a bat stationary as the ball hits the bat and bounces back at the same speed in the opposite direction.

The ball is in contact with the bat for $1.0 \times 10^{-3}$ s.




The table shows the change in momentum of the ball and the magnitude of the force applied to the ball.

Which row is correct?

| | $\frac{\text{change of momentum}}{\text{kg m/s}}$ | force applied to ball / N |
|---|---|---|
| **A** | 4.5 | 4500 |
| **B** | 4.5 | 9000 |
| **C** | 9.0 | 4500 |
| **D** | 9.0 | 9000 |

0625/22/M/J/25 Q7

Date: _____ Score: _____

**6** Diagram 1 shows a ball X of mass 0.15 kg travelling horizontally with a velocity of 2.1 m/s towards ball Y travelling at 0.4 m/s in the opposite direction.

Diagram 2 shows the velocities of the balls immediately after they collide.

diagram 1

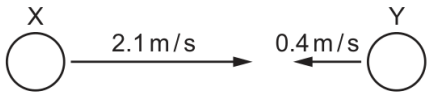
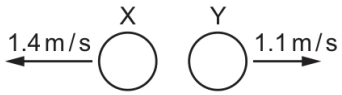



diagram 2




What is the mass of ball Y?

**A** 0.15 kg **B** 0.19 kg **C** 0.35 kg **D** 0.79 kg

0625/22/O/N/25 Q6

**8** A railway truck of mass 8000 kg moves with a velocity of 1.0 m/s to the right.

It collides with a railway truck of mass 12 000 kg moving at 2.0 m/s to the left.

The two trucks become joined to each other when they collide and move off together.

What is the velocity of the two trucks after the collision?

**A** 1.6 m/s to the left

**B** 0.80 m/s to the left

**C** 0.80 m/s to the right

**D** 1.6 m/s to the right

0625/23/M/J/25 Q8

**8** A ball is at rest on the ground. A boy kicks the ball. The boy's boot is in contact with the ball for a time of 0.040 s.

The average force on the ball is 200 N. The ball leaves the boy's boot with a speed of 20 m / s.

Which row gives the impulse of the boot on the ball and the average acceleration of the ball during the time of 0.040 s?

| | impulse on ball / Ns | average acceleration of ball / $m/s^2$ |
|---|---|---|
| **A** | 8 | 0.8 |
| **B** | 8 | 500 |
| **C** | 5000 | 0.8 |
| **D** | 5000 | 500 |

0625/23/O/N/25 Q8

* 0000800000004 *



**2** Trolley A and trolley B are on a horizontal, frictionless bench. Trolley A moves to the right with a constant velocity $u = 0.44\,\text{m/s}$. Trolley B is stationary.

Fig. 2.1 shows trolley A before it collides with trolley B.

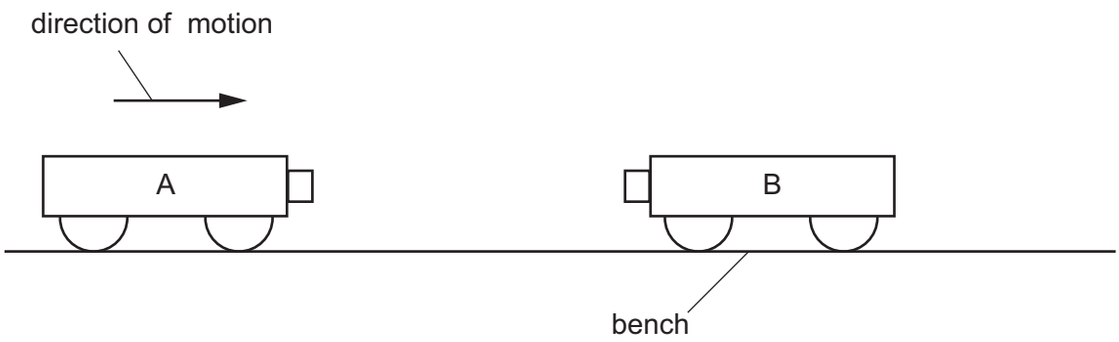



**Fig. 2.1** (not drawn to scale)

**(a)** State the momentum of trolley B before the collision. Explain your answer.

statement .....

explanation .....

.....

[1]

**(b)** After the collision, the two trolleys are joined together and travel with a constant velocity $v = 0.18\,\text{m/s}$ to the right. The mass of trolley A is 0.75 kg.

Calculate the mass of trolley B.

mass of trolley B = ..... [3]

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**(c) (i)** The trolleys move onto a rough surface which exerts a constant force *F* on the trolleys and brings them to rest in 2.6 s.

Calculate *F*.

*F* = ..... [2]

**(ii)** A different rough surface exerts a smaller resistive force on the trolleys. State how this affects the time taken to bring the trolleys to rest. Explain your answer.

statement .....

explanation .....

.....
[1]

[Total: 7]

**3** **(a)** Define momentum.

..... [1]

**(b)** Fig. 3.1 shows two toy trains, A and B, on a track.

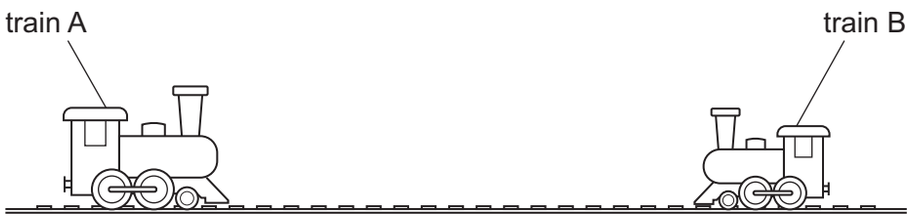


**Fig. 3.1**

The mass of train A is 0.45 kg and the mass of train B is 0.21 kg. The trains do **not** have motors.

Train A travels towards train B and they collide.

Immediately before the collision:

- the velocity of train A is 0.34 m / s to the right
- the velocity of train B is 0.12 m / s to the left.

The trains stick together when they collide.

Calculate the velocity of the trains immediately after the collision.

velocity = .....

direction .....

[4]

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**(c)** A different train is travelling with a momentum of 0.26 kg m / s.

The train slows down and stops after 2.1 s.

Calculate the average force acting on the train during this time.

average force = ..... [2]

[Total: 7]

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**1** **(a)** **(i)** State the difference between a scalar quantity and a vector quantity.

.....

..... [1]

**(ii)** In the list below, draw a line under each of the quantities that is a scalar quantity.

**acceleration** **mass** **momentum** **electric field strength**

**energy** **temperature**

[2]

**(b)** A tennis player throws a tennis ball into the air. The mass of the tennis ball is $5.8 \times 10^{-2}$ kg. The tennis ball leaves the tennis player's hand and reaches a maximum height of 5.0 m above the point of release.

**(i)** Show that the initial velocity of the tennis ball is 9.9 m / s upwards.

[3]

**(ii)** Calculate the momentum of the tennis ball as it leaves the tennis player's hand.

momentum = ..... [2]

[Total: 8]

**2** Fig. 2.1 shows a space vehicle which consists of a capsule and a nose cone.

The space vehicle is moving at a velocity of 7800 m / s. The mass of the space vehicle is 840 kg.

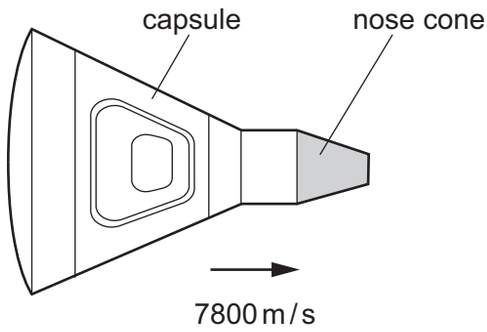


**Fig. 2.1**

**(a)** Show that the momentum of the space vehicle is approximately $6.55 \times 10^6$ kg m / s.

[1]

**(b)** The capsule ejects the nose cone, as shown in Fig. 2.2.

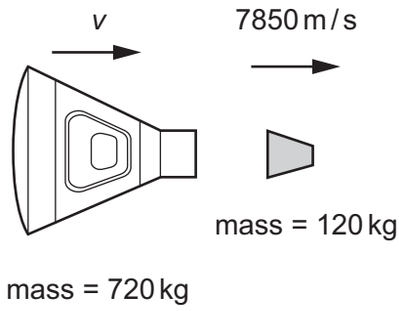


**Fig. 2.2** (not to scale)

Determine the velocity $v$ of the capsule after the nose cone is ejected. Give your answer to 3 significant figures.

velocity $v$ of the capsule = ..... [3]

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**(c)** A different space capsule returns to Earth.
Fig. 2.3 shows this capsule just before it lands in the sea.
The capsule travels at terminal velocity.

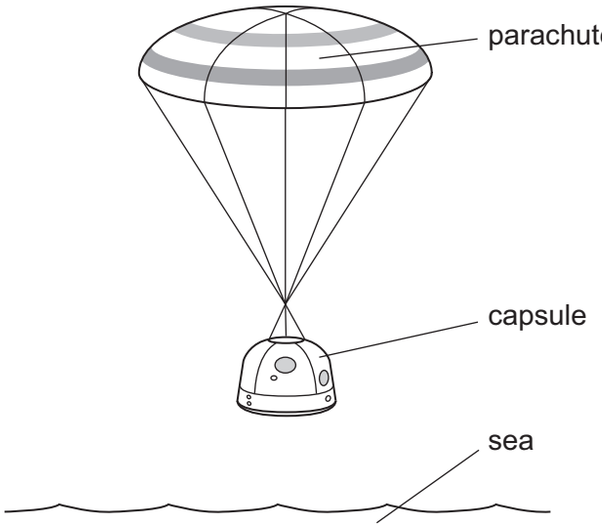


**Fig. 2.3**

The upward vertical force acting on the capsule is 120 kN.

Calculate the mass of the capsule.

mass of the capsule = ..... [2]

[Total: 6]

**2** A cricket ball is travelling at a speed of 25 m / s horizontally through the air. The mass of the ball is 0.14 kg.

A player catches the ball and the ball comes to rest.

**(a)** Calculate the change in momentum of the ball during the catch.

change in momentum = ..... [2]

**(b)** The player's hands are in contact with the moving ball for 9.7 ms.

Calculate the horizontal force exerted by her hands to bring the ball to rest.

force = ..... [3]

[Total: 5]

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# 1.7 Energy, work and power

10 MCQ + 5 theory questions · Paper 2 & Paper 4

· State that energy may be stored as kinetic, gravitational potential, chemical, elastic (strain), nuclear, electrost
· Describe how energy is transferred between stores during events and processes, including examples of transfer by fo
· Know the principle of the conservation of energy and apply this principle to simple examples including the interpre
· Recall and use the equation for kinetic energy $E_k = \frac{1}{2}mv^2$
· Recall and use the equation for the change in gravitational potential energy $\Delta E_p = mg\Delta h$
· Know the principle of the conservation of energy and apply this principle to complex examples involving multiple st
· Understand that mechanical or electrical work done is equal to the energy transferred

**8** A machine is very efficient.

What does this mean?

**A** It produces a large amount of power.

**B** It uses very little energy.

**C** It wastes very little energy.

**D** It works very quickly.

0625/21/O/N/25 Q8

**9** A car is travelling at a speed of 8.0 m / s.

The kinetic energy stored in the car is 32 000 J.

What is the kinetic energy stored in the car when its speed increases to 16 m / s?

**A** 16 000 J **B** 32 000 J **C** 64 000 J **D** 130 000 J

0625/22/F/M/25 Q9

**10** The Sun is **not** the primary source of energy for which resource?

**A** oil

**B** tidal

**C** wind

**D** wave

0625/22/F/M/25 Q10

**12** Four students determine the efficiency of four different electrical devices. The teacher knows that one student has made a mistake in the calculation.

Which value for efficiency is **not** correct?

**A** 12% **B** 36% **C** 78% **D** 102%

0625/22/F/M/25 Q12

**8** A stone is thrown vertically upwards at a speed of 5.0 m / s.

All of the energy initially in the kinetic store of the stone is transferred to its gravitational potential store when the stone reaches its maximum height.

Which equation gives the maximum height $h$ reached by the stone?

**A** $h = \frac{9.8}{2 \times 5.0^2}$

**B** $h = \frac{5.0^2 \times 2}{9.8}$

**C** $h = \frac{5.0^2}{2 \times 9.8}$

**D** $h = \frac{2 \times 9.8}{5.0^2}$

0625/22/M/J/25 Q8

Date: _____ Score: _____

**9** Three energy resources are listed.

1 geothermal

2 wind

3 nuclear fuel

Which resources get their energy from the Sun?

**A** 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 2 only

0625/22/M/J/25 Q9

**7** A mass is suspended from a vertical spring. The mass is pulled downwards causing the spring to extend further. The mass is held in this position and then released.

Which energy changes occur during the initial movement of the mass when the mass is released?

| | gravitational potential energy store | kinetic energy store | elastic potential energy store |
|---|---|---|---|
| **A** | decreases | decreases | increases |
| **B** | decreases | increases | increases |
| **C** | increases | decreases | decreases |
| **D** | increases | increases | decreases |

0625/22/O/N/25 Q7

**8** Which energy resource stores kinetic energy?

**A** coal

**B** nuclear fission

**C** solar

**D** wind

0625/22/O/N/25 Q8

Date: _____ Score: _____

**9** Which device transfers energy from a chemical energy store to a kinetic energy store?

**A** an a.c. generator

**B** a battery-powered torch

**C** a car engine

**D** a wind-up mechanical clock

0625/23/O/N/25 Q9

**10** For which energy resource is the Sun the main source?

**A** geothermal

**B** nuclear

**C** tidal

**D** wind

0625/23/O/N/25 Q10

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**3** Fig. 3.1 shows a side view of part of a concrete track at a skateboard park.

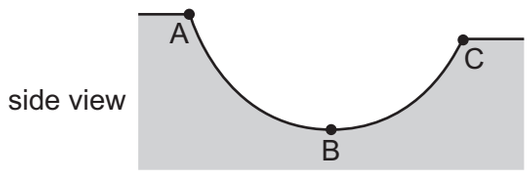


**Fig. 3.1**

**(a)** A skateboarder is initially at rest at point A. The skateboarder then travels through point B and comes to rest at point C.

Describe the transfer of energy as the skateboarder travels from A to B to C along the concrete track.

.....

.....

..... [2]

**(b)** **(i)** B is at ground level and C is at 2.8 m above ground level. The mass of the skateboarder is 65 kg.

Calculate the work done on the skateboarder as she travels from B to C.

work done = ..... [2]

**(ii)** The skateboarder falls off the skateboard at B. She hits the track and comes to rest after a few milliseconds.

State the equation that defines the force *F* with which the skateboarder hits the track. State the meaning of any symbols you use.

.....

..... [2]

[Total: 6]

**3** Fig. 3.1 shows a simplified diagram of a solar cell.

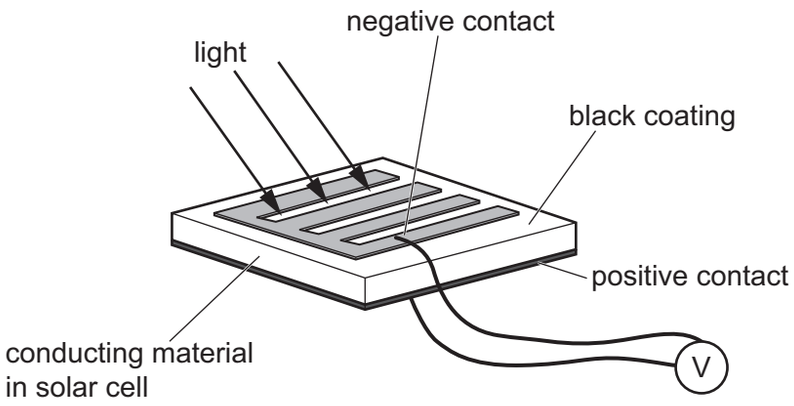


**Fig. 3.1**

**(a)** Describe the energy transfer in the solar cell.

..... [2]

**(b)** Suggest how the black coating allows the solar cell to transfer more energy.

.....

..... [1]

**(c)** 0.72 kW of light is incident on the solar cell in Fig. 3.1. The cell has an efficiency of 75%.

**(i)** Calculate the output power of the cell.

output power = ..... [2]

**(ii)** State the meaning of the term kilowatt-hour (kWh).

.....

..... [1]

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**(iii)** Energy is produced by each solar cell for an average of 6 hours per day. A household uses approximately 7400 kWh of electrical energy per year.

Calculate the number of solar cells needed to produce energy for one household. Give your answer as a whole number of solar cells.

number of solar cells = ..... [3]

[Total: 9]

**3** Fig. 3.1 shows black solar panels installed on the roof of a house and a large rechargeable battery.

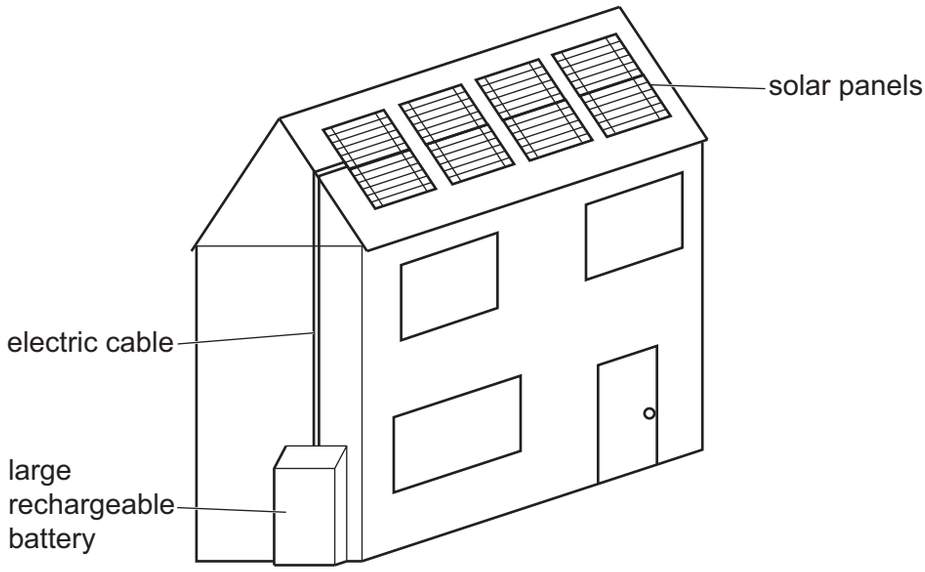


**Fig. 3.1** (not to scale)

The solar panels produce electricity and give a maximum power output of 3.5 kW. The efficiency of the solar panels is 16%.

**(a)** State and explain **one** advantage of using black solar panels.

.....

.....

..... [2]

**(b)** Calculate the power received by the solar panels from the Sun.

power = ..... [3]

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**(c)** The solar panels produce direct current (d.c.) and household appliances use alternating current (a.c.).

State the difference between alternating current and direct current.

.....

..... [1]

**(d)** Suggest **one** advantage of storing energy in the large rechargeable battery.

.....

..... [1]

**(e)** Calculate the charge that flows into the battery when there is a current of 4.0A for 2.0 hours.

charge = ..... [3]

[Total: 10]

**3** Fig. 3.1 shows an archer aiming an arrow at a target.

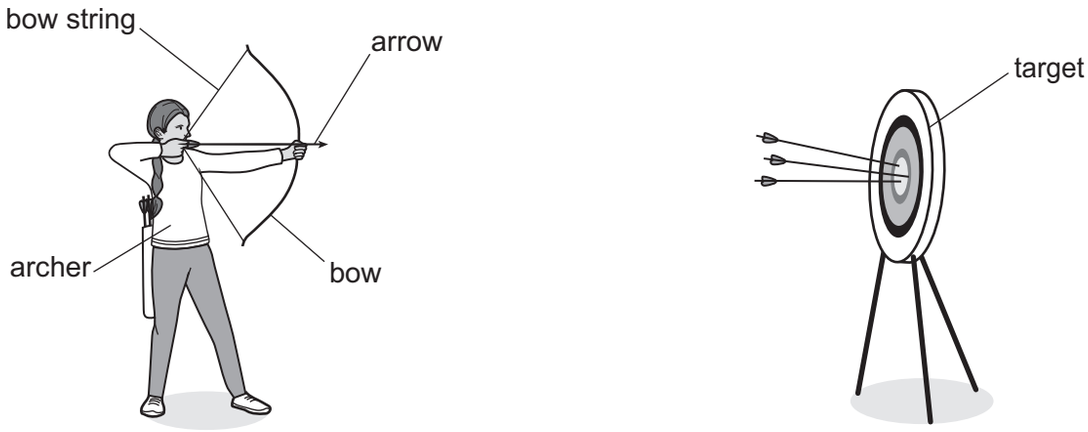


**Fig. 3.1**

**(a)** The archer pulls back on the bow string, doing a total of 110 J of work. Her hand moves a distance of 0.45 m. The bow is bent and stores energy.

Show that the average force applied by the archer in pulling the string back is approximately 240 N.

[2]

**(b)** The archer releases the bow string. All the energy in **(a)** is transferred to the arrow.

The arrow moves off at an initial speed *v*. The mass of the arrow is 0.030 kg.

**(i)** Calculate the initial speed *v* of the arrow.

initial speed *v* = ..... [3]

**(ii)** Explain why the speed of the arrow as it hits the target is less than the value in **(b)(i)**.

.....

..... [2]

[Total: 7]

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**3** **(a)** Fig. 3.1 shows a Sankey diagram for the power transfer in a coal-fired power station.

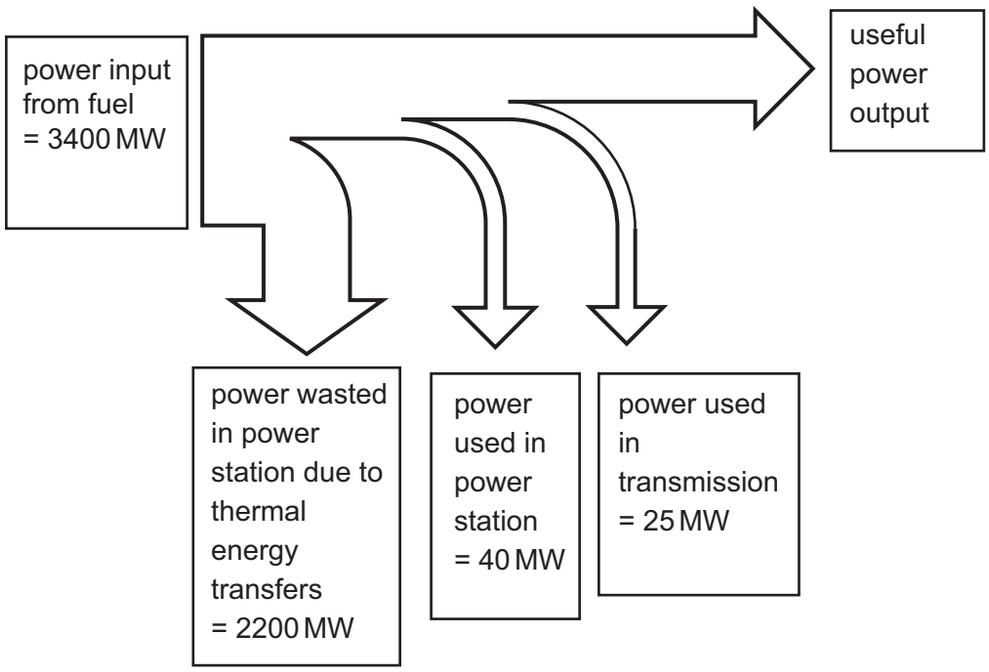



**Fig. 3.1**

**(i)** Show that the useful power output from the power station is approximately 1100 MW.

[2]

**(ii)** Calculate the efficiency of the power station.

efficiency = ..... [2]

**(b)** Describe how useful energy may be obtained from geothermal resources.

.....

.....

.....

.....

..... [2]

[Total: 6]

# 1.8 Pressure

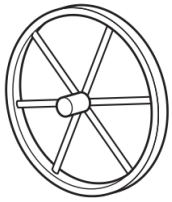
4 MCQ + 1 theory questions · Paper 2 & Paper 4

· Define pressure as force per unit area; recall and use the equation p = F/A
· Describe how pressure varies with force and area in the context of everyday examples
· Describe, qualitatively, how the pressure beneath the surface of a liquid changes with depth and density of the liq
· Recall and use the equation for the change in pressure beneath the surface of a liquid $\Delta p = \rho g \Delta h$
· Thermal physics

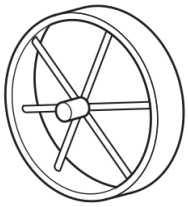
Date: _____ Score: _____

**9** A farmer has two carts. The carts have the same weight, but one has four narrow wheels and the other has four wide wheels.

narrow wheel



wide wheel



Which cart wheels sink less into soft ground and what is the reason?

| | cart wheels | reason |
|---|---|---|
| **A** | narrow | greater pressure on the ground |
| **B** | narrow | less pressure on the ground |
| **C** | wide | greater pressure on the ground |
| **D** | wide | less pressure on the ground |

0625/21/M/J/25 Q9

**9** Diagram 1 shows a conical flask containing water.

Diagram 2 shows a beaker with the same base area as the flask and containing water of the same depth *d*.

diagram 1

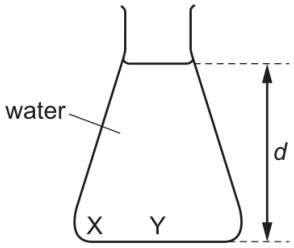
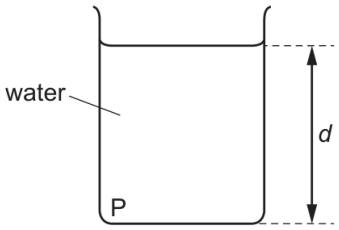



diagram 2




Which statement is correct?

**A** The pressure at P is equal to the pressure at X.

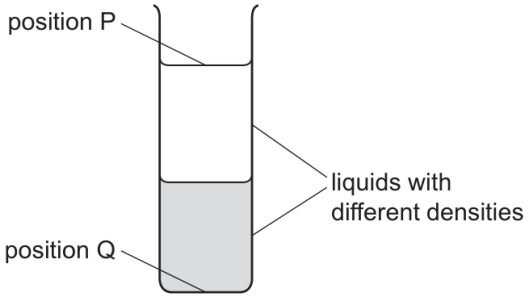
**B** The pressure at P is greater than the pressure at Y.

**C** The pressure at P is greater than the pressure at X.

**D** The pressure at Y is greater than the pressure at X.

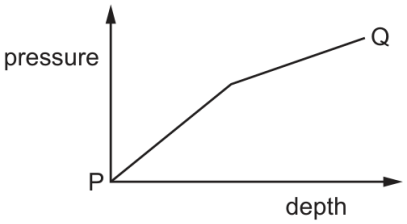
0625/21/O/N/25 Q9

**7** A tall cylinder is partly filled with two liquids, which do **not** mix. The two liquids have different densities. A student measures the pressure due to the liquids at different depths.

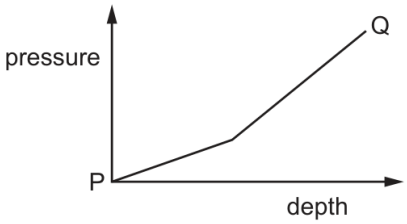



Which graph shows how the liquid pressure varies between positions P and Q?

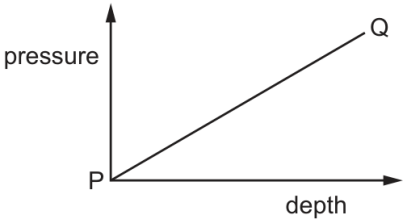
**A**



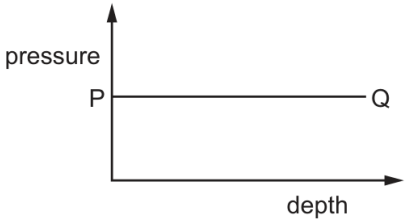

**B**




**C**




**D**

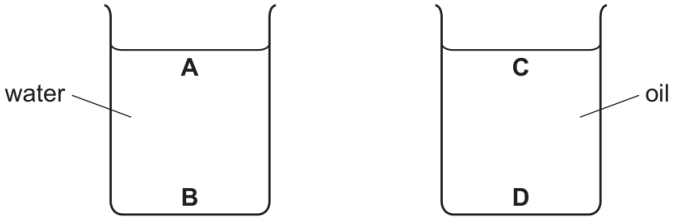



0625/22/F/M/25 Q7

**9** Two beakers are filled to the same depth, one with water and one with oil.

The density of water is 1000 kg / $m^3$. The density of oil is 920 kg / $m^3$.

In which position is the pressure the greatest?




0625/22/O/N/25 Q9

**4 (a) (i)** A rectangular glass tank contains 3.0 $m^3$ of liquid. The mass of the liquid is 2800 kg. Calculate the density of the liquid.

density = ..... [1]

**(ii)** A metal block is placed in the tank and it sinks to the bottom. The top surface of the block is 1.1 m below the surface of the liquid.

Fig. 4.1 shows the block on the bottom of the tank.

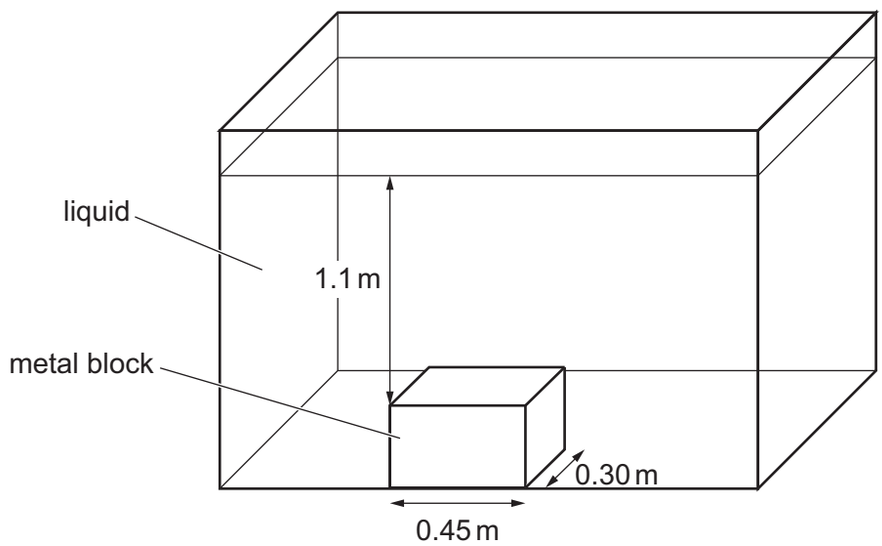



**Fig. 4.1**

Calculate the force exerted by the liquid on the top of the block.

force on top of the block = ..... [4]

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**(b)** Fig. 4.2 shows part of a steel railway track. There are small gaps between sections of the rail.

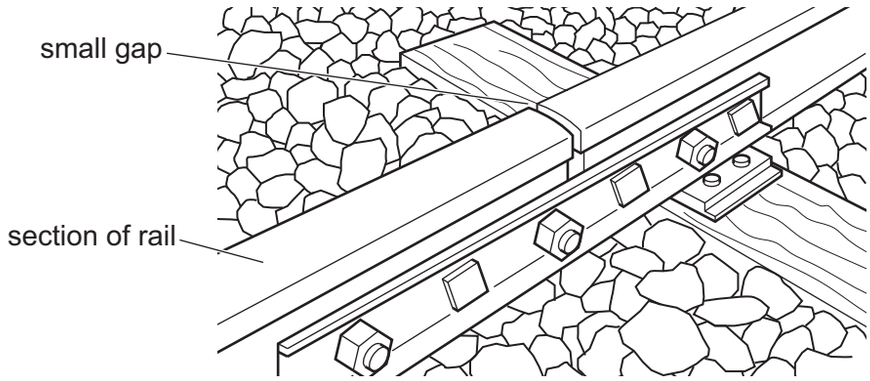


**Fig. 4.2**

State why gaps are needed between the sections of rail. Explain your answer in terms of particles.

statement .....

explanation .....

.....

[2]

**(c)** Fig. 4.3 shows a saucepan of boiling water on the heating ring of an electric cooker.

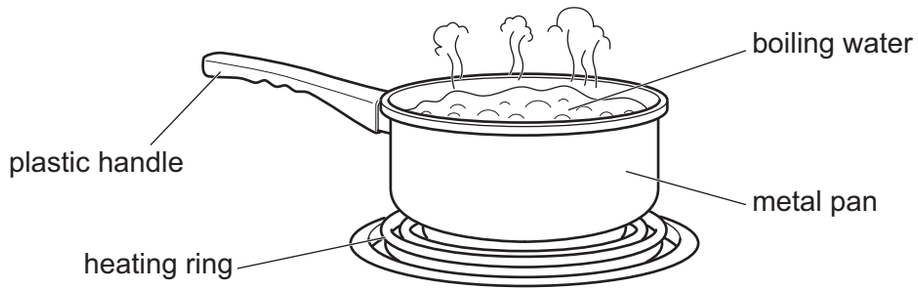


**Fig. 4.3**

The handle is made from plastic.
Explain why this material is suitable.

.....

.....

..... [2]

[Total: 9]

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# 2 THERMAL PHYSICS

54 questions · Paper 2 & Paper 4

# Vocabulary · Topic 2 THERMAL PHYSICS

词汇表（中英对照）

| | | | |
|---|---|---|---|
| **temperature** | 温度 | **convection** | 对流 |
| **thermal energy** | 热能 | **radiation** | 辐射 |
| **internal energy** | 内能 | **infrared** | 红外线 |
| **specific heat capacity** | 比热容 | **conductor** | 导体 |
| **melting** | 熔化 | **insulator** | 绝缘体 |
| **boiling** | 沸腾 | **Brownian motion** | 布朗运动 |
| **evaporation** | 蒸发 | **particle model** | 粒子模型 |
| **condensation** | 冷凝 | **absolute zero** | 绝对零度 |
| **solidification / freezing** | 凝固 | **kelvin** | 开尔文 |
| **conduction** | 传导 | **thermal expansion** | 热膨胀 |

# 2.1 Kinetic particle model of matter

16 MCQ + 1 theory questions · Paper 2 & Paper 4

· Know the distinguishing properties of solids, liquids and gases
· Know the terms for the changes in state between solids, liquids and gases (gas to solid and solid to gas transfers
· Describe the particle structure of solids, liquids and gases in terms of the arrangement, separation and motion of
· Describe the relationship between the motion of particles and temperature, including the idea that there is a lowes
· Describe the pressure and the changes in pressure of a gas in terms of the motion of its particles and their collis
· Know that the random motion of microscopic particles in a suspension is evidence for the kinetic particle model of
· Describe and explain this motion (sometimes known as Brownian motion) in terms of random collisions between the mic

**10** A hot gas and a cool liquid are at the same pressure.

Which row describes the separation and the motion of the particles of a hot gas compared to those of a cool liquid?

| | separation of gas particles | motion of gas particles |
|---|---|---|
| **A** | greater than a cool liquid | faster than a cool liquid |
| **B** | greater than a cool liquid | slower than a cool liquid |
| **C** | less than a cool liquid | faster than a cool liquid |
| **D** | less than a cool liquid | slower than a cool liquid |

0625/21/M/J/25 Q10

**11** A fixed mass of gas has a volume of 1.2 $m^3$ and is at a pressure of 200 kPa. The gas expands at constant temperature to a volume of 1.8 $m^3$.

What is the new pressure of the gas?

**A** 120 kPa **B** 130 kPa **C** 300 kPa **D** 600 kPa

0625/21/M/J/25 Q11

**14** A sealed container of gas is heated. The pressure of the gas increases.

Which statement explains this increase in pressure?

**A** The forces of the gas particles striking the walls of the container increase.

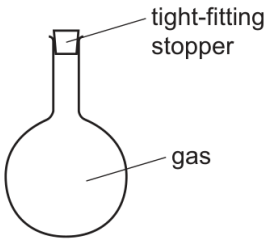
**B** The forces of attraction between the gas particles and the walls of the container increase.

**C** The gas particles collide with each other more frequently.

**D** The gas particles lose more energy when they strike the walls of the container.

0625/21/M/J/25 Q14

**10** A flask with a tight-fitting stopper contains a gas. The gas and flask are initially at room temperature.




The gas and flask are cooled.

Which row describes the average speed of the gas particles and the pressure of the gas as the temperature decreases?

| | average speed of the gas particles | pressure of the gas |
|---|---|---|
| **A** | decreases | decreases |
| **B** | decreases | stays the same |
| **C** | increases | increases |
| **D** | increases | stays the same |

0625/21/O/N/25 Q10

**11** Brownian motion is the random movement of microscopic particles in a liquid or gas.

Which statement about Brownian motion is correct?

**A** Brownian motion only occurs when the liquid or gas is heated.

**B** Much heavier fast-moving particles that **cannot** be seen collide with the microscopic particles.

**C** Much lighter fast-moving particles that **cannot** be seen collide with the microscopic particles.

**D** The microscopic particles move to avoid the fast-moving liquid or gas molecules.

0625/21/O/N/25 Q11

Date: _____ Score: _____

**13** A sample of gas is sealed inside a container of fixed volume.

The temperature of the gas is increased.

Which statement explains why the pressure of the gas increases?

**A** The gas particles collide with the walls of the container more frequently.

**B** The mass of the gas particles increases.

**C** The number of gas particles in the container increases.

**D** The time between collisions of the gas particles increases.

0625/22/F/M/25 Q13

**16** A fixed mass of gas is contained within a sealed syringe at a pressure of 100 kPa. The plunger is slowly moved until the pressure is 80 kPa. The final volume of the gas is 50 $cm^3$. The temperature of the gas does **not** change.

What is the volume of the gas before the plunger is moved?

**A** 0.25 $cm^3$ **B** 40 $cm^3$ **C** 63 $cm^3$ **D** 160 $cm^3$

0625/22/F/M/25 Q16

**10** A sample of a substance has a shape that is determined by its container.

What are the possible states of this sample?

**A** gas, liquid and solid

**B** gas and liquid only

**C** gas and solid only

**D** solid and liquid only

0625/22/M/J/25 Q10

**11** A stationary smoke particle is hit by a fast-moving nitrogen molecule.

Which row describes the motion of the smoke particle and of the nitrogen molecule after the collision?

| | smoke particle | nitrogen molecule |
|---|---|---|
| **A** | moves | rebounds |
| **B** | moves | stops |
| **C** | remains stationary | rebounds |
| **D** | remains stationary | stops |

0625/22/M/J/25 Q11

**12** A bubble of gas rises from the bottom of a lake to the top.

The table shows readings of the gas pressure inside the bubble and the volume of the bubble.

The temperature of the water in the lake is constant.

The values in three rows of the table are correct and one row is **not** correct.

Which row is **not** correct?

| | pressure / kPa | volume / $cm^3$ |
|---|---|---|
| **A** | 360 | 2.4 |
| **B** | 270 | 3.2 |
| **C** | 160 | 4.6 |
| **D** | 120 | 7.2 |

0625/22/M/J/25 Q12

**10** The pressure of the air in a car tyre rises after a long journey. This is caused by a change in the average speed of the air particles and a change in the temperature of the air in the tyre.

Which row describes the changes that have caused the air pressure in the tyre to rise?

| | average speed of the air particles | temperature of the air |
|---|---|---|
| **A** | decreases | decreases |
| **B** | decreases | increases |
| **C** | increases | decreases |
| **D** | increases | increases |

0625/22/O/N/25 Q10

**10** Which row describes the particle structure of a gas?

| | distance between the particles | motion of the particles |
|---|---|---|
| **A** | close together | vibrating |
| **B** | close together | random |
| **C** | far apart | vibrating |
| **D** | far apart | random |

0625/23/M/J/25 Q10

**11** The particles of a gas in a container of fixed volume gain energy in the kinetic store.

Which effect does this have on the gas?

**A** Both the pressure and the temperature of the gas increase.

**B** Only the temperature of the gas increases.

**C** Neither the pressure nor the temperature of the gas increases.

**D** Only the pressure of the gas increases.

0625/23/M/J/25 Q11

Date: _____ Score: _____

**12** Gas particles exert a pressure when they collide with the walls of a container.

Which statement about the gas particles is correct?

**A** They experience a change in force, which results in a pressure equal to momentum × area.

**B** They experience a change in force, which results in a pressure equal to $\frac{\text{momentum}}{\text{area}}$.

**C** They experience a change in momentum, which results in a pressure equal to force × area.

**D** They experience a change in momentum, which results in a pressure equal to $\frac{\text{force}}{\text{area}}$.

0625/23/M/J/25 Q12

**11** A football is inflated in a warm room and then taken outside where the temperature is much lower. The pressure and temperature inside the ball both decrease. The volume of the ball is unchanged.

Which row explains the decrease in pressure in terms of the movement of air particles?

| | kinetic energy of air particles | frequency of collisions between air particles and inside surface of ball |
|---|---|---|
| **A** | decreased | decreased |
| **B** | decreased | unchanged |
| **C** | unchanged | decreased |
| **D** | unchanged | unchanged |

0625/23/O/N/25 Q11

**12** A fixed mass of gas has a volume of 25 $cm^3$. The pressure of the gas is 100 kPa.

The volume of the gas is slowly decreased by 15 $cm^3$ at constant temperature.

What is the change in pressure of the gas?

**A** 67 kPa **B** 150 kPa **C** 170 kPa **D** 250 kPa

0625/23/O/N/25 Q12

**4** Fig. 4.1 shows gas trapped in a cylinder by a piston.

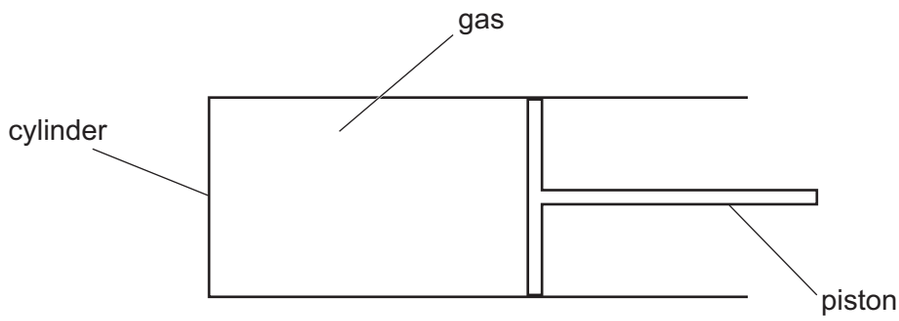



**Fig. 4.1**

**(a)** The volume of gas is 240 $cm^3$.
The piston is pushed to the left and is held in its new position.

**(i)** The pressure of the gas increases from $1.0 \times 10^5$ Pa to $1.4 \times 10^5$ Pa.

The temperature of the gas remains constant.

Calculate the volume of the gas when the piston is in its new position.

volume = ..... $cm^3$ [3]

**(ii)** The area of the piston in contact with the gas is $1.9 \times 10^{-3}$ $m^2$.

Calculate the force exerted on the piston by the gas when the piston is held in its new position.

force = ..... [2]

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**(iii)** The distance moved by the piston is 0.036 m. The average force exerted by the piston as it moves is 220 N.

Calculate the mechanical work done by the piston. State the equation you use.

work done = ..... [2]

**(b)** Explain, in terms of particles, why gases can be compressed but liquids cannot.

.....

.....

..... [1]

[Total: 8]

# 2.2 Thermal properties and temperature

17 MCQ + 3 theory questions · Paper 2 & Paper 4

· Describe, qualitatively, the thermal expansion of solids, liquids and gases at constant pressure
· Describe some of the everyday applications and consequences of thermal expansion
· Explain, in terms of the motion and arrangement of particles, the relative order of magnitudes of the expansion of
· Know that a rise in the temperature of an object increases its internal energy
· Describe an increase in temperature of an object in terms of an increase in the average kinetic energies of all of
· Define specific heat capacity as the energy required per unit mass per unit temperature increase; recall and use th
· Describe experiments to measure the specific heat capacity of a solid and a liquid

**12** The specific heat capacity of solid P is greater than that of solid Q.

What does this statement mean?

**A** Less energy is needed to raise the temperature by 1 °C of unit mass of solid P than unit mass of solid Q.

**B** Less energy is needed to melt unit mass of solid P than unit mass of solid Q.

**C** More energy is needed to raise the temperature by 1 °C of unit mass of solid P than unit mass of solid Q.

**D** More energy is needed to melt unit mass of solid P than unit mass of solid Q.

0625/21/M/J/25 Q12

**13** When water evaporates, what escapes from the surface of the water?

**A** individual atoms

**B** individual molecules

**C** individual protons

**D** tiny drops of water

0625/21/M/J/25 Q13

**12** The distance between two electricity pylons is 60 m. The cable that connects the two pylons is 62 m in length.

Why is the cable longer than the distance between the two pylons?

**A** to allow for contraction of the cable in cold weather

**B** to create a slope in the cable for electrons to flow down

**C** to keep the current low and the voltage high

**D** to reduce magnetic fields around the cable

0625/21/O/N/25 Q12

**13** A glass beaker contains a mass of 0.20 kg of water at a temperature of 20 °C.

The specific heat capacity of water is 4200 J / (kg °C).

A brass block of mass 0.15 kg is at a temperature of 170 °C. The block is carefully lowered into the beaker of water.

The final temperature of both the water and brass is 30 °C.

What is the specific heat capacity of the brass?

(Assume that no water is lost and no thermal energy has been transferred to the beaker or the surroundings.)

**A** 400 J / (kg °C)

**B** 660 J / (kg °C)

**C** 1200 J / (kg °C)

**D** 5600 J / (kg °C)

0625/21/O/N/25 Q13

**14** A student does an experiment to determine the specific heat capacity of a metal block.

She records the following measurements.

energy supplied = 12 kJ

initial temperature = 20 °C

final temperature = 45 °C

mass of metal block = 600 g

What is the specific heat capacity of the metal block?

**A** $8.0 \times 10^{-4}$ J / (kg °C)

**B** 0.80 J / (kg °C)

**C** 180 J / (kg °C)

**D** 800 J / (kg °C)

0625/22/F/M/25 Q14

**15** A metal block is left overnight in a cool, shady room. In the morning, the metal block is moved into warm surroundings.

Which statement about the metal block is correct in the morning?

**A** The internal energy of the metal block increases.

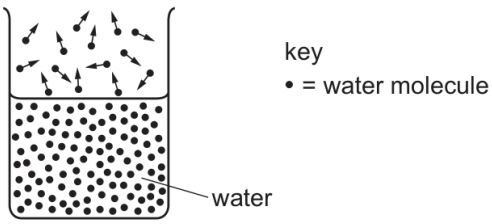
**B** The temperature of the metal block decreases.

**C** Convection transfers energy throughout the metal block.

**D** The metal contracts slightly.

0625/22/F/M/25 Q15

**17** The diagram shows the more energetic water molecules escaping from the surface of liquid water.




What is this process called?

**A** Brownian motion

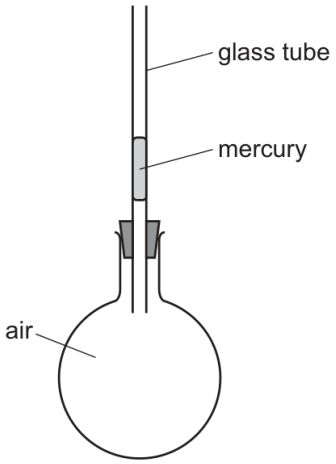
**B** condensation

**C** evaporation

**D** conduction

0625/22/F/M/25 Q17

**13** The diagram shows a glass flask sealed with a small volume of mercury in a glass tube. When the flask is gently warmed, the mercury rises up the tube.




What is the main cause of the movement of the mercury?

**A** expansion of air in the flask

**B** expansion of the glass flask

**C** expansion of the glass tube

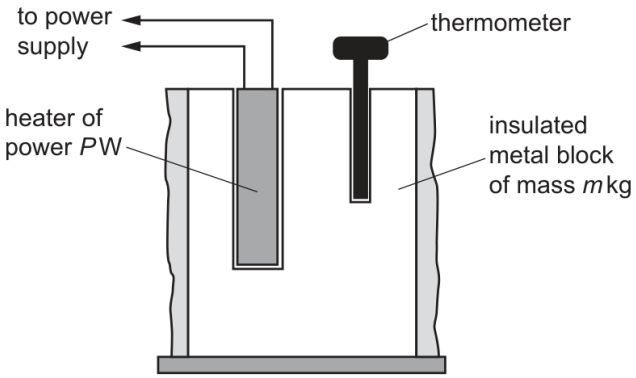
**D** expansion of the mercury

0625/22/M/J/25 Q13

Date: _____ Score: _____

**14** The diagram shows the equipment used in an experiment to determine the specific heat capacity of a metal.

In the experiment, a heater is switched on for $t$ s and the increase in temperature $\Delta\theta$ of a block of the metal is measured in °C.




Which equation gives the specific heat capacity of the metal?

**A** $c = \frac{P}{tm\Delta\theta}$ **B** $c = \frac{Pt}{m\Delta\theta}$ **C** $c = \frac{tm\Delta\theta}{P}$ **D** $c = \frac{m\Delta\theta}{Pt}$

0625/22/M/J/25 Q14

**11** In which range of temperatures is water a liquid?

**A** 0 K to 100 K

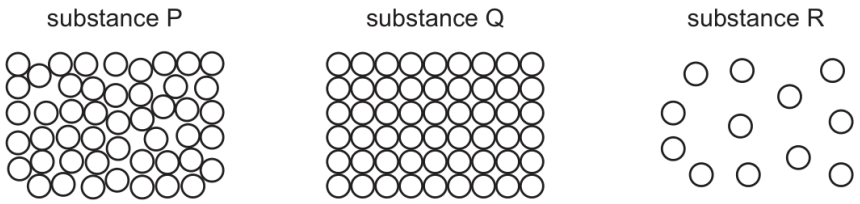
**B** 0 K to 273 K

**C** 100 K to 373 K

**D** 273 K to 373 K

0625/22/O/N/25 Q11

**12** The diagram represents the particle arrangements in samples of three different substances, P, Q and R.




The samples have equal volumes and the same temperature.

The samples are heated. The temperature increase of each sample is the same and **no** changes of state take place.

Which row shows possible percentage increases in volume for the three samples?

| | sample of substance P / % | sample of substance Q / % | sample of substance R / % |
|---|---|---|---|
| **A** | 0.066 | 1.9 | 7.2 |
| **B** | 0.066 | 7.2 | 1.9 |
| **C** | 1.9 | 0.066 | 7.2 |
| **D** | 1.9 | 7.2 | 0.066 |

0625/22/O/N/25 Q12

**13** A cup contains 0.25 kg of water at 60 °C. Some time later, the water has cooled to 20 °C.

How much energy has been transferred from the water to the surroundings?

(The specific heat capacity of water is 4200 J / (kg °C).)

**A** 21 000 J **B** 42 000 J **C** 63 000 J **D** 84 000 J

0625/22/O/N/25 Q13

**14** A pure substance is heated slowly. The temperature of the substance is measured as it starts to melt and when it finishes melting.

Which statement is correct?

**A** The temperature decreases slightly as the substance melts.

**B** The temperature fluctuates as the substance melts.

**C** The temperature increases as the substance melts.

**D** The temperature stays the same as the substance melts.

0625/22/O/N/25 Q14

**13** A solid changes into a liquid.

Which term describes this change in state?

**A** boiling

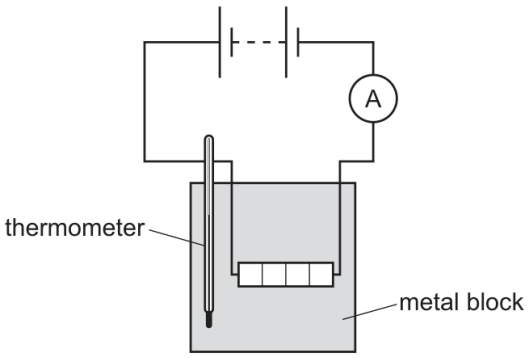
**B** condensation

**C** freezing

**D** melting

0625/23/M/J/25 Q13

**14** The diagram shows some of the apparatus used to determine the specific heat capacity of a metal block.




Which additional pieces of apparatus are required to find the specific heat capacity?

**A** stop-watch, voltmeter and balance

**B** resistor, stop-watch and ruler

**C** ruler, balance and force meter

**D** voltmeter, thermistor and insulation

0625/23/M/J/25 Q14

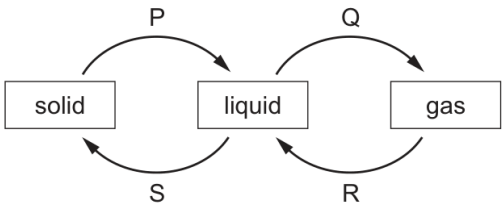
**13** The specific heat capacity of ice is 2000 J / kg °C.

Which thermal energy is required to raise the temperature of 100 g of ice by 5 °C?

**A** 40 J **B** 100 J **C** 1000 J **D** 4000 J

0625/23/O/N/25 Q13

**14** The diagram shows four labelled changes of state, P, Q, R and S, between solid, liquid and gas.




Which changes need an energy input?

**A** P and Q **B** Q and R **C** R and S **D** S and P

0625/23/O/N/25 Q14

**4** **(a)** A 12 V, 50 W immersion heater is used to heat 0.15 kg of water in a beaker. The water is initially at a room temperature of 20 °C. The specific heat capacity of water is 4200 J / (kg °C).

Calculate the energy supplied to raise the temperature of the water from 20 °C to 58 °C.

energy = ..... [3]

**(b)** The immersion heater is removed from the beaker.

One metal rod and one plastic rod are placed in the beaker of hot water as shown in Fig. 4.1. The rods are at room temperature (20 °C) before they are placed into the beaker.

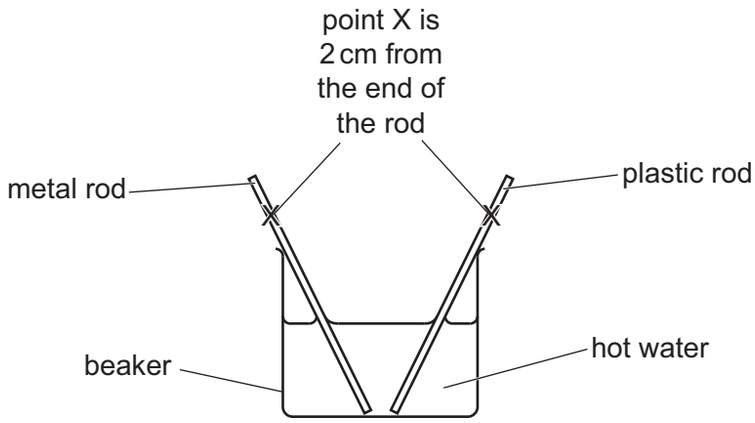



**Fig. 4.1**

Describe how the temperature of point X on each rod changes after the rods are placed in the beaker. Explain your answer.

.....

.....

.....

.....

..... [4]

[Total: 7]

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**5** **(a)** There is a large puddle of water on a road. The water in the puddle evaporates.

**(i)** Describe how evaporation from the puddle occurs. Use ideas about particles in your answer.

.....

.....

..... [2]

**(ii)** State and explain **one** change in the weather that causes a faster rate of evaporation.

Statement .....

Explanation .....

.....

.....

[2]

**(b)** A car travels on a dry road. The driver presses the brakes. The car travels a distance before it comes to rest. This distance is called the braking distance.

State and explain how the braking distance changes when the road is wet.

.....

.....

..... [1]

[Total: 5]

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**4** Fig. 4.1 shows a pan with a copper base on a hotplate. The hotplate heats the pan and the water.

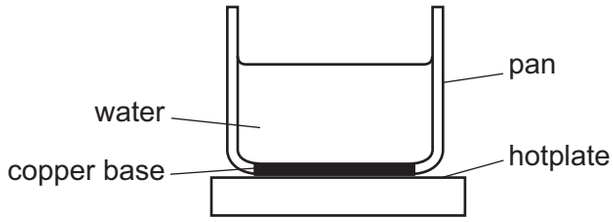



**Fig. 4.1**

**(a)** Explain how thermal energy is conducted through the copper base.

.....

.....

.....

..... [3]

**(b)** **(i)** Define, in words, specific heat capacity.

.....

.....

..... [2]

**(ii)** A student heats a metal object to 100 °C. The student places the metal object in an insulated cup containing water at 22 °C. The final temperature of the water and the metal object is 31 °C.

The specific heat capacity of water is 4.2 J / (g °C).
The mass of the water in the insulated cup is 50 g.
The mass of the metal object is 54 g.

Calculate the specific heat capacity of the metal.

specific heat capacity = ..... [3]

[Total: 8]

# 2.3 Transfer of thermal energy

15 MCQ + 2 theory questions · Paper 2 & Paper 4

· Describe experiments to demonstrate the properties of good thermal conductors and bad thermal conductors (thermal i
· Describe thermal conduction in all solids in terms of atomic or molecular lattice vibrations and also in terms of t
· Describe, in terms of particles, why thermal conduction is bad in gases and most liquids
· Know that there are many solids that conduct thermal energy better than thermal insulators but do so less well than
· Know that convection is an important method of thermal energy transfer in liquids and gases
· Explain convection in liquids and gases in terms of density changes and describe experiments to illustrate convecti
· Know that thermal radiation is infrared radiation and that all objects emit this radiation

**15** A student has a copper rod and a wooden rod. Both rods have the same dimensions.

The student puts one end of each rod into a beaker of crushed ice.

After a few minutes, the student picks up the rods by the ends **not** in the ice.

Which statement is correct?

**A** The copper rod feels colder than the wooden rod because copper is a better conductor than wood.

**B** The copper rod feels warmer than the wooden rod because copper is a better conductor than wood.

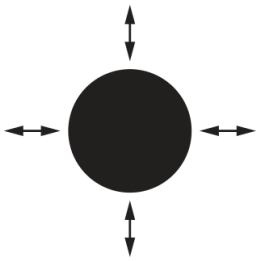
**C** The wooden rod feels colder than the copper rod because wood is a better conductor than copper.

**D** The wooden rod feels warmer than the copper rod because wood is a better conductor than copper.

0625/21/M/J/25 Q15

**16** The diagram shows a black object X at constant temperature.

The arrows represent the absorption and emission of thermal radiation at the surface.



Object Y is at the same constant temperature as X and is identical except that it has a white surface.

Which row correctly compares object X with object Y?

| | rate of absorption for X | rate of emission for X |
|---|---|---|
| **A** | higher | higher |
| **B** | lower | higher |
| **C** | higher | lower |
| **D** | lower | lower |

0625/21/M/J/25 Q16

**14** Which statement about convection is **not** correct?

**A** It enables water in a pan on a cooker to get evenly heated.

**B** It happens in liquids and gases.

**C** It occurs because cooler liquids rise above warmer liquids.

**D** It occurs because the density of a liquid decreases when it is heated.

0625/21/O/N/25 Q14

**15** Iron is a better conductor of thermal energy than rubber.

Which statement explains this?

**A** Electrons in iron can move from one part of the iron to another. Electrons in rubber **cannot** move in this way.

**B** Electrons in iron vibrate with more energy than those in rubber.

**C** Positive particles in iron can move from one part of the iron to another. Positive particles in rubber **cannot** move in this way.

**D** Positive particles in iron vibrate with more energy than those in rubber.

0625/21/O/N/25 Q15

**16** Two metal cans are identical, except that one has a shiny silver outer surface and the other has a dull black outer surface. They each have the same mass of hot water at the same temperature sealed inside them. They are both in a vacuum in the darkness of outer space.

How does the temperature of the water in each one change?

**A** The temperature of the water does **not** change.

**B** The water in the black can cools more slowly than that in the silver can.

**C** The water in the silver can cools more slowly than that in the black can.

**D** They both cool down at the same rate.

0625/21/O/N/25 Q16

**11** An experiment is done to compare the rate of conduction of thermal energy in metal and plastic spoons.

An ice cube is placed on one end of each spoon and the other end is heated gently.

Which row has the correct prediction and correct explanation?

| | prediction | explanation |
|---|---|---|
| **A** | The ice cube would melt more quickly on the metal spoon. | Metal is a good conductor. |
| **B** | The ice cube would melt more quickly on the metal spoon. | Plastic is a good conductor. |
| **C** | The ice cube would melt more quickly on the plastic spoon. | Metal is a poor conductor. |
| **D** | The ice cube would melt more quickly on the plastic spoon. | Plastic is a poor conductor. |

0625/22/F/M/25 Q11

Date: _____ Score: _____

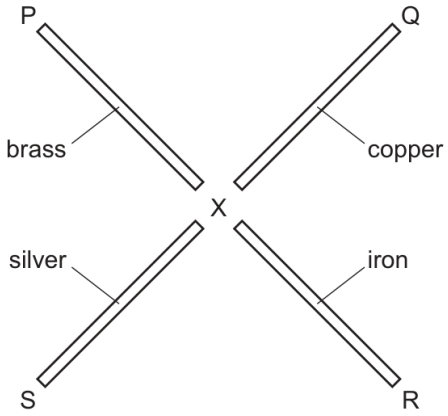
**18** A hot object emits thermal radiation.

Which row describes two changes that both reduce the rate of emission of radiation?

| | surface area | surface temperature |
|---|---|---|
| **A** | increased | increased |
| **B** | increased | reduced |
| **C** | reduced | increased |
| **D** | reduced | reduced |

0625/22/F/M/25 Q18

**15** The diagram shows four rods. Each rod is made of a different metal.




Wax is used to attach small metal balls at the rod ends P, Q, R and S.

Each rod is the same size. They are heated uniformly by a Bunsen burner at point X.

As the rods warm up, the wax melts and the balls fall off.

Why does the ball on the silver rod fall first?

**A** Silver is the best conductor of heat.

**B** Silver is the worst conductor of heat.

**C** Silver is the best radiator of heat.

**D** Silver is the worst radiator of heat.

0625/22/M/J/25 Q15

**16** Three solid copper spheres, X, Y and Z, are heated to the same temperature.

| sphere | diameter | colour of surface |
|---|---|---|
| X | 2*d* | black |
| Y | *d* | white |
| Z | 2*d* | white |

Which row gives the order for the rate of emission of radiation?

| | highest rate of emission | second highest rate of emission | lowest rate of emission |
|---|---|---|---|
| **A** | X | Y | Z |
| **B** | X | Z | Y |
| **C** | Z | X | Y |
| **D** | Z | Y | X |

0625/22/M/J/25 Q16

**15** A lamp has a metal filament that glows when heated by an electric current.

The middle of the filament has a very high temperature. The ends of the filament are connected to the base of the lamp and are cooler.

Which statement is correct?

**A** Some thermal energy is conducted to the base of the lamp.

**B** The filament radiates energy equally at all points along its length.

**C** The lamp transfers all of the input energy as visible light.

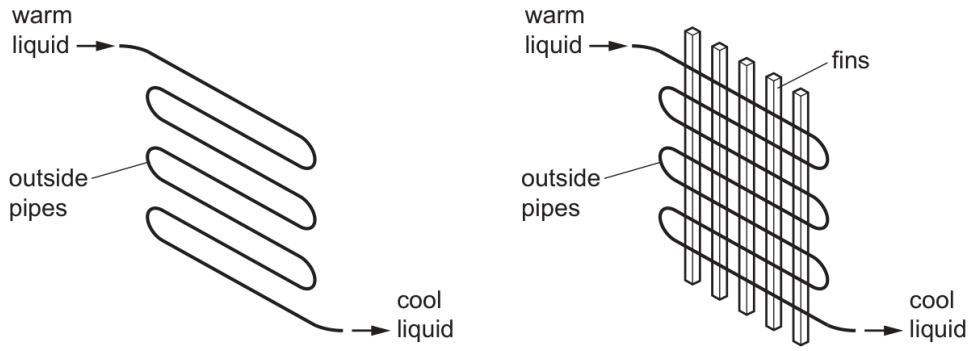
**D** When the potential difference (p.d.) across the filament is halved, the power output is halved.

0625/22/O/N/25 Q15

Date: _____ Score: _____

**16** A liquid passes through pipes inside a refrigerator. The liquid warms as the inside of the refrigerator cools. The warm liquid is then passed through pipes on the outside of the refrigerator to cool it before it goes back through the refrigerator.

The diagrams show two designs for the outside cooling pipes. One design has copper rods, called fins, attached to the pipes.




The outside pipes and fins can be painted silver or black.

Which design of outside pipes will cool the liquid most quickly?

**A** painted black with fins

**B** painted black without fins

**C** painted silver with fins

**D** painted silver without fins

0625/22/O/N/25 Q16

**15** Four different surfaces are at the same high temperature.

Which surface emits thermal radiation at the slowest rate?

| | colour of surface | texture of surface | surface area / $cm^2$ |
|---|---|---|---|
| **A** | black | dull | 10 |
| **B** | black | dull | 100 |
| **C** | white | shiny | 10 |
| **D** | white | shiny | 100 |

0625/23/M/J/25 Q15

**16** An object has a constant temperature.

Which statement is correct?

**A** There is **no** energy transfer to or from the object.

**B** The rate at which the object receives energy is the same as the rate at which it transfers energy away.

**C** The rate at which the object receives energy is smaller than the rate at which it transfers energy away.

**D** The rate at which the object receives energy is greater than the rate at which it transfers energy away.

0625/23/M/J/25 Q16

**15** Which row about the movement of electrons and ions during conduction of thermal energy through a metal is correct?

| | electrons | ions |
|---|---|---|
| **A** | move through metal | vibrate about fixed points |
| **B** | vibrate about fixed points | move through metal |
| **C** | vibrate about fixed points | vibrate about fixed points |
| **D** | move through metal | move through metal |

0625/23/O/N/25 Q15

**16** A scientist measures the air temperature at different heights from the floor in a cave. The results are recorded in the table.

| height / m | temperature / °C |
|---|---|
| 0 | 10 |
| 10 | 11 |
| 20 | 13 |
| 30 | 14 |
| 40 | 16 |

Why does altering the height affect the temperature of the air?

**A** The particles in warm air have less energy than the particles in cool air.

**B** The particles in cool air are further apart than the particles in warm air.

**C** Warm air is less dense than cool air.

**D** Cool air rises above warm air.

0625/23/O/N/25 Q16

**4** Fig. 4.1 shows a heater used to warm the air in a room.

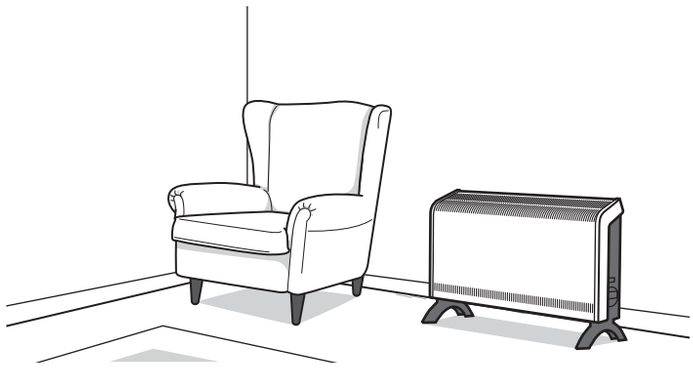


**Fig. 4.1**

**(a) (i)** State the main method of thermal energy transfer throughout the air in the room.

..... [1]

**(ii)** Explain how the heater warms **all** the air in the room.

.....

.....

.....

.....

..... [3]

**(b)** The power of the heater is 2.0 kW when it is connected to the mains supply with an e.m.f. of 230 V.

**(i)** Show that the current in the heater is approximately 8.7 A.

[2]

**(ii)** The plug connecting the heater to the mains supply is fitted with a fuse.

Fuse ratings of 3A, 5A, 10A and 13A are available.

State which fuse is used. Explain your answer.

fuse .....

explanation .....

.....

..... [2]

[Total: 8]

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**4** **(a)** Fig. 4.1 shows two metal containers, A and B.

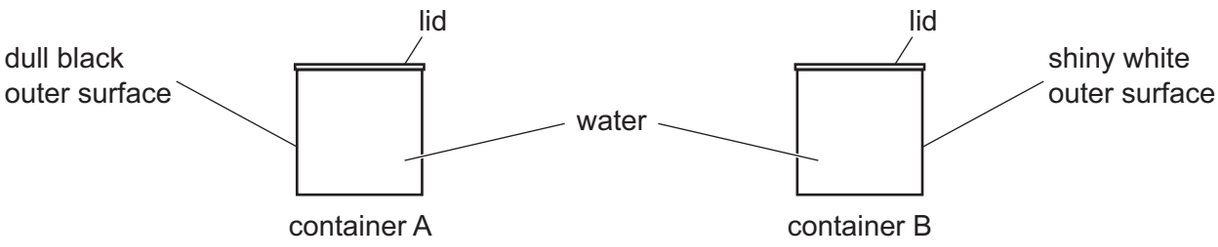


**Fig. 4.1**

Container A has a dull black outer surface and container B has a shiny white outer surface. A and B both contain very hot water at time = 0. The temperature of the water in each container is recorded every minute.

**(i)** State which container has the larger temperature change over the first 5 minutes. Explain your answer.

.....

.....

..... [2]

**(ii)** After 30 minutes, the temperature of the water in container A remains constant.

State, in terms of energy transfers, why the temperature remains constant.

.....

..... [1]

**(iii)** The experiment is repeated without the lids on the containers.

State the name of the additional method of energy transfer that occurs.

..... [1]

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**(b)** Fig. 4.2 shows a metal pan with a wooden handle.

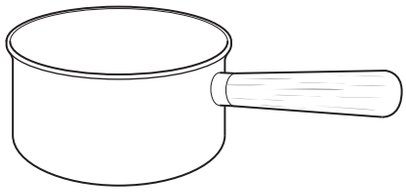


**Fig. 4.2**

Explain why there is a much greater rate of transfer of energy through the metal than through the wood.

.....

.....

.....

..... [3]

[Total: 7]

# 3 WAVES

59 questions · Paper 2 & Paper 4

# Vocabulary · Topic 3 WAVES

词汇表（中英对照）

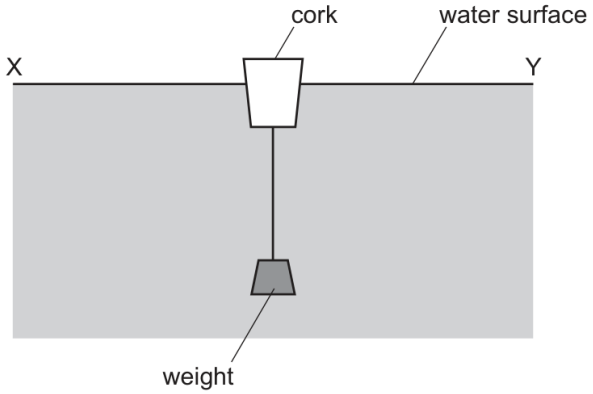
| | | | |
|---|---|---|---|
| **wave** | 波 | **ultraviolet** | 紫外线 |
| **wavelength** | 波长 | **X-rays** | X 射线 |
| **frequency** | 频率 | **gamma rays** | 伽马射线 |
| **amplitude** | 振幅 | **sound** | 声音 |
| **wavefront** | 波前 | **ultrasound** | 超声波 |
| **wave speed** | 波速 | **echo** | 回声 |
| **transverse wave** | 横波 | **critical angle** | 临界角 |
| **longitudinal wave** | 纵波 | **total internal reflection** | 全反射 |
| **reflection** | 反射 | **refractive index** | 折射率 |
| **refraction** | 折射 | **lens** | 透镜 |
| **diffraction** | 衍射 | **converging lens** | 凸透镜 |
| **dispersion** | 色散 | **diverging lens** | 凹透镜 |
| **electromagnetic spectrum** | 电磁波谱 | **focal length** | 焦距 |
| **radio waves** | 无线电波 | **real image** | 实像 |
| **microwaves** | 微波 | **virtual image** | 虚像 |
| **infrared** | 红外线 | **magnifying glass** | 放大镜 |
| **visible light** | 可见光 | | |

# 3.1 General properties of waves

11 MCQ + 2 theory questions · Paper 2 & Paper 4

· Know that waves transfer energy without transferring matter
· Describe what is meant by wave motion as illustrated by vibrations in ropes and springs, and by experiments using w
· Describe the features of a wave in terms of wavefront, wavelength, frequency, crest (peak), trough, amplitude and w
· Recall and use the equation for wave speed v = f λ
· Know that for a transverse wave, the direction of vibration is at right angles to the direction of propagation and
· Know that for a longitudinal wave, the direction of vibration is parallel to the direction of propagation and under
· Describe how waves can undergo: (a) reflection at a plane surface (b) refraction due to a change of speed (c) diffr

**17** The diagram shows a cork with a weight attached so that the cork floats upright in water.




Transverse waves travel across the water from X to Y.

In which direction do the waves make the cork move?

**A** → ← right and left

**B** ↑↓ up and down

**C** → only to the right

**D** ← only to the left

0625/21/M/J/25 Q17

**18** Parallel waves in a ripple tank are diffracted as they pass through a gap in a barrier.

Three changes that can be made to this arrangement are listed.

1 Decrease the wavelength of the waves.

2 Increase the amplitude of the waves.

3 Decrease the size of the gap.

Which changes will cause the shape of the diffracted waves to be less curved?

**A** 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

0625/21/O/N/25 Q18

**19** The frequency of a wave is doubled. The speed of the wave does **not** change.

What happens to the wavelength of the wave?

**A** It becomes four times as large.

**B** It does **not** change.

**C** It doubles.

**D** It halves.

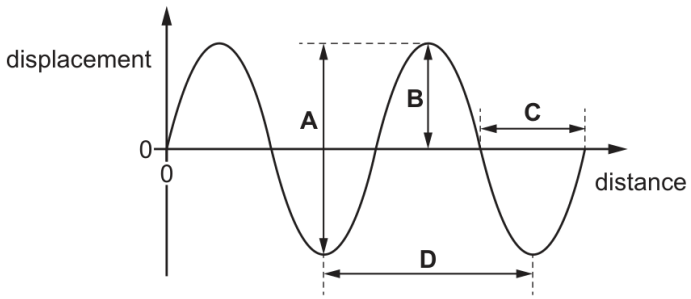
0625/22/F/M/25 Q19

**17** Which wave will diffract the most when passing through a gap that is 0.85 m wide?

| | wave speed / m/s | wave frequency / Hz |
|---|---|---|
| **A** | 300 | 280 |
| **B** | 300 | 400 |
| **C** | 350 | 280 |
| **D** | 350 | 400 |

0625/22/M/J/25 Q17

**18** Which arrow on the graph shows the amplitude of the wave?




0625/22/M/J/25 Q18

**19** Which row describes a seismic S-wave?

| | type of wave | direction of vibration |
|---|---|---|
| **A** | longitudinal | parallel to the direction of travel of the wavefront |
| **B** | longitudinal | perpendicular to the direction of travel of the wavefront |
| **C** | transverse | parallel to the direction of travel of the wavefront |
| **D** | transverse | perpendicular to the direction of travel of the wavefront |

0625/22/M/J/25 Q19

Date: _____ Score: _____

**17** An earthquake-monitoring station records the arrival of 16 complete wavelengths of an earthquake wave in a time of 20 s.

The speed of the earthquake wave is 6.0 km / s.

What is the wavelength of the earthquake wave?

**A** $1.3 \times 10^{-4}$ m **B** $2.1 \times 10^{-4}$ m **C** $4.8 \times 10^{3}$ m **D** $7.5 \times 10^{3}$ m

0625/22/O/N/25 Q17

**18** Which statement is correct?

**A** Seismic S-waves are longitudinal.

**B** Sound waves are longitudinal.

**C** The amplitude is always greater than the wavelength in a longitudinal wave.

**D** The vibration is at right-angles to the direction of propagation in a longitudinal wave.

0625/22/O/N/25 Q18

**17** Which row correctly describes light waves?

| | wave type | direction of vibration |
|---|---|---|
| **A** | longitudinal | parallel to direction of wave travel |
| **B** | longitudinal | perpendicular to direction of wave travel |
| **C** | transverse | parallel to direction of wave travel |
| **D** | transverse | perpendicular to direction of wave travel |

0625/23/M/J/25 Q17

**18** A narrow beam of light consists of two waves of different wavelengths.

The beam of light hits a sharp edge.

Which effect will be seen beyond the sharp edge?

**A** Both waves continue along the same straight line as before.

**B** Both waves bend sharply and continue along a new straight line.

**C** The wave of longer wavelength bends round the sharp edge more than the wave of shorter wavelength.

**D** The wave of shorter wavelength bends round the sharp edge more than the wave of longer wavelength.

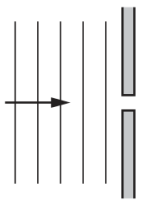
0625/23/M/J/25 Q18

**18** When water waves pass through a gap, they diffract.

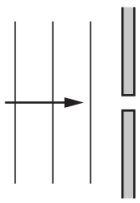
The diagrams show wavefronts approaching a gap.

In which diagram will the diffraction be least?

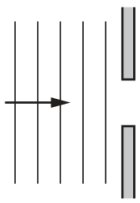
**A**



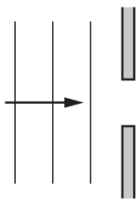
**B**



**C**



**D**



0625/23/O/N/25 Q18

**7** **(a)** Fig. 7.1 is a scale drawing of light waves approaching a narrow slit.

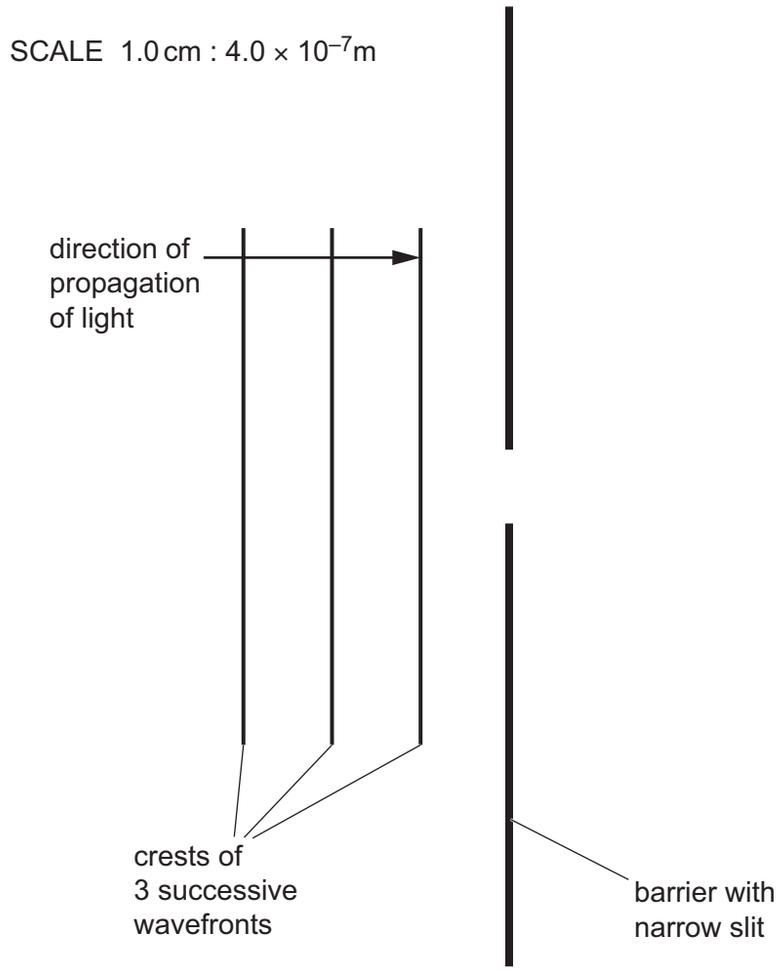



**Fig. 7.1**

**(i)** Name the wave effect produced by the narrow slit.

..... [1]

**(ii)** Using Fig. 7.1, determine the wavelength of the light. Give your answer to **two** significant figures.

wavelength = ..... [2]

**(iii)** On Fig. 7.1, draw **three** wavefronts that have passed through the narrow slit. [3]

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**(b)** A foghorn emits a sound with frequency 380 Hz. The sound is heard by a ship 2.5 km away from the foghorn. The speed of sound in air is 330 m / s.

**(i)** Show that the wavelength of the sound is approximately 0.9 m.
State any equation you use in words or symbols.

[2]

**(ii)** Calculate the time it takes for sound to travel to the ship from the foghorn.

time = ..... [2]

[Total: 10]

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**6** **(a)** Fig. 6.1 shows successive crests of a water wave approaching a boundary.

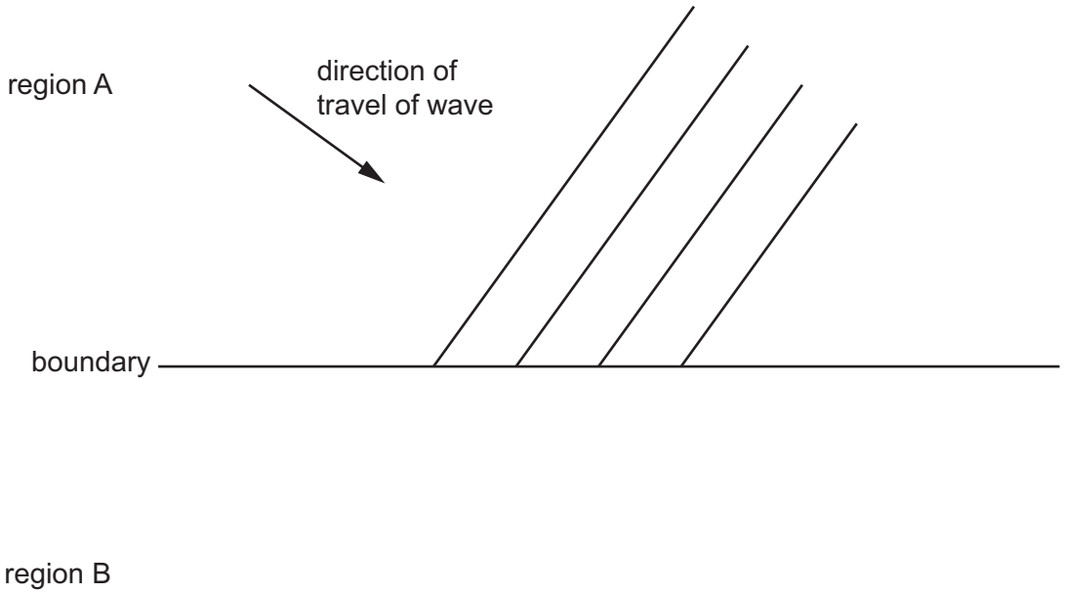



**Fig. 6.1**

**(i)** The speed of the waves in region B is lower than the speed of the waves in region A.

On Fig. 6.1, draw the crests of the waves in region B. [3]

**(ii)** State the wave effect that occurs as the wave crosses the boundary between region A and region B.

..... [1]

**(b)** **(i)** Light can be totally internally reflected when striking the boundary between two different regions.

State **two** conditions that are necessary for total internal reflection to occur.

1 .....

.....

2 .....

.....
[2]

**(ii)** State **two** advantages of using optical fibres for transmitting high speed broadband.

1 .....

2 .....
[2]

[Total: 8]

# 3.2 Light

22 MCQ + 5 theory questions · Paper 2 & Paper 4

· Define and use the terms normal, angle of incidence and angle of reflection
· Describe the formation of an optical image by a plane mirror and give its characteristics, i.e. same size, same dis
· State that for reflection, the angle of incidence is equal to the angle of reflection; recall and use this relation
· Use simple constructions, measurements and calculations for reflection by plane mirrors
· Define and use the terms normal, angle of incidence and angle of refraction
· Describe an experiment to show refraction of light by transparent blocks of different shapes
· Describe the passage of light through a transparent material (limited to the boundaries between two mediums only)

**18** A thin converging lens is used as a magnifying glass.

How does the image compare with the object?

**A** real and inverted

**B** real and upright

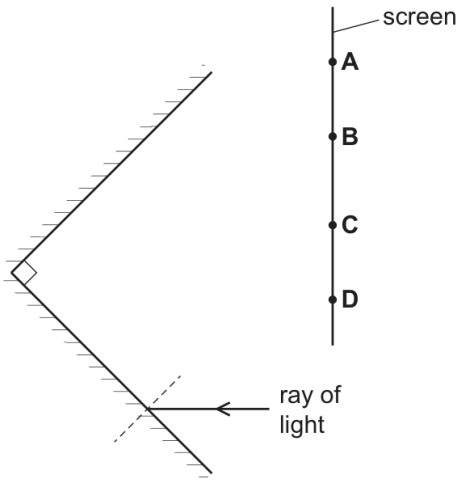
**C** virtual and inverted

**D** virtual and upright

0625/21/M/J/25 Q18

**19** The diagram shows two plane mirrors at 90° to each other. A ray of light is incident on one of the mirrors. The ray reflects off both mirrors before reaching a screen.

At which labelled point does the ray reach the screen?




0625/21/M/J/25 Q19

**20** The speed of light in a material is 50% of the speed of light in air.

What is the refractive index of the material?

**A** 0.5 **B** 2.0 **C** $6.0 \times 10^8$ **D** $1.5 \times 10^8$

0625/21/M/J/25 Q20

**21** A beam of white light is split into a spectrum of seven colours.

Which name is given to this process?

**A** diffraction

**B** dispersion

**C** reflection

**D** refraction

0625/21/M/J/25 Q21

**17** A ray of light reflects from a plane mirror.

The angle between the incident ray and the surface of the mirror is 35°.

Which statement is correct?

**A** The angle of incidence is 35°.

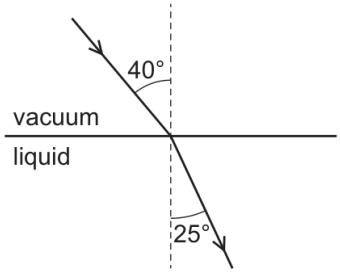
**B** The angle of incidence is 55°.

**C** The angle of reflection is 70°.

**D** The angle of reflection is 110°.

0625/21/O/N/25 Q17

**19** A beam of light passes through a vacuum and then enters a transparent liquid. The diagram shows the path it takes.




The light travels through the vacuum at a speed of $3.0 \times 10^8\,\text{m/s}$.

What is the speed of light in the liquid?

**A** $1.9 \times 10^8\,\text{m/s}$

**B** $2.0 \times 10^8\,\text{m/s}$

**C** $4.6 \times 10^8\,\text{m/s}$

**D** $4.8 \times 10^8\,\text{m/s}$

0625/21/O/N/25 Q19

**20** Which statement about a thin converging lens is correct?

**A** All rays of light refracted by the lens pass through the principal focus.

**B** All rays initially parallel to the principal axis of the lens are refracted through the principal focus.

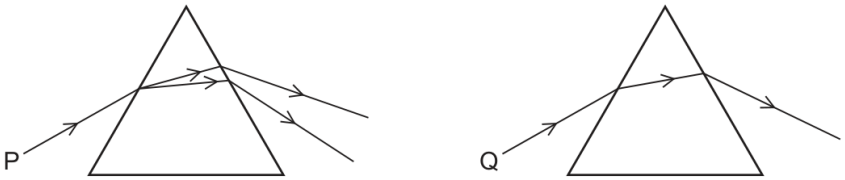
**C** The focal length of the lens is the distance between the image and the principal focus.

**D** The focal length of the lens is the distance between the object and the image.

0625/21/O/N/25 Q20

Date: _____ Score: _____

**21** The diagram shows two narrow beams of light, P and then Q, passing through the same prism. Beams P and Q are both incident on the same point on the prism and at the same angle.




Which row correctly compares beams P and Q?

| | dispersion | nature of light |
|---|---|---|
| **A** | only Q is dispersed | P is monochromatic and Q is white light |
| **B** | only Q is dispersed | P is white light and Q is monochromatic |
| **C** | only P is dispersed | P is monochromatic and Q is white light |
| **D** | only P is dispersed | P is white light and Q is monochromatic |

0625/21/O/N/25 Q21

**21** Which statement about the image produced by a plane mirror is **not** correct?

**A** The image is real.

**B** The image is upright.

**C** The image is the same size as the object.

**D** The distance from the image to the mirror is the same as the distance from the object to the mirror.

0625/22/F/M/25 Q21

**22** A ray of light is incident on a boundary between air and water.

Under which conditions is the ray of light totally internally reflected?

**A** The ray, initially in the air, is incident on the air–water boundary at any angle of incidence.

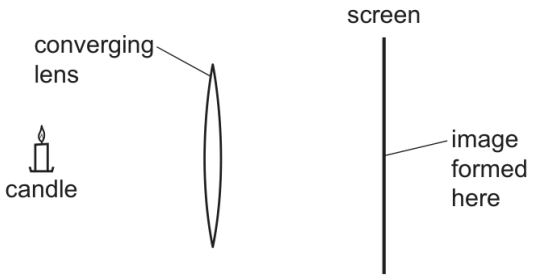
**B** The ray, initially in the air, is incident on the air–water boundary at an angle of incidence greater than the critical angle.

**C** The ray, initially in the water, is incident on the water–air boundary at any angle of incidence.

**D** The ray, initially in the water, is incident on the water–air boundary at an angle of incidence greater than the critical angle.

0625/22/F/M/25 Q22

**23** A converging lens is used to make an image on a screen.




Which type of image is formed on the screen?

**A** real and inverted

**B** real and upright

**C** virtual and inverted

**D** virtual and upright

0625/22/F/M/25 Q23

**20** The image of an object is formed by a plane mirror.

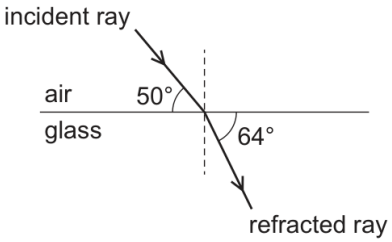
Which row gives the correct properties of the image?

| | real or virtual | distance of the image from the mirror |
|---|---|---|
| **A** | real | closer to the mirror than the object |
| **B** | real | the same distance from the mirror as the object |
| **C** | virtual | closer to the mirror than the object |
| **D** | virtual | the same distance from the mirror as the object |

0625/22/M/J/25 Q20

**21** A ray of light, travelling in air, is incident on an air–glass boundary.

The angles between the incident ray, the refracted ray and the boundary are shown.

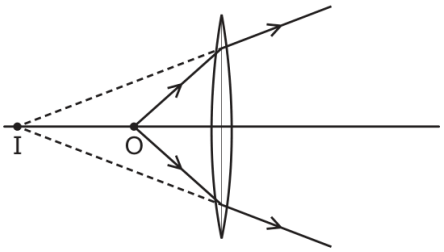



Which equation is used to calculate the refractive index $n$ of the glass?

**A** $n = \frac{\sin 40°}{\sin 26°}$ **B** $n = \frac{\sin 40°}{\sin 64°}$ **C** $n = \frac{\sin 50°}{\sin 26°}$ **D** $n = \frac{\sin 50°}{\sin 64°}$

0625/22/M/J/25 Q21

**22** A small object O is placed near a converging lens, as shown. The lens forms an image I.




Which statement is correct?

**A** The image I is diminished.

**B** The image I is inverted.

**C** The image I is real.

**D** The object O is closer to the lens than its principal focus.

0625/22/M/J/25 Q22

**19** A water droplet is falling towards a horizontal plane mirror.

At one moment, the droplet is a distance of 1.5 m above the mirror and falling at a speed of 2.0 m / s.

What is the distance between the droplet and its image and what is the speed of the image at this moment?

| | distance between water droplet and its image / m | speed of image / m / s |
|---|---|---|
| **A** | 1.5 | 2.0 |
| **B** | 1.5 | 4.0 |
| **C** | 3.0 | 2.0 |
| **D** | 3.0 | 4.0 |

0625/22/O/N/25 Q19

**20** Total internal reflection may occur when light reaches an air–glass boundary.

Under which conditions is light totally internally reflected?

| | medium in which light travels towards the boundary | angle of incidence |
|---|---|---|
| **A** | air | greater than the critical angle |
| **B** | air | less than the critical angle |
| **C** | glass | greater than the critical angle |
| **D** | glass | less than the critical angle |

0625/22/O/N/25 Q20

**21** A long-sighted person **cannot** see objects clearly that are a short distance away because the image is blurred.

Where is the image formed relative to the retina and which type of lens is used to correct long-sightedness?

| | position of image | lens used for correction |
|---|---|---|
| **A** | behind retina | converging |
| **B** | behind retina | diverging |
| **C** | in front of retina | converging |
| **D** | in front of retina | diverging |

0625/22/O/N/25 Q21

Date: _____ Score: _____

**19** An object is placed 30 cm in front of a plane mirror.

Which statement describes the image of the object?

**A** The image is the same size as the object and 30 cm from the object.

**B** The image is the same size as the object and 60 cm from the object.

**C** The image is smaller than the object and 30 cm from the object.

**D** The image is smaller than the object and 60 cm from the object.

0625/23/M/J/25 Q19

**20** Which statement about total internal reflection at a glass–air boundary is correct?

**A** The angle of incidence inside the glass must be equal to the critical angle.

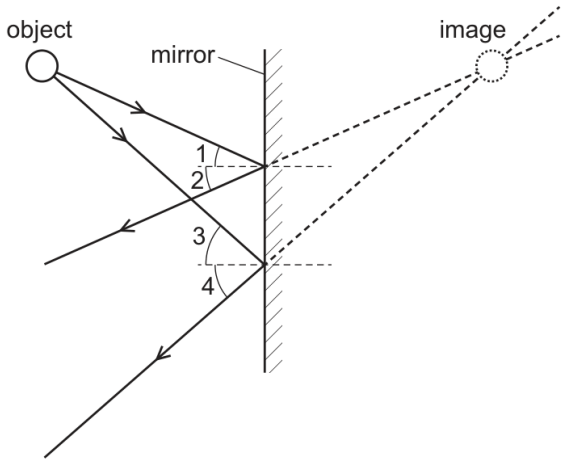
**B** The angle of incidence inside the glass must be less than the critical angle.

**C** The critical angle is proportional to the refractive index of the glass.

**D** The speed of light in glass must be less than the speed of light in air.

0625/23/M/J/25 Q20

**19** The diagram shows a scale drawing of the formation of an image by the reflection of light from a plane mirror.




What are the relationships between the labelled angles?

**A** angle 1 = angle 2, and angle 3 = angle 4

**B** angle 1 = angle 3, and angle 2 = angle 4

**C** angle 1 = angle 4, and angle 3 = angle 2

**D** angle 1 = angle 2 = angle 3 = angle 4

0625/23/O/N/25 Q19

**20** The speed of light in a vacuum is $3.0 \times 10^8$ m/s.

The speed of light in glass is $2.2 \times 10^8$ m/s.

A light ray in a vacuum enters glass with an angle of incidence of 40°.

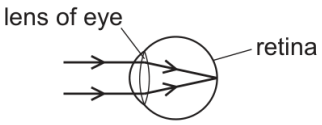
What is the angle of refraction within the glass?

**A** 28° **B** 29° **C** 55° **D** 61°

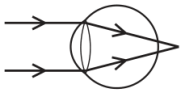
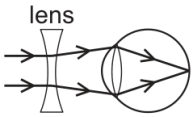
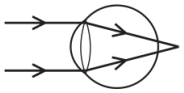
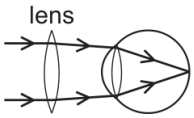
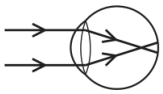
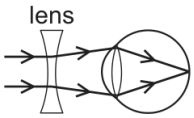
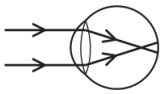
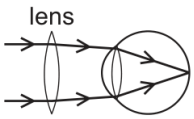
0625/23/O/N/25 Q20

Date: _____ Score: _____

**21** The diagram shows a human eye that correctly focuses the light on the retina.




Which row shows a short-sighted eye and use of the correct lens to focus the image?

| | ray diagram for short-sighted eye | corrected vision |
|---|---|---|
| **A** |  |   |
| **B** |  |   |
| **C** |  |   |
| **D** |  |   |

0625/23/O/N/25 Q21

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**5** A ray of light is incident on a soap film. Fig. 5.1 shows a magnified image of a small part of the soap film. The ray of light is refracted as it enters the soap film.

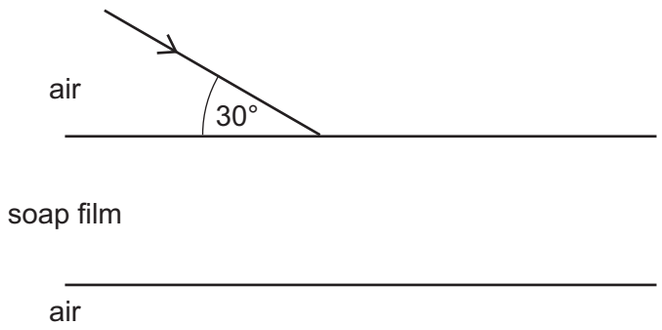


**Fig. 5.1**

The refractive index of the soap film is 1.28.

**(a)** Define refractive index in terms of the speed of light.

.....

..... [1]

**(b)** **(i)** Show that the angle of refraction as the light enters the soap film is approximately 43°.

[2]

**(ii)** On Fig. 5.1, carefully draw the refracted light ray in the soap film and label the angle of refraction. [2]

**(c)** The ray of light is monochromatic red light with a wavelength of 680 nm in air.

**(i)** Define monochromatic.

.....

..... [1]

**(ii)** Calculate the frequency of the light.

frequency = ..... [3]

[Total: 9]

**6** Fig. 6.1 shows part of an optical fibre used in high-speed broadband communication.

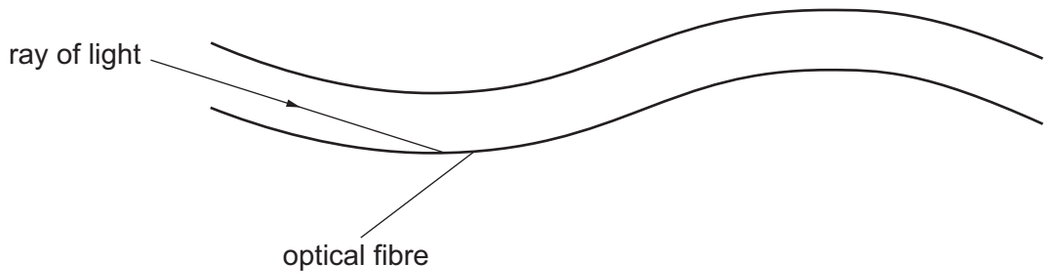


**Fig. 6.1**

**(a)** State **two** advantages of using optical fibres in high-speed data transmission compared to electrical signals sent on copper wires.

.....

..... [2]

**(b)** **(i)** The optical fibre is made of glass with a refractive index of 1.4.

Calculate the critical angle *c*.

*c* = ..... [3]

**(ii)** State the meaning of critical angle.

.....

..... [1]

**(iii)** On Fig. 6.1, label the angle of incidence of the ray of light as it hits the wall of the glass fibre. Draw the continuation of the ray until it leaves the glass fibre. [2]

[Total: 8]

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**6** Fig. 6.1 shows an object O which is 5.0 cm away from the centre of a thin, converging lens L. The focal length of L is 3.0 cm. Fig. 6.1 is drawn to full scale.

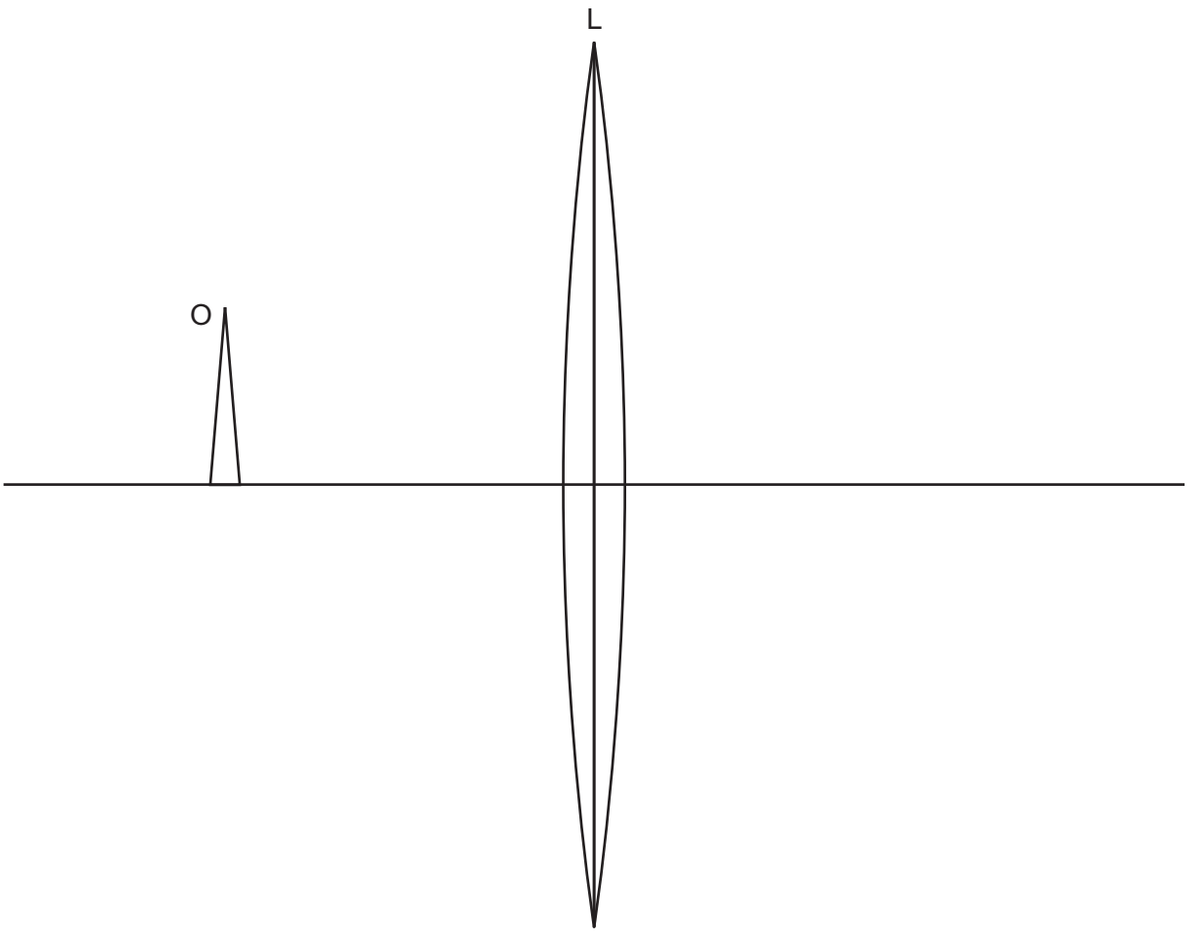


**Fig. 6.1**

**(a) (i)** On Fig. 6.1, label the principal axis with a P. [1]

**(ii)** On Fig. 6.1, place a letter X at a focal point. [1]

**(iii)** On Fig. 6.1, draw two rays from O to locate the tip of the image produced by the lens. [2]

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**(iv)** In Table 6.1, place a tick in the right-hand column next to **all** the terms that describe the image in **(a)(iii)**.

**Table 6.1**

| | |
|---|---|
| diminished | |
| enlarged | |
| inverted | |
| real | |
| same size | |
| upright | |
| virtual | |

[3]

**(b)** The object moves closer to L. The new distance between L and the object is less than the focal length of L.

Describe how the new image is different from the image in **(a)(iv)**.

.....

.....

..... [2]

[Total: 9]

**6** **(a)** Fig. 6.1 is a full-size diagram of a lens and an image I of an object.

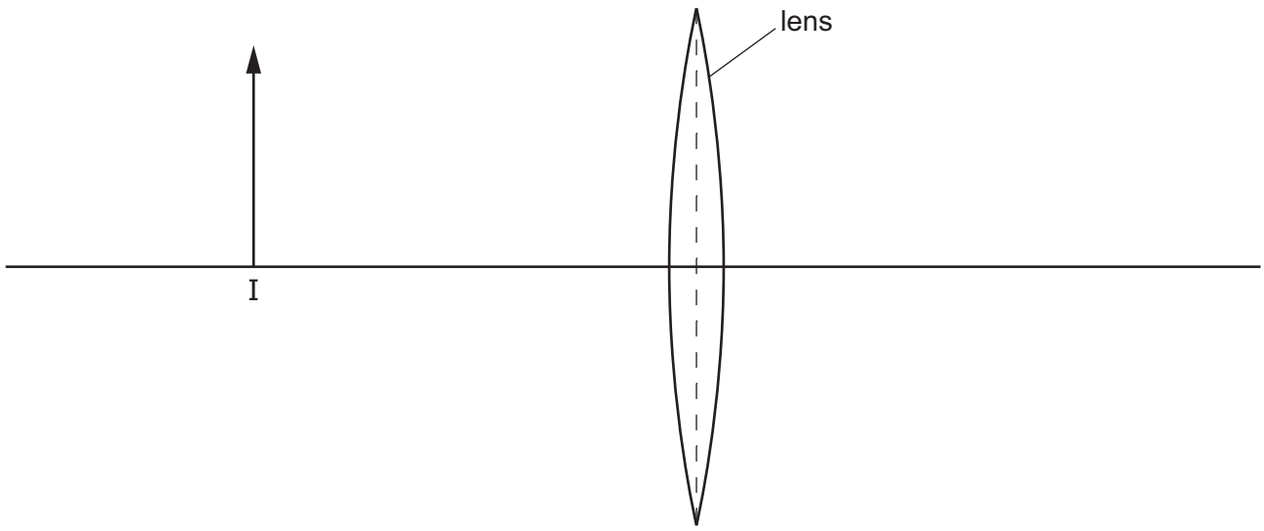


**Fig. 6.1**

The focal length of the lens is 3.0 cm.

**(i)** On Fig. 6.1, mark **two** points with labels F, to show the positions of the principal focuses (focal points) of the lens. [1]

**(ii)** On Fig. 6.1, draw **two** rays from the image I to locate the object. Draw the object and label it O. [3]

**(iii)** The object is moved so that it is 6 cm to the left of the lens.

State **one** characteristic of the image formed when the object is in this position.

..... [1]

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**(b)** Fig. 6.2 shows a simplified diagram of an eye with rays from a near object. The eye needs correction for long-sightedness.

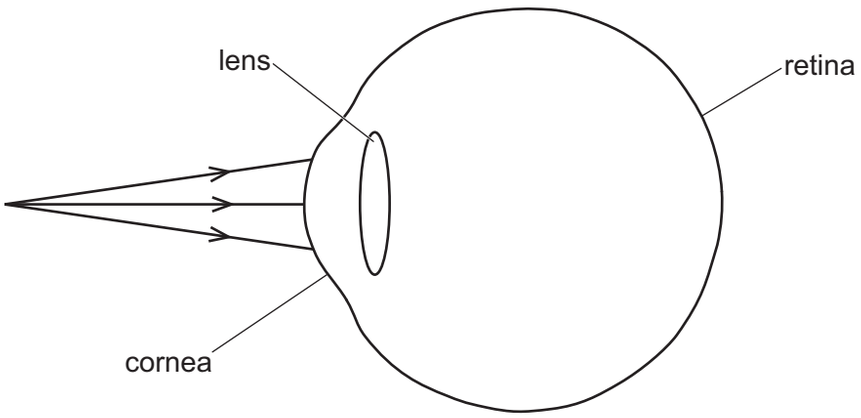


**Fig. 6.2**

**(i)** On Fig. 6.2, draw the path of the rays inside the eye to show the effect of long-sightedness. [1]

**(ii)** A lens is used to correct the long-sightedness.

Draw a lens suitable for this correction.

lens to correct long-sightedness

[1]

[Total: 7]

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**5** Fig. 5.1 shows four rays of red light, P, Q, R and S, coming from a spotlight in a swimming pool.

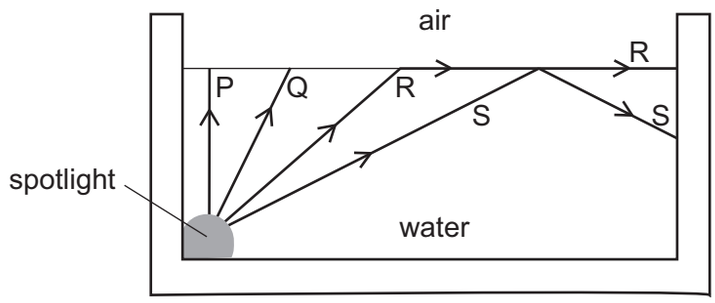


**Fig. 5.1**

**(a)** Define, in words, refractive index for a ray of light travelling from air to water.

..... [1]

**(b)** On Fig. 5.1, draw the path of rays P and Q at the water–air boundary. [2]

**(c)** The angle of incidence for ray R is 49°.

Calculate the refractive index of the water.

refractive index = ..... [2]

**(d)** Explain why ray S is totally internally reflected at the water surface.

.....

..... [2]

[Total: 7]

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# 3.3 Electromagnetic spectrum

6 MCQ + 1 theory questions · Paper 2 & Paper 4

· Know the main regions of the electromagnetic spectrum in order of frequency and in order of wavelength
· Know that all electromagnetic waves travel at the same high speed in a vacuum
· Describe typical uses of the different regions of the electromagnetic spectrum including: (a) radio waves; radio an
· Describe the harmful effects on people of excessive exposure to electromagnetic radiation, including: (a) microwave
· Know that the speed of electromagnetic waves in a vacuum is 3.0 × 108 m / s and is approximately the same in air co
· Know that communication with artificial satellites is mainly by microwaves: (a) some satellite phones use low orbit
· Know that many important systems of communications rely on electromagnetic radiation including: (a) mobile phones (

Date: _____ Score: _____

**22** Satellites are sometimes used to receive and re-transmit television signals.

Which row shows the type of satellite and the electromagnetic waves used?

| | type of satellite | electromagnetic waves used |
|---|---|---|
| **A** | low orbit | infrared |
| **B** | low orbit | microwaves |
| **C** | geostationary | infrared |
| **D** | geostationary | microwaves |

0625/21/M/J/25 Q22

**22** Ultraviolet radiation and infrared radiation are both parts of the electromagnetic spectrum.

However, they differ in wavelength and frequency.

Which row describes how they differ?

| | ultraviolet | infrared |
|---|---|---|
| **A** | longer wavelength than infrared | higher frequency than ultraviolet |
| **B** | longer wavelength than infrared | lower frequency than ultraviolet |
| **C** | shorter wavelength than infrared | higher frequency than ultraviolet |
| **D** | shorter wavelength than infrared | lower frequency than ultraviolet |

0625/21/O/N/25 Q22

**23** What is the approximate speed of light in air?

**A** 300 m / s

**B** 300 km / s

**C** $3.0 \times 10^8$ m / s

**D** $3.0 \times 10^8$ km / s

0625/22/M/J/25 Q23

**22** What is the speed of infrared radiation in glass with a refractive index of 1.5?

**A** 340 m / s

**B** $2.0 \times 10^8$ m / s

**C** $3.0 \times 10^8$ m / s

**D** $4.5 \times 10^8$ m / s

0625/22/O/N/25 Q22

**21** What is the speed of radio waves in a vacuum and what is the speed of gamma rays in a vacuum?

| | speed of radio waves | speed of gamma rays |
|---|---|---|
| **A** | less than $3.0 \times 10^8$ m / s | less than $3.0 \times 10^8$ m / s |
| **B** | less than $3.0 \times 10^8$ m / s | more than $3.0 \times 10^8$ m / s |
| **C** | $3.0 \times 10^8$ m / s | $3.0 \times 10^8$ m / s |
| **D** | more than $3.0 \times 10^8$ m / s | more than $3.0 \times 10^8$ m / s |

0625/23/M/J/25 Q21

**22** Visible light, X-rays and microwaves are all components of the electromagnetic spectrum.

Which statement about the waves is correct?

**A** In a vacuum, microwaves travel faster than visible light and have a shorter wavelength than visible light.

**B** In a vacuum, microwaves travel at the same speed as visible light and have a shorter wavelength than visible light.

**C** In a vacuum, X-rays travel faster than visible light and have a shorter wavelength than visible light.

**D** In a vacuum, X-rays travel at the same speed as visible light and have a shorter wavelength than visible light.

0625/23/O/N/25 Q22

**5 (a) (i)** Electromagnetic waves have many uses.

On Fig. 5.1 draw **one** line from each use to the region of the electromagnetic (e-m) spectrum it uses.

| Use | Region of e-m spectrum |
|---|---|
| security marking | gamma rays |
| Bluetooth | ultraviolet |
| optical fibres | infrared |
| detection of cancer | radio waves |

**Fig. 5.1**

[2]

**(ii)** State **one** advantage of using microwaves compared with radio waves to transmit mobile (cell) phone signals.

.....

..... [1]

**(b)** Describe the difference between transverse waves and longitudinal waves.

.....

.....

.....

..... [2]

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**(c)** Fig. 5.2 shows a simplified diagram of seismic P-waves travelling through the Earth. The paths of the waves are curved.
Seismic waves are produced by rock movements in earthquakes.

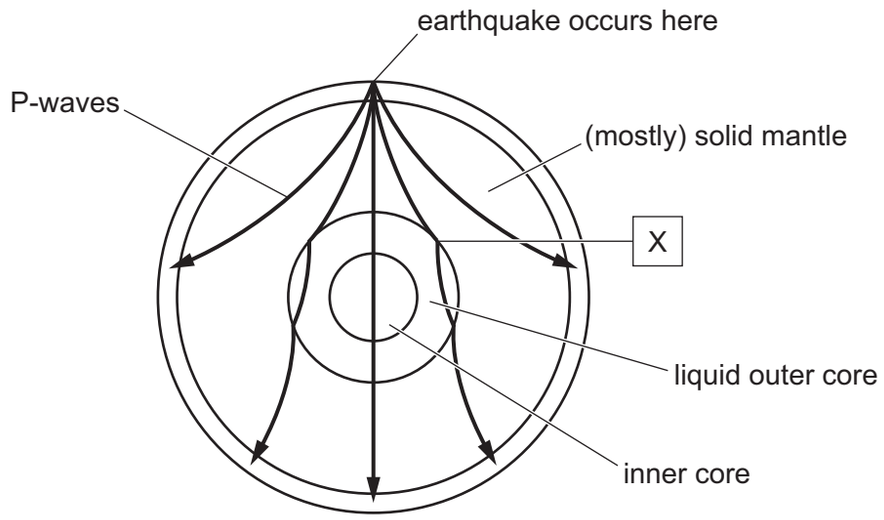


**Fig. 5.2** (not to scale)

**(i)** State the type of wave that P-waves can be modelled as.

..... [1]

**(ii)** At point X, the P-wave travels from the solid mantle to the liquid core. There is a sudden change in direction of the path of the wave.

Explain the change in direction of the P-wave at point X.

.....

.....

..... [2]

**(d)** The Earth's surface absorbs incoming radiation from the Sun and also emits thermal radiation from its surface.

Over the past 50 years, scientists estimate that the average temperature of the Earth's surface has increased by approximately 0.75 °C.

Explain what is causing this average temperature rise.

.....

.....

.....

..... [2]

[Total: 10]

# 3.4 Sound

9 MCQ + 3 theory questions · Paper 2 & Paper 4

· Describe the production of sound by vibrating sources
· Describe the longitudinal nature of sound waves
· State the approximate range of frequencies audible to humans as 20 Hz to 20 000 Hz
· Know that a medium is needed to transmit sound waves
· Know that the speed of sound in air is approximately 330–350 m / s
· Describe a method involving a measurement of distance and time for determining the speed of sound in air
· Describe how changes in amplitude and frequency affect the loudness and pitch of sound waves

**23** A sound wave travels through the air and strikes a solid block of steel.

Some of the sound is reflected from the steel, and some is transmitted through the steel.

How does the speed and the wavelength of the sound in steel compare to the speed and the wavelength of the sound in air?

| | speed in steel | wavelength in steel |
|---|---|---|
| **A** | less than in air | less than in air |
| **B** | less than in air | greater than in air |
| **C** | greater than in air | less than in air |
| **D** | greater than in air | greater than in air |

0625/21/M/J/25 Q23

**23** A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does **not** change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

| | amplitude | frequency |
|---|---|---|
| **A** | constant | constant |
| **B** | constant | varying |
| **C** | varying | constant |
| **D** | varying | varying |

0625/21/O/N/25 Q23

**20** Which statement about ultrasound is correct?

**A** It has a higher frequency than audible sound and it is a longitudinal wave.

**B** It has a higher frequency than audible sound and it is a transverse wave.

**C** It has a lower frequency than audible sound and it is a longitudinal wave.

**D** It has a lower frequency than audible sound and it is a transverse wave.

0625/22/F/M/25 Q20

**24** A ship sends an ultrasound pulse to the seabed to measure the depth of the sea.

The speed of ultrasound waves in sea water is 1500 m / s.

The pulse takes 4.6 seconds to travel to the seabed and back to the ship.

What is the depth of the sea?

**A** 330 m **B** 650 m **C** 3500 m **D** 6900 m

0625/22/M/J/25 Q24

**25** A dolphin has a range of audible frequencies of 150 Hz–150 kHz.

Which range of frequencies can be heard both by humans with good hearing and by dolphins?

**A** 20 Hz–150 Hz

**B** 20 Hz–150 kHz

**C** 20 kHz–150 kHz

**D** 150 Hz–20 kHz

0625/22/M/J/25 Q25

**23** Sound travels through air as a series of compressions and rarefactions.

Which statement correctly compares a compression with a rarefaction?

**A** In a compression, the wavelength is longer than in a rarefaction.

**B** In a compression, the wavelength is shorter than in a rarefaction.

**C** In a compression, the density of the air is greater than in a rarefaction.

**D** In a compression, the density of the air is lower than in a rarefaction.

0625/22/O/N/25 Q23

**22** A student makes a sound in front of a large wall.

She hears an echo at time $t$ after making the sound.

The speed of sound in air is $v$.

Which expression is used to calculate the distance between the student and the wall?

**A** $v \times t$ **B** $v \div t$ **C** $(v \times t) \div 2$ **D** $(v \div t) \div 2$

0625/23/M/J/25 Q22

**23** What is a compression?

**A** a point in a longitudinal wave where the particles of the medium are close together

**B** a point in a longitudinal wave where the particles of the medium are far apart

**C** a point in a transverse wave where the particles of the medium are close together

**D** a point in a transverse wave where the particles of the medium are far apart

0625/23/M/J/25 Q23

**23** Both the amplitude and the frequency of a sound wave decrease.

What happens to the sound that is heard?

**A** The sound is louder and has a higher pitch.

**B** The sound is louder and has a lower pitch.

**C** The sound is quieter and has a higher pitch.

**D** The sound is quieter and has a lower pitch.

0625/23/O/N/25 Q23

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**5** A dolphin communicates with other dolphins underwater by emitting sounds in the range 7–15 kHz.

**(a)** State the value of the speed of sound in air and state how the speed of sound in water differs from the speed of sound in air.

speed of sound in air ..... m/s

speed of sound in water .....

[1]

**(b)** State and explain if humans with normal hearing can hear all the sounds emitted by the dolphin.

statement .....

explanation .....

.....

[2]

**(c)** Complete Table 5.1 to describe differences in loudness and pitch of two different dolphin sounds.

**Table 5.1**

| Frequency / kHz | amplitude | loudness | pitch |
|---|---|---|---|
| 14 | large | | |
| 8 | small | | |

[2]

**(d)** Complete the sentences to describe how sound is transmitted through water.

Sound waves are made of vibrating ..... which produce compressions and rarefactions. A compression is a region of ..... and a rarefaction is a region of ..... . The sound waves travel ..... to the direction of the vibrations.

[3]

[Total: 8]

**5** **(a)** Describe an experiment to determine the speed of sound in air.

State the apparatus required, the measurements taken and how to calculate the speed of sound in air from the measurements.

You may draw a **labelled** diagram as part of your answer.

.....

.....

.....

.....

.....

.....

.....

[4]

**(b)** Bluetooth is used to connect a mobile (cell) phone to headphones.

**(i)** State the name of the region of the electromagnetic spectrum used by Bluetooth.

..... [1]

**(ii)** State the speed of electromagnetic waves in air.

speed = ..... [1]

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**(iii)** The frequency of a Bluetooth network is 2.48 GHz.

Calculate the wavelength of the waves used in the Bluetooth network.

wavelength = ..... [3]

[Total: 9]

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**6** Ultrasound is an example of a longitudinal wave.

**(a)** Define the term ultrasound.

..... [1]

**(b)** Describe what is meant by a longitudinal wave.

.....

..... [1]

**(c)** Ultrasound is used to locate objects below the surface of the sea.

**(i)** Describe how ultrasound is used to locate an object below the surface of the sea.

You may draw a labelled diagram as part of your answer.

.....

.....

.....

.....
[3]

**(ii)** State **one** other use of ultrasound.

.....

..... [1]

[Total: 6]

# 4 ELECTRICITY AND MAGNETISM

80 questions · Paper 2 & Paper 4

# Vocabulary · Topic 4 ELECTRICITY AND MAGNETISM

词汇表（中英对照）

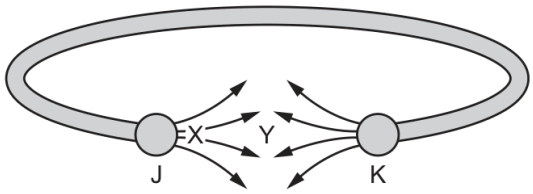
| | |
|---|---|
| **electric charge** | 电荷 |
| **current** | 电流 |
| **potential difference (p.d.)** | 电势差 |
| **electromotive force (e.m.f.)** | 电动势 |
| **resistance** | 电阻 |
| **voltage** | 电压 |
| **circuit** | 电路 |
| **series** | 串联 |
| **parallel** | 并联 |
| **resistor** | 电阻器 |
| **diode** | 二极管 |
| **thermistor** | 热敏电阻 |
| **LDR** | 光敏电阻 |
| **ammeter** | 电流表 |
| **voltmeter** | 电压表 |
| **fuse** | 保险丝 |
| **magnet** | 磁铁 |
| **magnetic field** | 磁场 |
| **magnetic field lines** | 磁感线 |
| **pole** | 磁极 |
| **electromagnet** | 电磁铁 |
| **solenoid** | 螺线管 |
| **electromagnetic induction** | 电磁感应 |
| **generator** | 发电机 |
| **transformer** | 变压器 |
| **step-up** | 升压 |
| **step-down** | 降压 |
| **motor** | 电动机 |
| **commutator** | 换向器 |
| **brushes** | 电刷 |
| **earthing** | 接地 |

# 4.1 Simple phenomena of magnetism

4 MCQ + 1 theory questions · Paper 2 & Paper 4

· Describe the forces between magnetic poles and between magnets and magnetic materials, including the use of the ter
· Describe induced magnetism
· State the differences between the properties of temporary magnets (made of soft iron) and the properties of permane
· State the difference between magnetic and non-magnetic materials
· Describe a magnetic field as a region in which a magnetic pole experiences a force
· Draw the pattern and direction of magnetic field lines around a bar magnet
· State that the direction of a magnetic field at a point is the direction of the force on the N pole of a magnet at

**24** The diagram shows a bracelet containing two magnets. The arrows represent the pattern and the direction of the magnetic field due to the magnets.



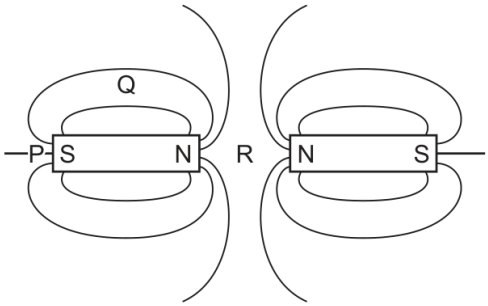

Which row identifies the magnetic pole at J, the magnetic pole at K and where the magnetic field is the strongest?

| | magnetic pole at J | magnetic pole at K | where the magnetic field is the strongest |
|---|---|---|---|
| **A** | N | N | X |
| **B** | N | N | Y |
| **C** | S | S | X |
| **D** | S | S | Y |

0625/21/M/J/25 Q24

Date: _____ Score: _____

**24** The diagram shows part of the magnetic field around two magnets.




Which list gives the order of magnetic field strengths at the positions P, Q and R, from weakest to strongest?

**A** P → Q → R

**B** P → R → Q

**C** Q → P → R

**D** R → Q → P

0625/21/O/N/25 Q24

**24** Which statements about magnets are correct?

1 Permanent magnets are made of steel.

2 Electromagnets are temporary magnets.

3 A magnet can be used to induce magnetism in an iron bar.

**A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

0625/22/O/N/25 Q24

Date: _____ Score: _____

**24** Which row describes the magnetic field around a weak magnet?

| | direction of field | spacing of field lines |
|---|---|---|
| **A** | north pole to south pole | close together |
| **B** | north pole to south pole | far apart |
| **C** | south pole to north pole | close together |
| **D** | south pole to north pole | far apart |

0625/23/M/J/25 Q24

**6** Fig. 6.1 shows a person using a magnetic window cleaner. The part on the outside of the window is attracted to the inside part through the glass window.

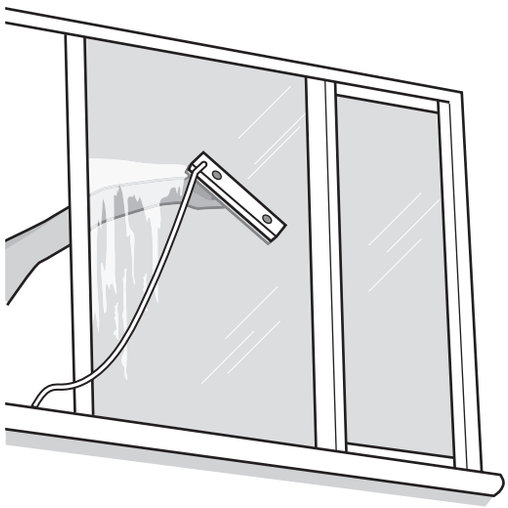


**Fig. 6.1**

Each part of the window cleaner contains two magnets. Fig. 6.2 shows the magnetic field between the parts of the window cleaner.

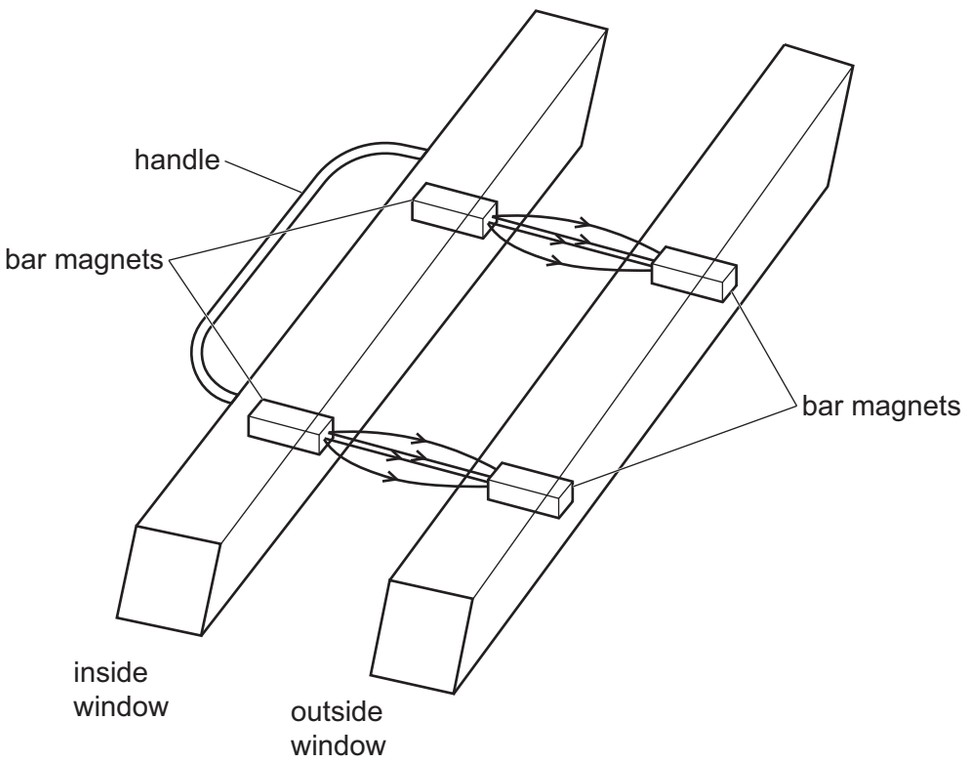



**Fig. 6.2**

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**(a)** Glass is **not** a magnetic material.

State the difference between magnetic and non-magnetic materials.

.....

..... [1]

**(b)** Suggest a suitable material for the magnets in the window cleaner. Explain your answer.

.....

..... [1]

**(c)** Label the poles of the magnets in Fig. 6.2. [1]

**(d)** State how the field lines in Fig. 6.2 show different strengths of the magnetic field between the magnets.

.....

..... [1]

[Total: 4]

# 4.2 Electrical quantities

21 MCQ + 3 theory questions · Paper 2 & Paper 4

· State that there are positive and negative charges
· State that positive charges repel other positive charges, negative charges repel other negative charges, but positi
· Describe simple experiments to show the production of electrostatic charges by friction and to show the detection o
· Explain that charging of solids by friction involves only a transfer of negative charge (electrons)
· Describe an experiment to distinguish between electrical conductors and insulators
· Recall and use a simple electron model to explain the difference between electrical conductors and insulators and g
· State that charge is measured in coulombs

**25** Which row describes an electric field and its direction?

| | description of electric field | direction of electric field |
|---|---|---|
| **A** | region in which an electric charge experiences a force | direction of the force on a positive charge |
| **B** | region in which an electric charge experiences a force | direction of the force on a negative charge |
| **C** | region in which an electric charge produces a current | direction of the force on a positive charge |
| **D** | region in which an electric charge produces a current | direction of the force on a negative charge |

0625/21/M/J/25 Q25

**26** Copper wire is available in fixed lengths but in various diameters *d*. Each wire has a different resistance *R*.

Which relationship between *R* and *d* is correct?

**A** *R* is directly proportional to *d*.

**B** *R* is directly proportional to $d^2$.

**C** *R* is inversely proportional to *d*.

**D** *R* is inversely proportional to $d^2$.

0625/21/M/J/25 Q26

**25** A fully charged battery stores a charge of 18 000 C. The battery is charged with a current of 830 mA.

How long does it take to fully charge a completely discharged battery?

**A** about 10 minutes

**B** about 3.5 hours

**C** about 4 hours

**D** about 6 hours

0625/21/O/N/25 Q25

**26** What is the definition of electromotive force?

**A** the electrical work done by a source in moving a unit charge around a complete circuit

**B** the electrical work done by a unit charge passing through a component

**C** the mechanical work done by a source in moving a unit charge around a complete circuit

**D** the mechanical work done by a unit charge passing through a component

0625/21/O/N/25 Q26

**27** A cell passes a current of 2.0 A in a circuit for 30 s. In this time, the cell transfers 120 J of energy.

What is the electromotive force (e.m.f.) of the cell?

**A** 0.50 V **B** 1.5 V **C** 2.0 V **D** 8.0 V

0625/21/O/N/25 Q27

**24** A negatively charged rod is brought near to a metal sphere. A positive charge develops on the surface of the metal sphere closest to the negatively charged rod.

Why is this?

**A** Electrons in the sphere are attracted by the negatively charged rod.

**B** Electrons in the sphere are repelled by the negatively charged rod.

**C** Positive charges in the sphere are attracted by the negatively charged rod.

**D** Positive charges in the sphere are repelled by the negatively charged rod.

0625/22/F/M/25 Q24

**25** The electromotive force (e.m.f.) of a cell is defined in terms of which quantities?

**A** the energy transferred by the cell in moving charge around the circuit

**B** the energy transferred by the cell in moving voltage around the circuit

**C** the power transferred by the cell in moving charge around the circuit

**D** the power transferred by the cell in moving voltage around the circuit

0625/22/F/M/25 Q25

**26** A resistor of resistance $R$ is connected to a battery of electromotive force (e.m.f.) $V$.

There is a current $I$ in the resistor.

Power $P$ is dissipated by the resistor.

In time $t$, the energy transferred is $E$.

Which expression is correct?

**A** $E = IVt$ **B** $E = PIt$ **C** $P = VIR$ **D** $P = \frac{V}{R}$

0625/22/F/M/25 Q26

Date: _____ Score: _____

**27** A copper wire of length 80 cm and cross-sectional area 0.20 $mm^2$ has resistance $R$.

Which length of copper wire of cross-sectional area 0.40 $mm^2$ has resistance $R$?

**A** 10 cm **B** 20 cm **C** 40 cm **D** 160 cm

0625/22/F/M/25 Q27

**26** A positively charged balloon is placed against a wall. The charge makes the balloon stick to the wall.

Which statement explains this?

**A** Positive charges on the balloon attract electrons in the wall.

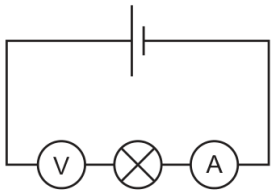
**B** Positive charges on the balloon attract protons in the wall.

**C** Positive charges on the balloon repel electrons in the wall.

**D** Positive charges on the balloon repel protons in the wall.

0625/22/M/J/25 Q26

**27** A student is trying to determine the resistance of a lamp. He sets up the **incorrect** circuit shown.




Which change would allow him to take the measurements needed to calculate the resistance of the lamp?

**A** Disconnect the ammeter and move it to the left of the lamp.

**B** Swap the positions of the ammeter and the voltmeter in the circuit.

**C** Disconnect the ammeter and connect it in parallel with the lamp.

**D** Disconnect the voltmeter and connect it in parallel with the lamp.

0625/22/M/J/25 Q27

**28** A metallic wire X has a resistance.

What is the difference in resistance for a wire of the same material as X that is:

- shorter than X but the same diameter
- the same length as X but with a smaller diameter?

| | resistance of shorter wire | resistance of wire with smaller diameter |
|---|---|---|
| **A** | greater | greater |
| **B** | greater | smaller |
| **C** | smaller | greater |
| **D** | smaller | smaller |

0625/22/M/J/25 Q28

**25** In a circuit, two identical lamps are in parallel. The parallel pair of lamps is in series with a resistor and a battery.

The battery does 120 J of work moving 50 C of charge around the circuit in time *t*.

In each lamp, 40 J of work is done by the charge that passes through it in time *t*.

The electromotive force (e.m.f.) of the battery is *E* and the potential difference (p.d.) across one lamp is *V*.

Which statement about *E* and *V* is correct?

**A** *E* is 2.4 V and *V* is 0.80 V.

**B** *E* is 2.4 V and *V* is 1.6 V.

**C** *E* is 4.8 V and *V* is 2.4 V.

**D** *E* is 4.8 V and *V* is 4.8 V.

0625/22/O/N/25 Q25

**26** Two wires, X and Y, are made from the same metal and have the same resistance.

Which row identifies a possible pair of values for X and for Y?

| | length of X / cm | diameter of X / mm | length of Y / cm | diameter of Y / mm |
|---|---|---|---|---|
| **A** | 50 | 0.40 | 200 | 0.10 |
| **B** | 50 | 0.40 | 200 | 0.20 |
| **C** | 50 | 0.40 | 200 | 0.80 |
| **D** | 50 | 0.40 | 200 | 1.60 |

0625/22/O/N/25 Q26

**25** A 0.60 m length of resistance wire with a diameter of 0.25 mm has a resistance of 6.0 Ω.

A second piece of resistance wire of the same material with a diameter of 0.50 mm also has a resistance of 6.0 Ω.

What is the length of the second piece of wire?

**A** 0.15 m **B** 0.30 m **C** 1.20 m **D** 2.40 m

0625/23/M/J/25 Q25

**26** A circuit contains a cell with an electromotive force (e.m.f.) of 2.0 V. The current in the circuit is 2.0 A.

How much energy is transferred by the cell in 2.0 minutes?

**A** 2.0 J **B** 4.0 J **C** 8.0 J **D** 480 J

0625/23/M/J/25 Q26

**27** What is the electromotive force (e.m.f.) of a cell?

**A** the charge that passes through the cell per unit time

**B** the energy supplied per unit charge as charge passes through the cell

**C** the total charge flowing through the cell

**D** the total energy stored in the cell

0625/23/M/J/25 Q27

**25** Why does a plastic rod become negatively charged when it is rubbed with a cloth?

**A** The rod gains electrons.

**B** The rod gains protons.

**C** The rod loses electrons.

**D** The rod loses protons.

0625/23/O/N/25 Q25

**26** There is an electric current in a circuit with a battery. This question refers to the positive and negative terminals of the battery.

Which row correctly describes the directions of the conventional current and the electron flow in the circuit?

| | direction of conventional current | direction of electron flow |
|---|---|---|
| **A** | from the negative to the positive terminal | from the negative to the positive terminal |
| **B** | from the negative to the positive terminal | from the positive to the negative terminal |
| **C** | from the positive to the negative terminal | from the negative to the positive terminal |
| **D** | from the positive to the negative terminal | from the positive to the negative terminal |

0625/23/O/N/25 Q26

**27** Which electrical quantity is defined in terms of the energy supplied by a source in driving charge round a complete circuit?

**A** current

**B** electromotive force

**C** potential difference

**D** power

0625/23/O/N/25 Q27

**28** A torch has a simple circuit with a 3.0 V battery and a lamp. There is a 20 mA current in the lamp.

How much energy is transferred to the lamp in 5.0 minutes?

**A** 0.30 J **B** 18 J **C** 60 J **D** 0.30 kJ

0625/23/O/N/25 Q28

* 0000800000006 *



**3** Fig. 3.1 shows a mains electric heater used to heat a small room.

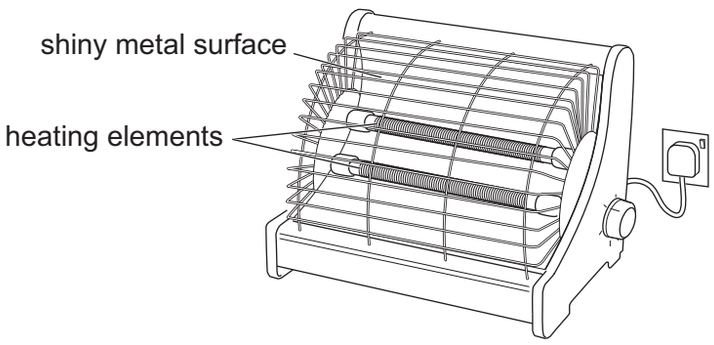



**Fig. 3.1**

**(a)** State the region of the electromagnetic spectrum which radiates thermal energy from the heater.

..... [1]

**(b)** Explain why the shiny metal surface behind the heating elements increases the thermal energy radiated into the room.

.....

.....

..... [2]

**(c)** The metal outer casing of the heater is earthed. State why this is an important safety feature.

.....

..... [1]

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**(d)** The mains voltage is 230 V. The two identical heating elements are connected in parallel. Each heating element has a resistance of 89 Ω.

**(i)** Calculate the current in **one** heating element.

current = ..... [2]

**(ii)** Show that the electrical power of the heater is approximately 1200 W. State any equation you use in words or symbols.

[2]

**(iii)** The heater is 95% efficient at converting electrical work done to thermal energy.

Calculate the thermal energy emitted by the heater in **(d)(ii)** in 60 s. Give your answer to **two** significant figures.

thermal energy = ..... [3]

[Total: 11]

**7** **(a)** State what is meant by an electric field.

.....

..... [1]

**(b)** A plastic rod is rubbed with a cloth. The plastic rod becomes negatively charged and the cloth becomes positively charged.

**(i)** Explain why.

.....

.....

..... [2]

**(ii)** The negatively charged plastic rod is suspended by an insulating thread. Another negatively charged plastic rod is brought close to the suspended rod.

State what happens to the suspended plastic rod.

..... [2]

**(c)** **(i)** Define the kilowatt-hour (kW h) in words.

.....

..... [1]

**(ii)** A small lamp illuminates an electric oven. The lamp has an output power of 25 W and operates for 220 hours in one year. The p.d. across the lamp is 230 V.

**1.** Calculate the energy transferred by the lamp in one year. Give your answer in kW h.

energy = ..... kW h [2]

**2.** Calculate the current in the lamp.

current = ..... [2]

[Total: 10]

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**8** **(a)** **(i)** State what is meant by an electric field.

.....

..... [1]

**(ii)** Fig. 8.1 shows a negative point charge.

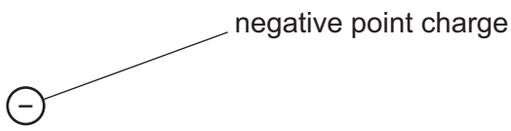



**Fig. 8.1**

On Fig. 8.1, draw **four** field lines to show the pattern and the direction of the electric field due to the negative point charge. [2]

**(b)** The potential difference (p.d.) of a lightning strike is $2.9 \times 10^8$ V. The energy transferred by the lightning strike is $4.5 \times 10^5$ MJ.

Calculate the charge that flows.

charge = ..... [3]

**(c)** The current in a lamp is 2.0 A. The lamp is switched on for a time of 120 s.

Calculate the charge that flows.

charge = ..... [2]

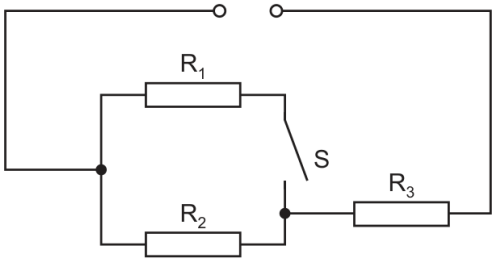
[Total: 8]

# 4.3 Electric circuits

15 MCQ + 6 theory questions · Paper 2 & Paper 4

· Draw and interpret circuit diagrams containing cells, batteries, power supplies, generators, potential dividers, sw
· Draw and interpret circuit diagrams containing diodes and light-emitting diodes (LEDs) and know how these component
· Know that the current at every point in a series circuit is the same
· Know how to construct and use series and parallel circuits
· Calculate the combined e.m.f. of several sources in series
· Calculate the combined resistance of two or more resistors in series
· State that, for a parallel circuit, the current from the source is larger than the current in each branch

**27** The diagram shows three 20 Ω resistors, $R_1$, $R_2$ and $R_3$, and a switch S connected to a d.c. power supply.




Switch S is open.

The potential difference (p.d.) across $R_3$ is 9.0 V.

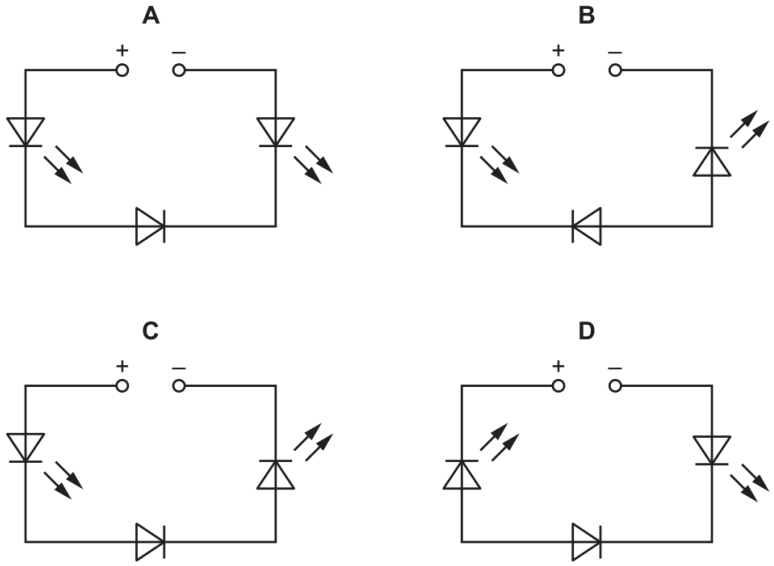
What is the p.d. across $R_3$ when switch S is closed?

**A** 3.0 V **B** 6.0 V **C** 12 V **D** 27 V

0625/21/M/J/25 Q27

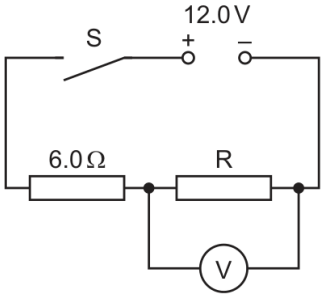
**28** The circuits shown all include a d.c. power supply, one diode and two light-emitting diodes (LEDs).

In which circuit will both LEDs be turned on?




0625/21/M/J/25 Q28

**28** The diagram shows two resistors in series connected to a power supply.




The reading on the voltmeter is 3.0 V when the switch is closed.

What is the resistance of resistor R?

**A** 0.67 Ω **B** 1.2 Ω **C** 1.5 Ω **D** 2.0 Ω

0625/21/O/N/25 Q28

**29** Lamps can be connected in series or in parallel to a power supply.

What is an advantage of connecting the lamps in parallel rather than in series?

**A** All of the lamps can be switched on and off using the same switch.

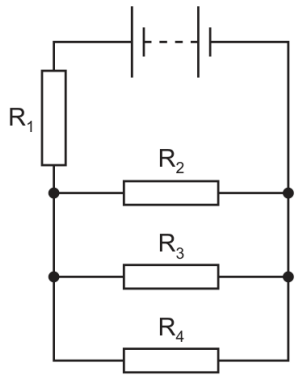
**B** Fewer connecting leads are needed to make the circuit.

**C** When one lamp stops working, the others may continue to work.

**D** Less power is transferred from the supply.

0625/21/O/N/25 Q29

**28** Four identical resistors are connected to a battery as shown. Each resistor has a value of 120 Ω.




Resistor $R_4$ is removed.

Which effect does removing this resistor have on the effective resistance of the circuit?

**A** The effective resistance does **not** change.

**B** The effective resistance decreases by 20 Ω.

**C** The effective resistance increases by 20 Ω.

**D** The effective resistance decreases by 120 Ω.

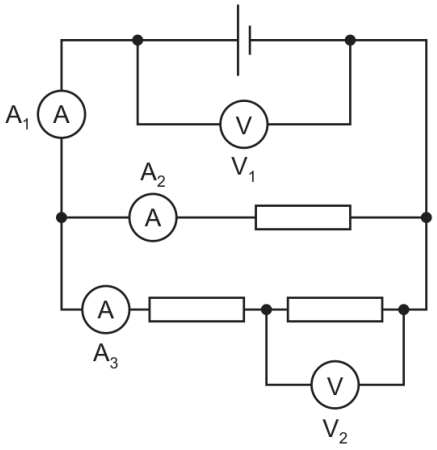
0625/22/F/M/25 Q28

**29** The resistors in the circuit shown are identical.

The reading on voltmeter $V_2$ is 1.5 V.

The reading on ammeter $A_2$ is 0.50 A.

The reading on ammeter $A_3$ is 0.25 A.



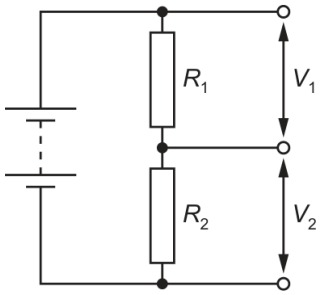

What are the readings on ammeter $A_1$ and voltmeter $V_1$?

| | ammeter $A_1$ / A | voltmeter $V_1$ / V |
|---|---|---|
| **A** | 0.25 | 1.5 |
| **B** | 0.25 | 3.0 |
| **C** | 0.75 | 1.5 |
| **D** | 0.75 | 3.0 |

0625/22/F/M/25 Q29

Date: _____ Score: _____

**29** The diagram shows the circuit for a potential divider.




Which equation is used to calculate $V_1$?

**A** $V_1 = \dfrac{R_1}{V_2 \times (R_1 + R_2)}$

**B** $V_1 = \dfrac{V_2 \times R_1}{R_2}$

**C** $V_1 = \dfrac{V_2 \times R_1}{R_1 + R_2}$

**D** $V_1 = \dfrac{R_1}{V_2 \times R_2}$

0625/22/M/J/25 Q29

**30** Which electrical device is used to convert a.c. to d.c.?

**A** diode

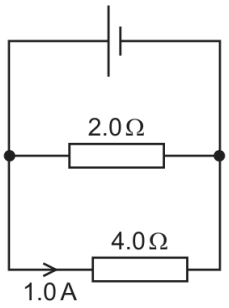
**B** relay

**C** thermistor

**D** transformer

0625/22/M/J/25 Q30

**31** A cell is connected to a parallel combination of a 2.0 Ω resistor and a 4.0 Ω resistor. The current in the 4.0 Ω resistor is 1.0 A.

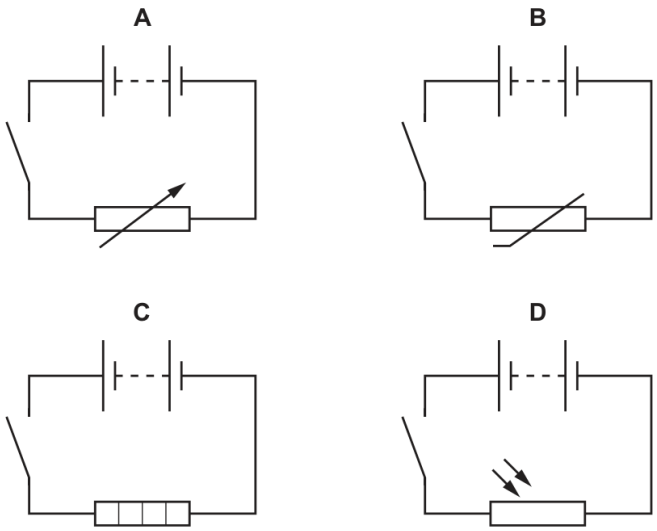



What is the current in the cell?

**A** 1.0 A **B** 1.5 A **C** 2.0 A **D** 3.0 A

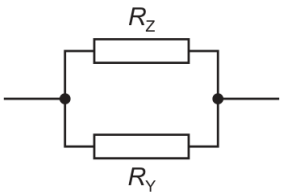
0625/22/M/J/25 Q31

**27** Which circuit contains a light-dependent resistor (LDR)?




0625/22/O/N/25 Q27

**28** A teacher asks three students to write down an equation to determine the combined resistance $R_C$ of resistors with resistances $R_Y$ and $R_Z$.




Which students are correct?

student 1 $\frac{1}{R_C} = \frac{1}{R_Y} + \frac{1}{R_Z}$

student 2 $\frac{1}{R_C} = \frac{1}{R_Y + R_Z}$

student 3 $R_C = \frac{1}{R_Y} + \frac{1}{R_Z}$

**A** 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

0625/22/O/N/25 Q28

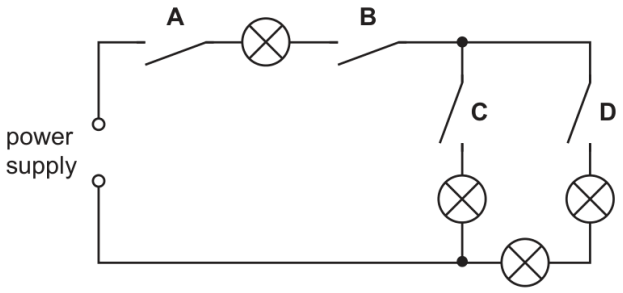
Date: _____ Score: _____

**28** Four lamps and four switches are connected to a power supply, as shown.

When all the switches are closed, all the lamps are lit.

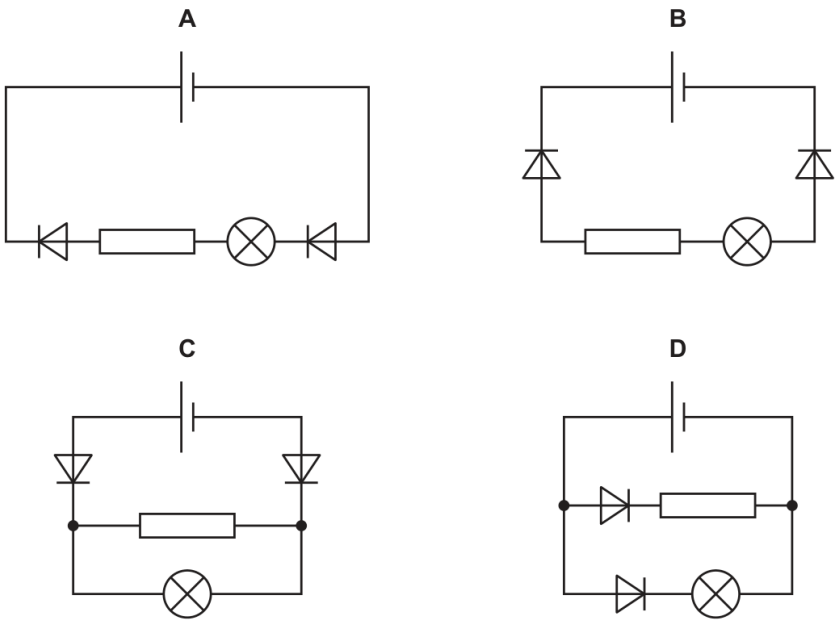
When one of the switches is then opened, only **one** lamp goes out.

Which switch is opened?




0625/23/M/J/25 Q28

**29** In which circuit does the lamp light?

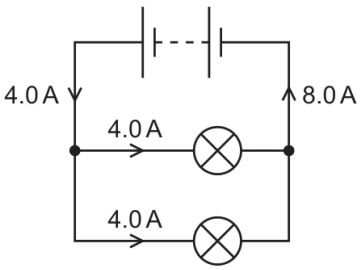



0625/23/O/N/25 Q29

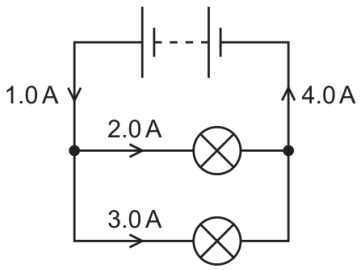
Date: _____ Score: _____

**30** Which circuit shows possible correct values for the currents?

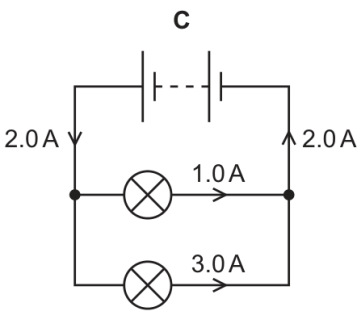
**A**



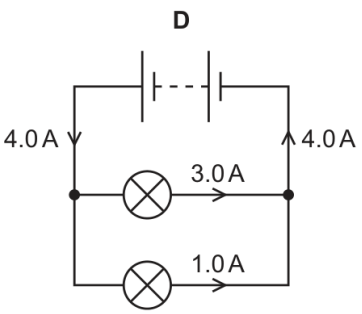

**B**




**C**




**D**

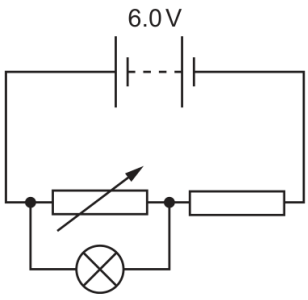



0625/23/O/N/25 Q30

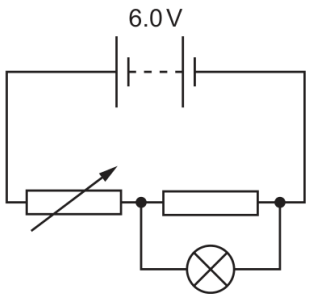
**31** The circuit diagrams show circuits that can control the potential difference (p.d.) across a lamp.

Which circuit allows the p.d. across the lamp to be varied from 0 V to 6.0 V?

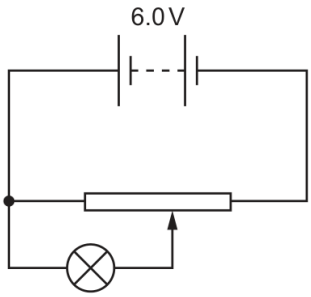
**A**



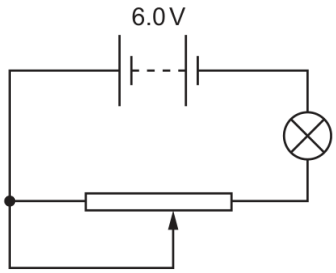

**B**




**C**




**D**




0625/23/O/N/25 Q31

**7** **(a)** Fig. 7.1 shows a sketch of the current–voltage graph for an electrical component.

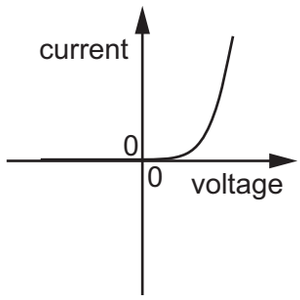



**Fig. 7.1**

**(i)** Name the electrical component.

Explain how you identified the component from the graph in Fig. 7.1.

name .....

explanation .....

.....

[2]

**(ii)** Draw the circuit symbol for this component.

[1]

**(b)** Fig. 7.2 shows an electric circuit for two identical electric heaters, A and B, connected to a mains supply of 230 V.

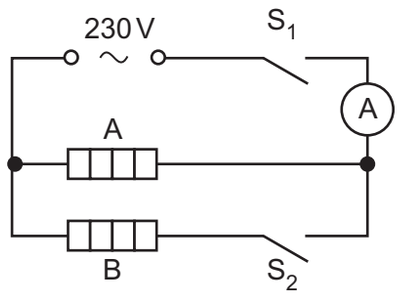



**Fig. 7.2**

$S_1$ is closed. $S_2$ is open. The reading on the ammeter is 3.9 A.

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**(i)** Calculate the resistance of heater A.

resistance = ..... [2]

**(ii)** Calculate the energy transferred by heater A in 5.0 minutes.

energy = ..... [3]

**(iii)** $S_1$ remains closed and $S_2$ is closed.

Determine the reading on the ammeter. Show your working.

ammeter reading = ..... [2]

[Total: 10]

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**7** Fig. 7.1 shows a circuit containing a 6.0 V battery of cells and three identical resistors.

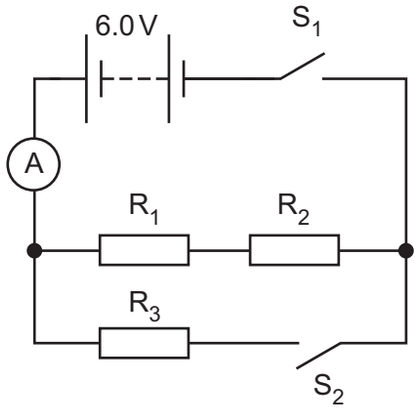


**Fig. 7.1**

**(a)** $S_1$ is closed and $S_2$ is open. The current in the ammeter is 0.080 A.

Calculate the resistance of $R_1$.

resistance = ..... [2]

**(b)** $S_1$ and $S_2$ are both closed.

**(i)** Determine the reading on the ammeter. Show your working.

ammeter reading = ..... [3]

**(ii)** Explain in terms of work done and potential difference why there is a larger heating effect in $R_3$ than in $R_1$.

.....

.....

..... [2]

[Total: 7]

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**5** A light-dependent resistor (LDR) has a low resistance in high light intensity and a high resistance in the dark.

**(a)** Sketch a graph of resistance (*y*-axis) against light intensity (*x*-axis) for an LDR.

[2]

**(b)** Fig. 5.1 shows part of the electric circuit used to turn on a light when it is dark.

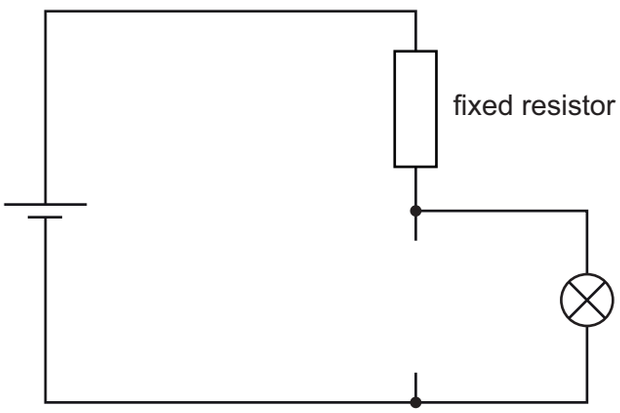


**Fig. 5.1**

**(i)** Complete the circuit in Fig. 5.1 with the symbol for a light-dependent resistor (LDR). [1]

**(ii)** Explain why the lamp is off in the light and the lamp is on in the dark. Use ideas about potential difference (p.d.) in your answer.

.....

.....

.....

..... [3]

[Total: 6]

**7** A circuit consists of an a.c. supply and two lamps. The lamps are connected in parallel.

**(a)** Draw the circuit diagram.

[2]

**(b)** The electromotive force (e.m.f.) of the a.c. supply is 230 V. When connected to the 230 V supply, the resistance of one lamp is 1200 Ω and the resistance of the other lamp is 800 Ω.

**(i)** Calculate the current in the 800 Ω lamp.

current = ..... [2]

**(ii)** Calculate the energy used by the 800 Ω lamp in 5.0 hours.

Give your answer in kWh.

energy = ..... kWh [3]

**(iii)** Calculate the combined resistance of the two lamps in this circuit.

resistance = ..... [2]

[Total: 9]

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**7** A student sets up the circuit shown in Fig. 7.1.

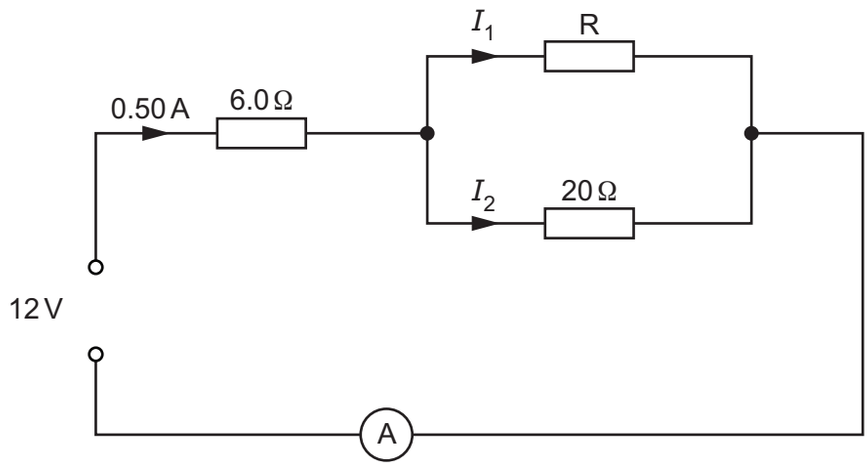


**Fig. 7.1**

**(a)** Determine the value of the current measured by the ammeter.

current = ..... [1]

**(b)** Calculate the potential difference (p.d.) across the 6.0 Ω resistor.

p.d. = ..... [2]

**(c)** Show that the p.d. across the 20 Ω resistor is 9.0 V.

[1]

**(d)** Calculate the resistance of resistor R.

resistance = ..... [3]

[Total: 7]

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**6** **(a)** Fig. 6.1 shows a circuit which includes an LDR (light-dependent resistor).

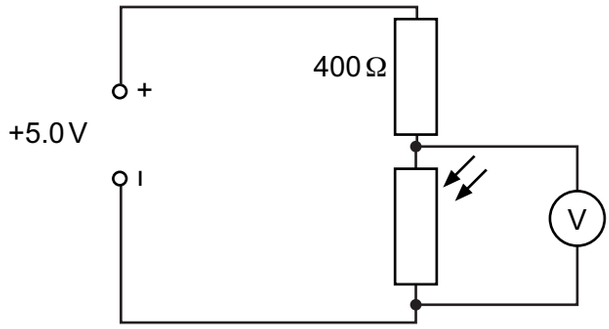


**Fig. 6.1**

The LDR is in a brightly lit room. The voltmeter reads 1.8 V.

**(i)** Calculate the current in the 400 Ω resistor.

current = ..... [3]

**(ii)** The light level in the room changes from bright to dark.

State and explain the effect on the voltmeter reading.

statement .....

explanation .....

.....

.....

.....

[3]

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**(b)** Fig. 6.2 shows two light-emitting diodes (LEDs) connected in parallel in a circuit. R is a red LED and G is a green LED.

The voltmeter across the LEDs shows a potential difference (p.d.) of 2.0 V.

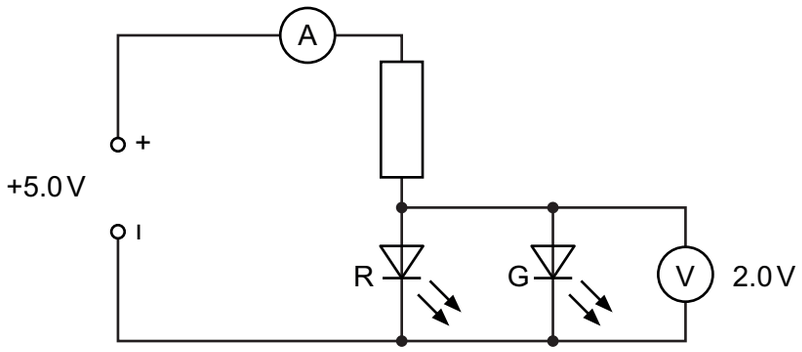



**Fig. 6.2**

**(i)** Fig. 6.3 shows a current-p.d. graph for the LEDs.

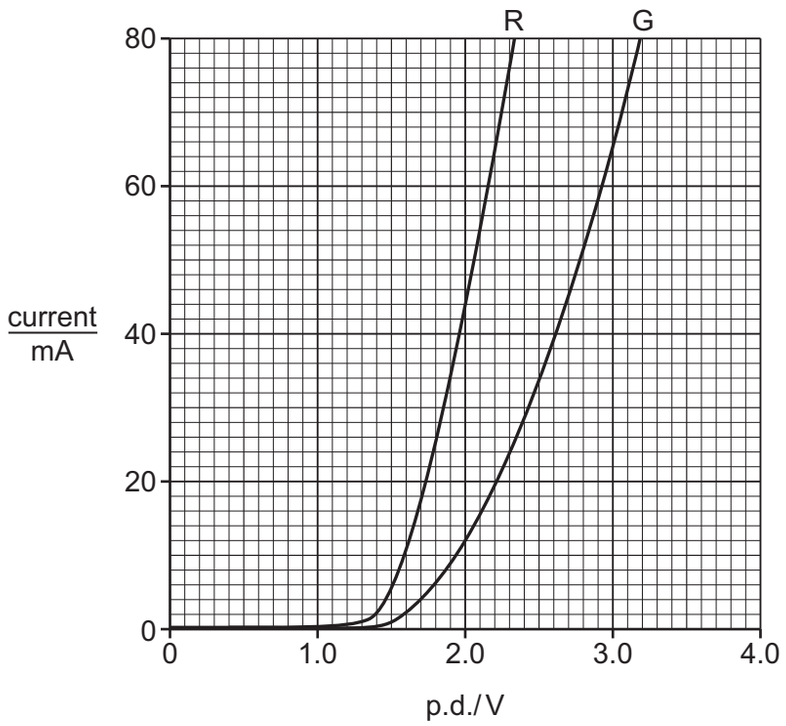



**Fig. 6.3**

Use Fig. 6.3 to determine the current in the ammeter.

current = ..... [2]

**(ii)** The connections to the power supply are reversed. State the reading on the ammeter.

current = ..... [1]

[Total: 9]

# 4.4 Electrical safety

2 MCQ + 0 theory questions · Paper 2 & Paper 4

· State the hazards of: (a) damaged insulation (b) overheating cables (c) damp conditions (d) excess current from ove
· Know that a mains circuit consists of a live wire (line wire), a neutral wire and an earth wire and explain why a s
· Explain the use and operation of trip switches and fuses and choose appropriate fuse ratings and trip switch settin
· Explain why the outer casing of an electrical appliance must be either non-conducting (double-insulated) or earthed
· State that a fuse without an earth wire protects the circuit and the cabling for a double- insulated appliance

**29** An electrical appliance has a symbol on it to indicate it is double-insulated.

Which statement is correct?

**A** A two-core, flexible cable with **no** earth wire is used to connect it to the mains supply.

**B** It needs an earth wire that is connected to a fuse.

**C** **No** fuse or circuit breaker is needed.

**D** The cable should be connected to the live and earth connectors of the plug but **not** the neutral.

0625/22/O/N/25 Q29

**29** In which mains electric wires are the fuse and the switch placed?

| | fuse | switch |
|---|---|---|
| **A** | earth | live |
| **B** | live | live |
| **C** | live | neutral |
| **D** | neutral | neutral |

0625/23/M/J/25 Q29

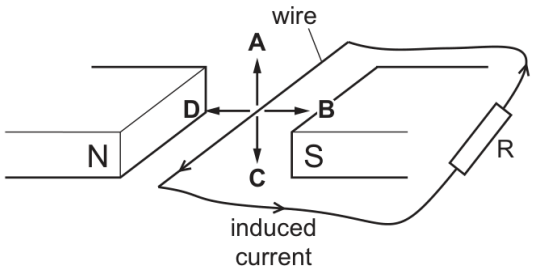
# 4.5 Electromagnetic effects

21 MCQ + 7 theory questions · Paper 2 & Paper 4

· Know that a conductor moving across a magnetic field or a changing magnetic field linking with a conductor can indu
· Describe an experiment to demonstrate electromagnetic induction
· State the factors affecting the magnitude of an induced e.m.f.
· Know that the direction of an induced e.m.f. opposes the change causing it
· State and use the relative directions of force, field and induced current
· Describe a simple form of a.c. generator (rotating coil or rotating magnet) and the use of slip rings and brushes w
· Sketch and interpret graphs of e.m.f. against time for simple a.c. generators and relate the position of the genera

**29** A wire connected to a resistor R is moved in a magnetic field. A current is induced in the direction shown.

In which direction is the wire moved?




0625/21/M/J/25 Q29

**30** Which component is **not** found in an a.c. generator?

**A** coil of wire

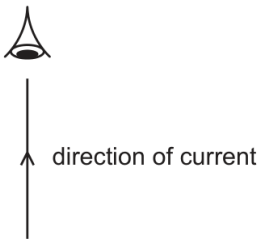
**B** split-ring commutator

**C** magnetic poles

**D** slip rings

0625/21/M/J/25 Q30

**31** There is a current in a straight metal wire in the direction shown.




Which row describes the direction and the pattern of the magnetic field due to the wire when looking from vertically above the wire?

| | direction | pattern |
|---|---|---|
| **A** | anticlockwise | circular |
| **B** | left to right | parallel lines |
| **C** | clockwise | circular |
| **D** | right to left | parallel lines |

0625/21/M/J/25 Q31

**32** A simple transformer contains an iron core.

What is the function of the iron core in the transformer?

**A** to conduct current from the primary coil to the secondary coil

**B** to enable the secondary coil to act as an electromagnet

**C** to ensure there is a high voltage between the primary coil and the secondary coil

**D** to link the magnetic field in the primary coil to the secondary coil

0625/21/M/J/25 Q32

**30** The lamps on a bicycle are powered by a simple electrical generator.

A permanent magnet rotates around a stationary coil of wire wound on an iron core.

Which statement is correct?

**A** The current in the lamps decreases as the speed of rotation increases.

**B** The current in the lamps decreases when the magnet is replaced by a stronger magnet.

**C** The output electromotive force (e.m.f.) will be higher if the gap between the magnet and the coil is bigger.

**D** The output e.m.f. will be higher if there are more turns on the coil.

0625/21/O/N/25 Q30

**31** A current-carrying coil in the magnetic field between the poles of two magnets experiences a turning effect.

Which change would decrease the turning effect?

**A** decreasing the distance between the magnets

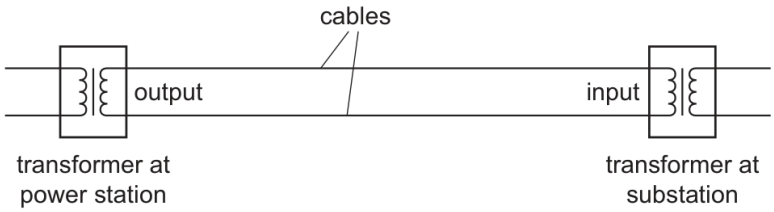
**B** increasing the current in the coil

**C** using stronger magnets

**D** decreasing the number of turns on the coil

0625/21/O/N/25 Q31

**32** Cables transmit electrical power from the output of the transformer at a power station to the input of another transformer at a substation.



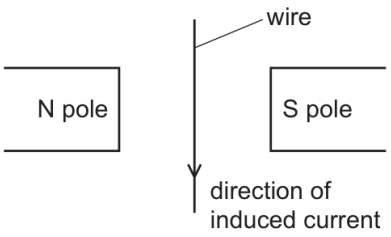

The power at the output of the transformer at the power station is 400 MW.

Which situation delivers the most power to the input of the transformer at the substation?

| | potential difference at power station transformer output / kV | diameter of cables |
|---|---|---|
| **A** | 200 | large |
| **B** | 200 | small |
| **C** | 400 | large |
| **D** | 400 | small |

0625/21/O/N/25 Q32

**30** A wire is moved in the region between the poles of a magnet.




A current is induced in the wire.

In which direction is the wire moved?

**A** horizontally from the N pole to the S pole

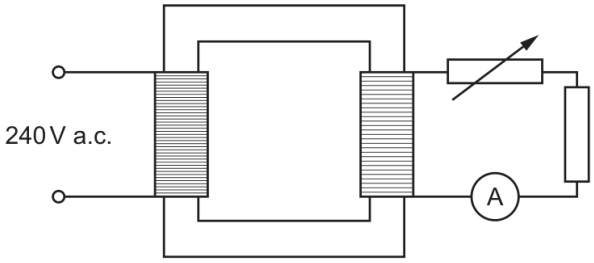
**B** horizontally from the S pole to the N pole

**C** perpendicularly into the plane of the paper

**D** perpendicularly out of the plane of the paper

0625/22/F/M/25 Q30

**31** A fixed resistor is connected to the secondary coil of a step-down transformer. The transformer is 100% efficient.




The variable resistor is adjusted so that the current in the fixed resistor decreases by 0.25 A.

What is the change to the current in the primary circuit?

**A** It decreases by 0.25 A.

**B** It decreases by less than 0.25 A.

**C** It decreases by more than 0.25 A.

**D** There is no change.

0625/22/F/M/25 Q31

**32** An electrical device changes the voltage of an electrical supply from 240 V a.c. to 20 V a.c.

What is this device?

**A** a generator

**B** a relay

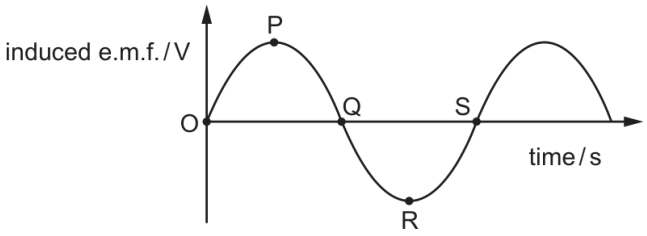
**C** a transformer

**D** a voltmeter

0625/22/F/M/25 Q32

Date: _____ Score: _____

**33** An a.c. generator is set up so that its coil of wire is perpendicular to the magnetic field. The coil of wire is rotated slowly to produce an alternating electromotive force (e.m.f.). A graph of the induced e.m.f. against time is plotted as the coil rotates.



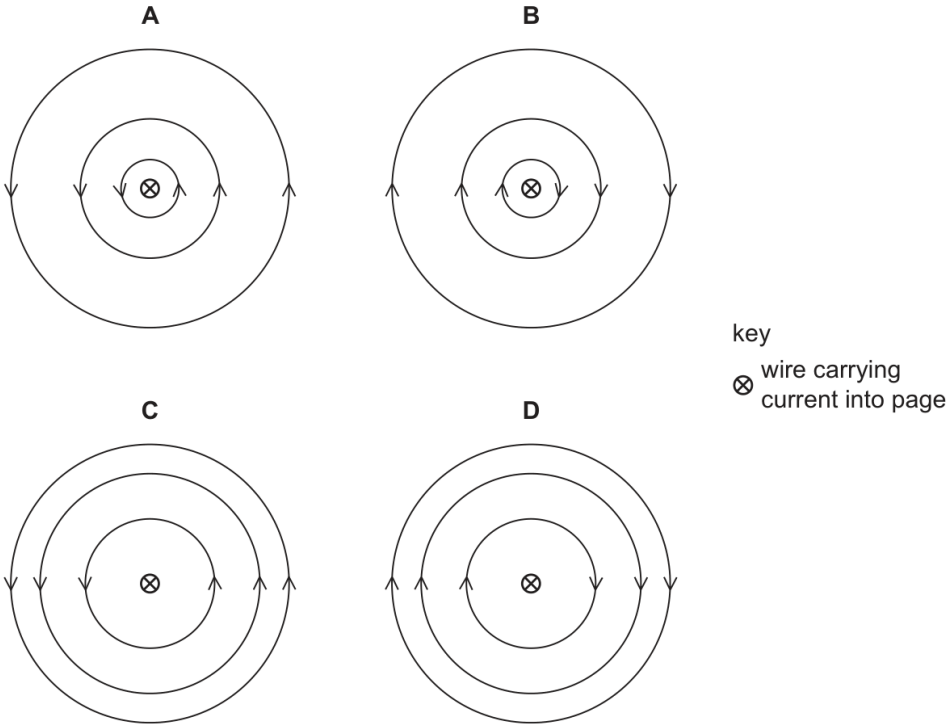

Which section of the graph represents a rotation of three-quarters of a turn?

**A** OP **B** OQ **C** OR **D** OS

0625/22/F/M/25 Q33

**32** When there is an electric current in a long straight wire, a magnetic field is created around the wire.

Which diagram shows the correct pattern and direction of magnetic field lines around a long straight wire carrying current into the page?




0625/22/M/J/25 Q32

**33** The primary voltage supplied to a transformer is 3.0 V. The secondary voltage is 5.0 V and the secondary current is 2.0 A. The transformer is 100% efficient.

What is the primary current in the transformer?

**A** 1.2 A **B** 2.0 A **C** 3.3 A **D** 7.5 A

0625/22/M/J/25 Q33

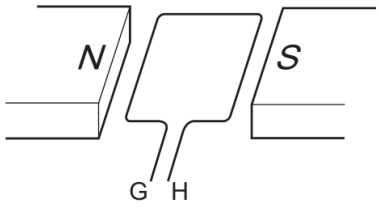
**30** A length of wire moves across the space between the poles of a magnet. A voltmeter measures the electromotive force (e.m.f.) generated between the two ends of the wire.

What gives the highest reading on the voltmeter?

| | speed of wire | strength of magnet |
|---|---|---|
| **A** | fast | weak |
| **B** | fast | strong |
| **C** | slow | weak |
| **D** | slow | strong |

0625/22/O/N/25 Q30

**31** The diagram shows a simple a.c. generator.




What is connected to G and H so that alternating current can be provided to an external circuit?

**A** cell

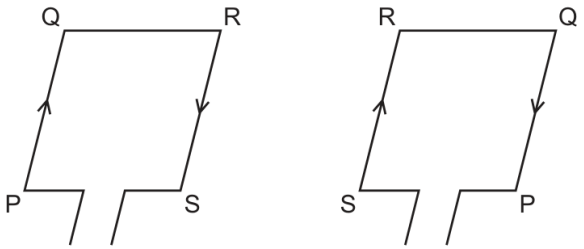
**B** commutator

**C** slip rings

**D** soft iron core

0625/22/O/N/25 Q31

**32** The diagrams show the coil in a d.c. electric motor in two positions. As the coil rotates through 180°, the direction of the current in the coil reverses from PQRS to SRQP.




Which part of the motor is responsible for reversing the current?

**A** the brushes

**B** the coil

**C** the magnets

**D** the split-ring commutator

0625/22/O/N/25 Q32

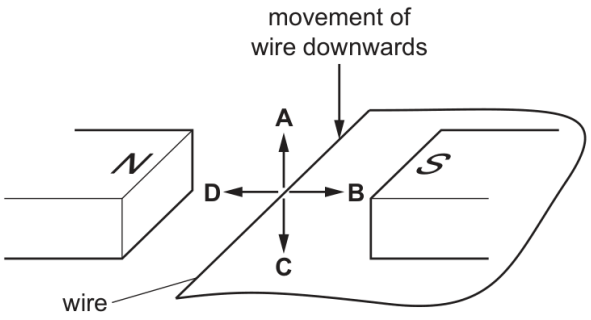
Date: _____ Score: _____

**30** The diagram shows a wire moving downwards in a magnetic field between two magnetic poles.

There is an induced current in the wire because of the movement.

The induced current produces a force on the wire.

In which direction is this force?




0625/23/M/J/25 Q30

**31** When the generator shown rotates, there is an a.c. electromotive force (e.m.f.) across the resistor between points P and Q.

Diagram 1 shows the position of the coil at time $t = 0$.

The coil rotates clockwise. At time $t_1$, the coil position is horizontal, as shown in diagram 2.

diagram 1

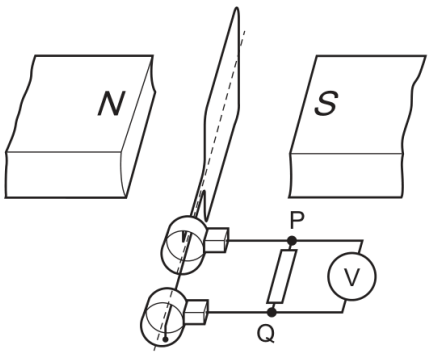
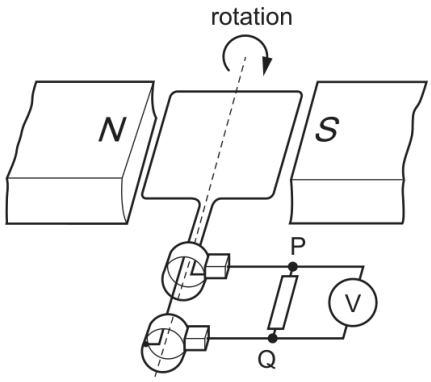
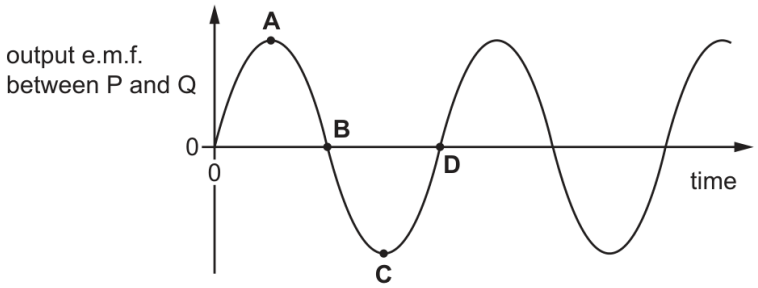



diagram 2




The graph shows the output e.m.f. between points P and Q plotted against time.

Which point shows the e.m.f. at time $t_1$?




0625/23/M/J/25 Q31

**32** What is the purpose of the split-ring commutator in an electric motor?

**A** to ensure that the magnetic field in the motor changes direction every half rotation

**B** to ensure that the magnetic field in the motor stays in the same direction at all times

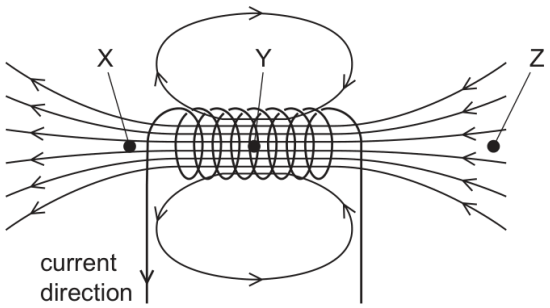
**C** to ensure that the turning effect on the motor changes direction every half rotation

**D** to ensure that the turning effect on the motor stays in the same direction at all times

0625/23/M/J/25 Q32

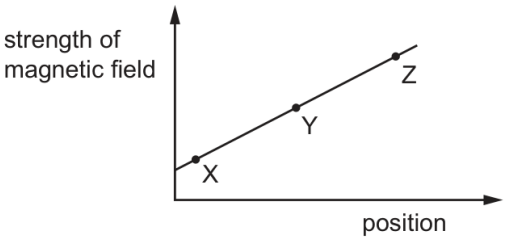
Date: _____ Score: _____

**24** The diagram shows the magnetic field due to a current in a solenoid.

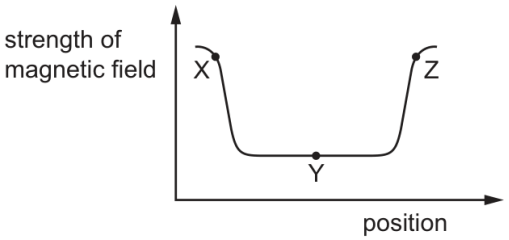



Which graph represents the strength of the magnetic field at positions X, Y and Z?

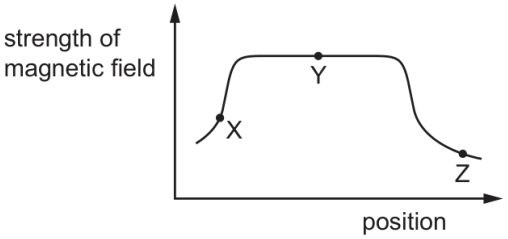
**A**



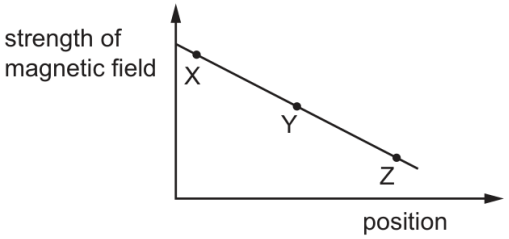

**B**




**C**

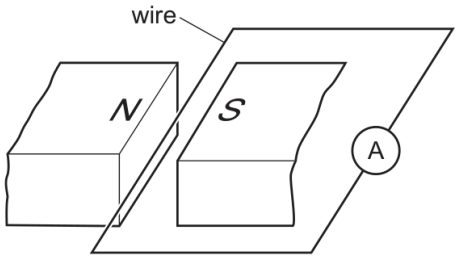



**D**




0625/23/O/N/25 Q24

**32** The diagram shows a wire between two magnetic poles. The wire is connected in a circuit with an ammeter.




The wire is moved downwards, towards the bottom of the page. A current is induced in the wire.

In which direction is the force on the wire caused by this current?

**A** towards the bottom of the page

**B** towards the left of the page

**C** towards the right of the page

**D** towards the top of the page

0625/23/O/N/25 Q32

**8** Fig. 8.1 shows a diagram of part of a simple a.c. generator.

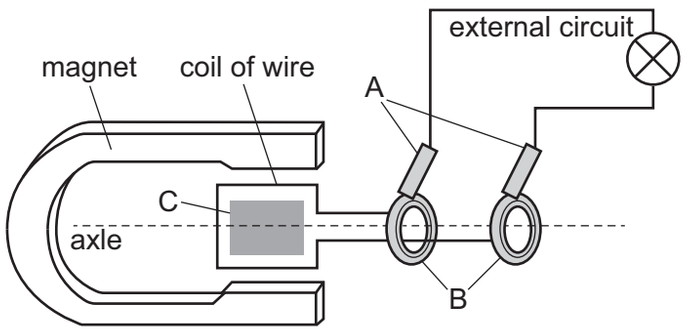



**Fig. 8.1**

**(a) (i)** Identify components A and B in Fig. 8.1.

A .....

B .....
[2]

**(ii)** Component C is made of soft iron.

Describe the effect of this component on the generator.

.....

..... [1]

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**(b)** The coil of the generator rotates at a constant speed of two complete revolutions per second. Sketch a graph of the e.m.f. generated against time on the axes in Fig. 8.2. The coil is in the position shown in Fig. 8.1 at time = 0.

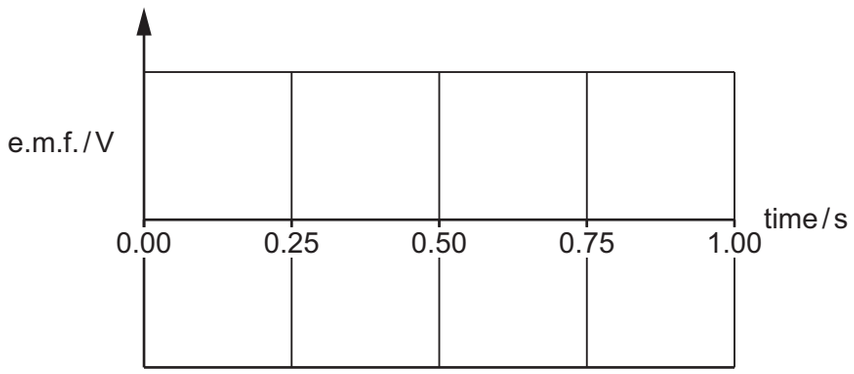


**Fig. 8.2**

[3]

**(c)** In power stations, transformers are used to step up the voltage of electricity generated before it is transmitted through cables over long distances.

**(i)** Explain the advantages of transmitting electricity at high voltages.

.....

.....

..... [2]

**(ii)** A power station generates electricity at 25 000 V. A transformer steps up the voltage to 300 000 V. The primary coil of the step-up transformer has 450 turns.

Calculate the number of turns $N_s$ on the secondary coil of the transformer.

$N_s$ = ..... [2]

[Total: 10]

**8** Fig. 8.1 shows a solenoid.

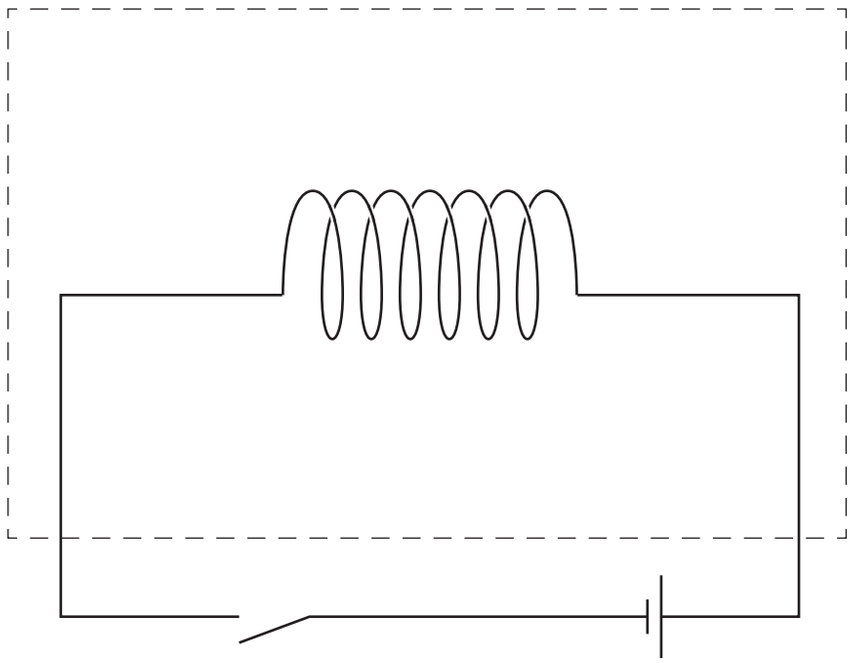

**Fig. 8.1**

**(a) (i)** Draw on Fig. 8.1 **four** complete magnetic field lines that show the pattern and direction of the magnetic field inside and outside the solenoid. [4]

**(ii)** Mark a point inside the box in Fig. 8.1 where the magnetic field is strong. Label this point B.

Explain how the diagram shows that the magnetic field is strong at B.

explanation .....

..... [1]

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**(b)** Fig. 8.2 shows a solenoid in an electric circuit for a bell.

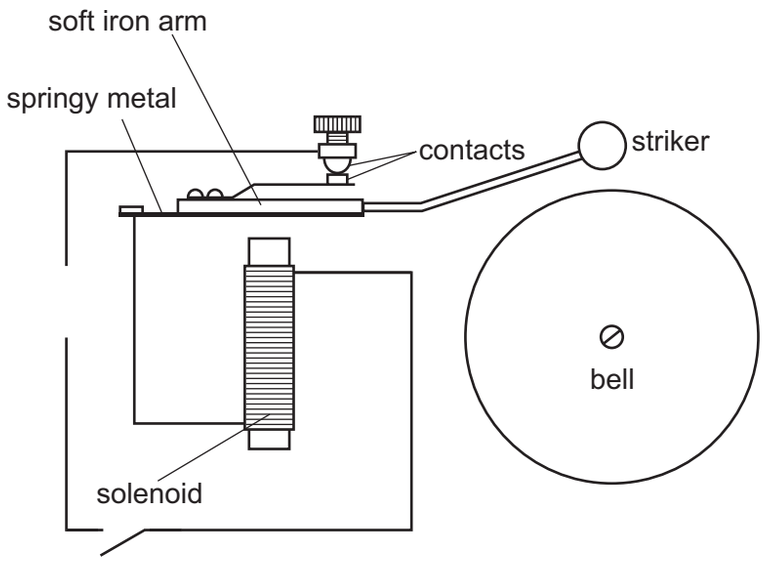


**Fig. 8.2**

**(i)** Complete the circuit in Fig. 8.2 with the symbol for a direct current (d.c.) power supply. [1]

**(ii)** Explain why the soft iron arm pivots, making the striker hit the bell when the switch is closed.

.....

.....

.....

..... [2]

**(iii)** Explain why the arm pivots back to its original position after the striker hits the bell.

.....

.....

.....

..... [2]

[Total: 10]

* 0000800000014 *



**8** Fig. 8.1 shows a metal rod suspended in the magnetic field produced by a pair of permanent magnets. The metal rod is connected to a cell and there is a current in the metal rod.

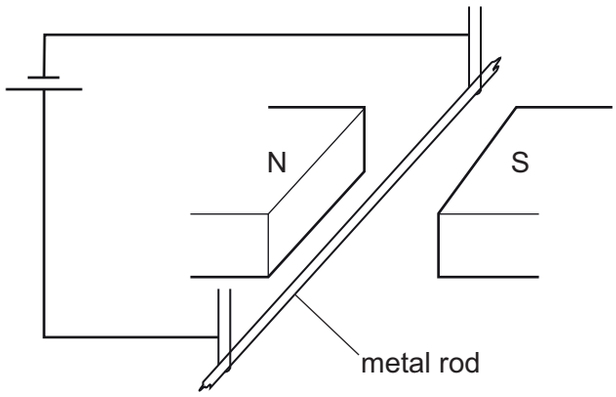


**Fig. 8.1**

**(a)** State the direction of the force on the metal rod due to the current.

Explain your answer.

direction of force .....

explanation .....

.....

.....
[3]

**(b)** The connections to the cell are reversed.

State how this change affects the force on the metal rod.

..... [1]

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**(c)** Two magnets and a cell are used to make a simple electric motor as shown in Fig. 8.2.

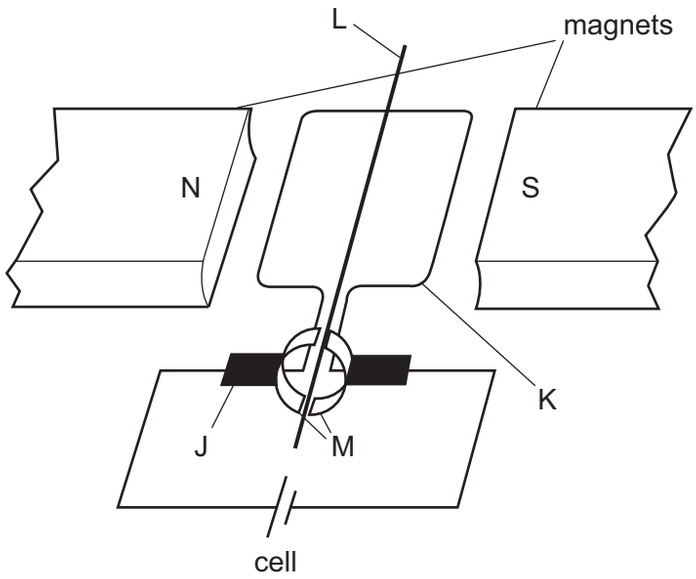


**Fig. 8.2**

Describe the function of parts J, K, L and M.

J .....

.....

K .....

.....

L .....

.....

M .....

.....

[4]

[Total: 8]

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**8** **(a)** Fig. 8.1 shows a simplified diagram of an a.c. generator.

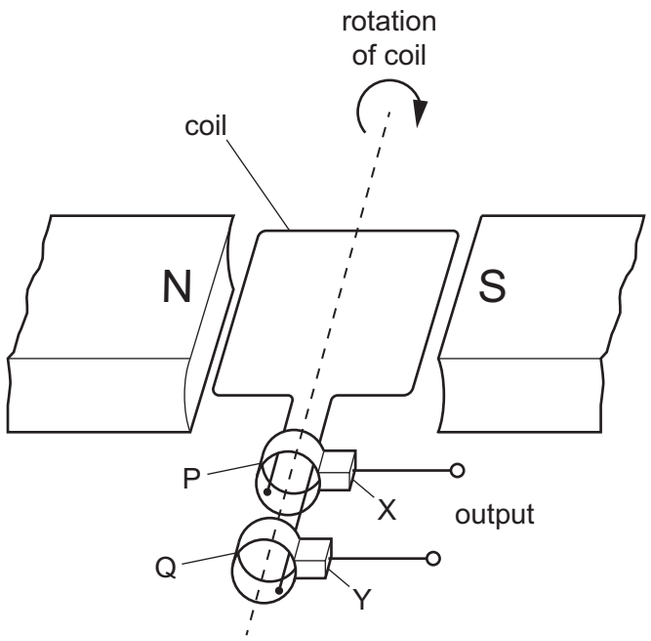


**Fig. 8.1**

**(i)** State the names of components:

P and Q .....

X and Y. .....
[2]

**(ii)** Explain why an electromotive force (e.m.f.) is only induced when the coil is turning.

.....

..... [1]

**(iii)** State **one** possible change that causes a larger e.m.f. to be induced.

..... [1]

**(b)** Fig. 8.2 shows a circuit diagram.

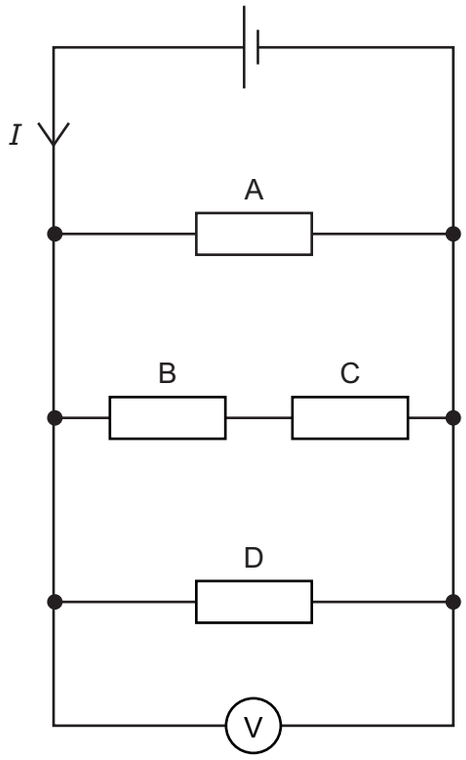


**Fig. 8.2**

Resistors A, B, C and D are identical. The current in resistor A is 2.4 A.

**(i)** State the value of the current in resistor B. Explain your answer.

current in resistor B .....

explanation .....

.....

[2]

**(ii)** Calculate the value of the current $I$.

current $I$ = ..... [1]

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**(iii)** The reading on the voltmeter is 5.0 V.

Calculate the resistance of resistor A.

resistance = ..... [2]

[Total: 9]

**8** **(a)** State what is meant by a magnetic field.

.....

..... [1]

**(b)** Fig. 8.1 shows an ammeter connected to a coil wound on a thin plastic cylinder. A small trolley is placed on a curved track which passes through the cylinder.

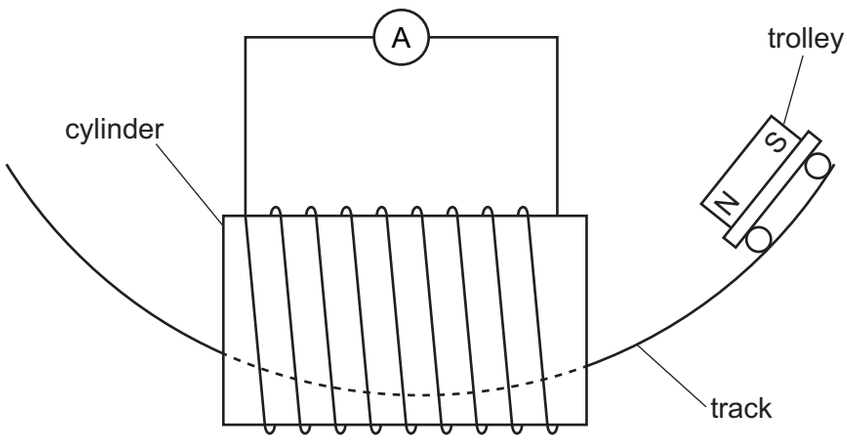


**Fig. 8.1**

The trolley is released from the position shown in Fig. 8.1. It travels through the coil from right to left. The trolley travels back from left to right. It has a lower maximum speed when it travels from left to right.

The plastic cylinder does not affect any magnetic field. A magnet is fixed to the trolley.

**(i)** State why there is a lower maximum speed when the trolley travels back from left to right.

.....

..... [1]

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**(ii)** Fig. 8.2 shows a close-up view of the ammeter.

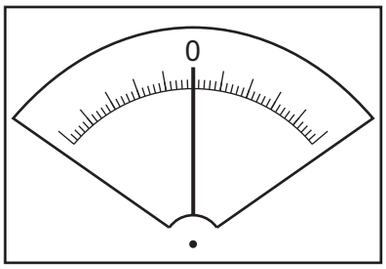


**Fig. 8.2**

As the trolley enters the coil moving from right to left, the ammeter needle deflects to the left and then returns to zero.

State and explain any changes to the deflection on the ammeter as the trolley enters the coil when moving back from left to right.

.....

.....

.....

.....

.....

..... [3]

[Total: 5]

**9** **(a)** Fig. 9.1 shows a transformer.

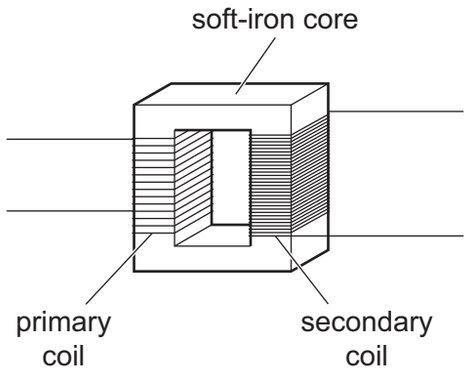


**Fig. 9.1**

**(i)** There is an alternating current in the primary coil.

Describe how an alternating current is produced in the secondary coil.

.....

.....

.....

..... [3]

**(ii)** A step-up transformer has a turns ratio of 1 : 20. The voltage across the primary coil is 12 V.

Calculate the voltage across the secondary coil.

voltage across secondary coil = ..... [2]

**(b)** The power lost in a cable is $1.25 \times 10^{-3}$ W. The resistance of the cable is 0.050 Ω.

Calculate the current in the cable.

current = ..... [2]

**(c)** State **two** advantages of high-voltage transmission.

1 .....

.....

2 .....

.....

[2]

[Total: 9]

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**7** Fig. 7.1 shows a simple a.c. generator.

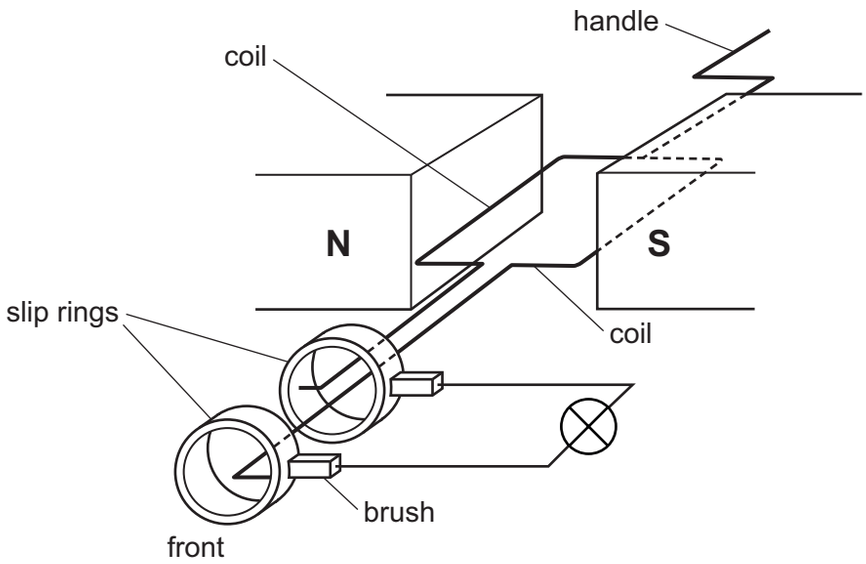


**Fig. 7.1**

**(a) (i)** Explain the function of the slip rings and brushes.

.....

.....

..... [1]

**(ii)** Describe how an alternating current is generated in the lamp.

.....

.....

.....

.....

.....

..... [3]

**(iii)** State **two** possible changes that cause a larger maximum current in the lamp.

1 .....

2 .....

[2]

**(b)** Fig. 7.2 is a graph to show how the electromotive force (e.m.f.) varies with time for the coil.

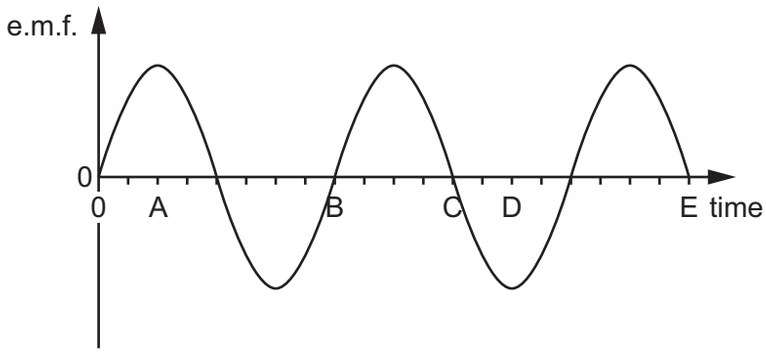


**Fig. 7.2**

**(i)** Determine how many revolutions the coil has made, from time = 0 to E.

number of revolutions = ..... [1]

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**(ii)** Fig. 7.3 shows the end view of the position of the plane of the coil at time = 0. The coil is rotating anti-clockwise.

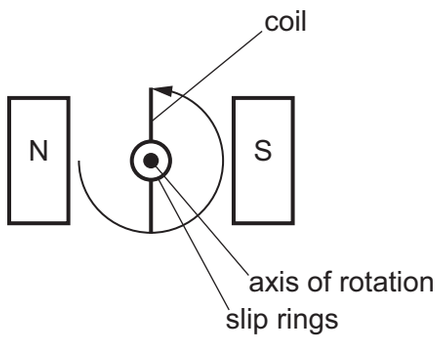


**Fig. 7.3**

On Fig. 7.4, draw **four** lines to match the coil position to the times A, B, C and D. The coil is viewed from the front.

Each coil position may be used once, more than once or not at all.

| time label from graph | coil position |
|---|---|
| A | |
| B | |
| C | |
| D | |

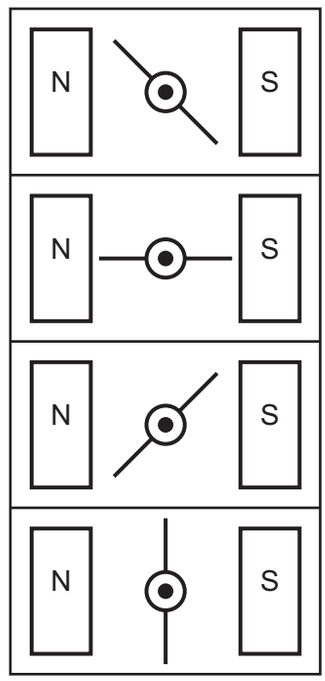


**Fig. 7.4**

[3]

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# 5 NUCLEAR PHYSICS

41 questions · Paper 2 & Paper 4

# Vocabulary · Topic 5 NUCLEAR PHYSICS

词汇表（中英对照）

| | | | |
|---|---|---|---|
| **atom** | 原子 | **radioactive decay** | 放射性衰变 |
| **nucleus** | 原子核 | **alpha particle** | α 粒子 |
| **proton** | 质子 | **beta particle** | β 粒子 |
| **neutron** | 中子 | **gamma radiation** | γ 辐射 |
| **electron** | 电子 | **half-life** | 半衰期 |
| **proton number** | 质子数 | **background radiation** | 背景辐射 |
| **nucleon number** | 核子数 | **ionising radiation** | 电离辐射 |
| **isotope** | 同位素 | **nuclear fission** | 核裂变 |
| **radioactivity** | 放射性 | **nuclear fusion** | 核聚变 |

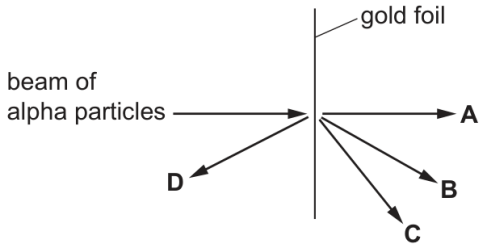
# 5.1 The nuclear model of the atom

7 MCQ + 1 theory questions · Paper 2 & Paper 4

· Describe the structure of an atom in terms of a positively charged nucleus and negatively charged electrons in orbi
· Know how atoms may form positive ions by losing electrons or form negative ions by gaining electrons
· Describe how the scattering of alpha (α) particles by a sheet of thin metal supports the nuclear model of the atom,
· Describe the composition of the nucleus in terms of protons and neutrons
· State the relative charges of protons, neutrons and electrons as +1, 0 and –1 respectively
· Define the terms proton number (atomic number) Z and nucleon number (mass number) A and be able to calculate the nu
· Use the nuclide notation A ZX

**33** In an alpha particle scattering experiment, a beam of alpha particles is fired at a very thin piece of gold foil.

In which direction do most of the alpha particles travel after hitting the foil?




0625/21/M/J/25 Q33

**34** An atom of boron contains 5 protons and 5 electrons. The nucleon number of the atom is 11.

How many neutrons are there in the atom?

**A** 5 **B** 6 **C** 11 **D** 16

0625/21/M/J/25 Q34

**33** $^{239}_{94}Pu$ is an isotope of plutonium.

The nucleus can absorb a neutron and then split to form two other nuclei.

Two nuclei that form are gold, $^{207}_{79}Au$, and phosphorus, $^{31}_{15}P$.

In this reaction, neutrons are also produced.

What is the number of neutrons produced when the nucleus splits?

**A** 1 **B** 2 **C** 4 **D** 5

0625/21/O/N/25 Q33

**34** When a thin gold foil is bombarded with alpha particles, some alpha particles are deflected through large angles.

Which statement explains this deflection?

**A** Most of the atom consists of empty space.

**B** All of the positive charge and most of the mass of the gold atom are concentrated in a small volume.

**C** Positive charge in the gold atom is spread evenly throughout the atom.

**D** All of the negative charge is concentrated at its centre.

0625/22/M/J/25 Q34

**33** The arrow shows the path of an $\alpha$-particle travelling towards the centre of a gold nucleus.

Which description of the path of the $\alpha$-particle after striking the gold nucleus is correct?

**A** The $\alpha$-particle is deflected through a very small angle towards the top of the page.

**B** The $\alpha$-particle is deflected back along its initial path.

**C** The $\alpha$-particle carries on in a straight line with **no** deflection.

**D** The $\alpha$-particle is captured by the nucleus and goes into orbit around it.

0625/22/O/N/25 Q33

**34** A teacher writes the nuclide equation for a fusion reaction.

$$^{2}_{1}H + {}^{X}_{1}H \rightarrow {}^{4}_{2}Q + {}^{1}_{0}n$$

What is the missing number X and the missing element Q?

| | X | Q |
|---|---|---|
| **A** | 2 | H |
| **B** | 2 | He |
| **C** | 3 | H |
| **D** | 3 | He |

0625/22/O/N/25 Q34

**34** Which observation provides evidence for the nuclear atom?

**A** attraction of opposite charges

**B** emission of $\gamma$-rays during the decay of a radioactive nuclide

**C** scattering of $\alpha$-particles by a thin metal foil

**D** scattering of $\gamma$-rays by a thin metal foil

0625/23/M/J/25 Q34

**9** **(a)** Describe the structure of an atom of helium-4, $^{4}_{2}He$.

.....

.....

.....

..... [3]

**(b)** The Sun is a medium-sized star powered by nuclear fusion reactions which release energy.

**(i)** State what happens during nuclear fusion reactions which form helium.

.....

.....

..... [2]

**(ii)** State **two** regions of the electromagnetic spectrum by which the Sun radiates most of its energy.

..... [2]

**(c)** Describe what happens to a star when most of the fuel in its centre has been converted to helium.

.....

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..... [2]

[Total: 9]

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# 5.2 Radioactivity

27 MCQ + 6 theory questions · Paper 2 & Paper 4

· Know what is meant by background radiation
· Know the sources that make a significant contribution to background radiation including: (a) radon gas (in the air)
· Know that ionising nuclear radiation can be measured using a detector connected to a counter
· Use count rate measured in counts / s or counts / minute
· Use measurements of background radiation to determine a corrected count rate
· Describe the emission of radiation from a nucleus as spontaneous and random in direction
· Identify alpha (α), beta (β) and gamma (γ) emissions from the nucleus by recalling: (a) their nature (b) their rela

**35** A detector placed near a radioactive isotope gives a measured count rate of 800 counts / minute. The half-life of the isotope is three hours.

The background count rate is 32 counts / minute.

What is the count rate on the detector after six hours?

**A** 192 counts / minute

**B** 200 counts / minute

**C** 224 counts / minute

**D** 232 counts / minute

0625/21/M/J/25 Q35

**36** Sodium-24 is a radioactive isotope that emits beta radiation.

Which nuclear equation shows how sodium-24 decays?

**A** $^{24}_{11}\text{Na} \rightarrow {}^{24}_{12}\text{Mg} + {}^{0}_{-1}\beta$

**B** $^{24}_{11}\text{Na} \rightarrow {}^{24}_{10}\text{Ne} + {}^{0}_{-1}\beta$

**C** $^{24}_{11}\text{Na} \rightarrow {}^{20}_{9}\text{F} + {}^{4}_{2}\beta$

**D** $^{24}_{11}\text{Na} \rightarrow {}^{20}_{13}\text{A}l + {}^{4}_{2}\beta$

0625/21/M/J/25 Q36

**37** Which safety precautions must be taken when using a source of gamma radiation?

1 Reduce the distance between the source and the person.

2 Reduce the time of exposure to the radiation.

3 Use a suitable shielding material between the source and the person.

**A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

0625/21/M/J/25 Q37

**34** Three statements about alpha ($\alpha$) particles and beta ($\beta$) particles are listed.

1 The magnitude of the charge on an $\alpha$-particle is twice the charge on a $\beta$-particle.

2 The mass of an $\alpha$-particle is large compared with the mass of a $\beta$-particle.

3 The kinetic energy of an $\alpha$-particle is small compared to the kinetic energy of a $\beta$-particle.

Which statements help to explain the different ionising effects of $\alpha$-particles and $\beta$-particles?

**A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

0625/21/O/N/25 Q34

**35** A beta-particle is a fast-moving electron.

Which statement explains how beta-particles are emitted from an atom?

**A** An electron is emitted as a beta-particle from an inner electron shell of the atom.

**B** An electron is emitted as a beta-particle from an outer electron shell of the atom.

**C** A neutron changes into a proton and a beta-particle is emitted from the nucleus.

**D** A proton changes into a neutron and a beta-particle is emitted from the nucleus.

0625/21/O/N/25 Q35

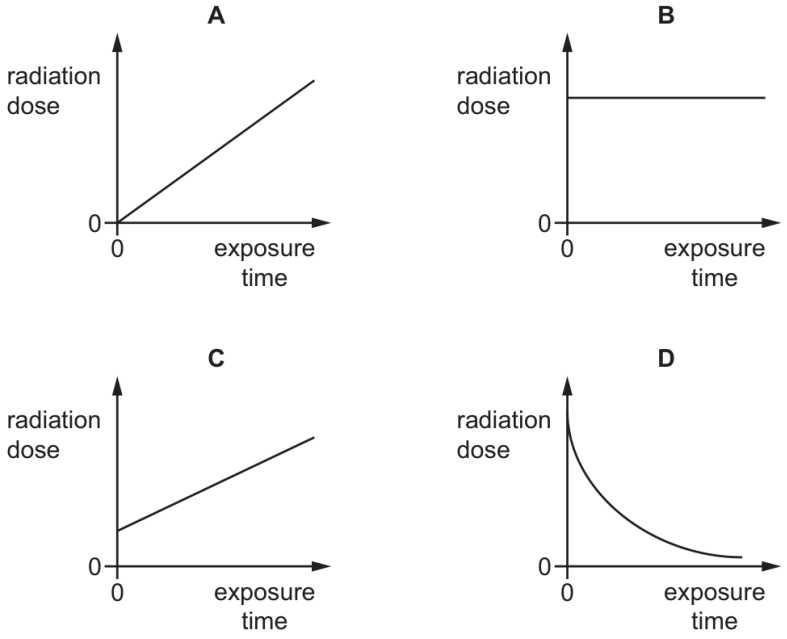
**36** Which properties must a radioisotope have for its radiation to make it suitable to kill bacteria in food?

| | half-life | type of radiation emitted |
|---|---|---|
| **A** | less than one minute | $\gamma$ only |
| **B** | several hours | $\alpha$ only |
| **C** | several hours | $\gamma$ only |
| **D** | several thousand years | $\alpha$ only |

0625/21/O/N/25 Q36

**37** A student tries to predict the effect of exposure time on the radiation dose received from a source of ionising radiation. Radiation dose measures the amount of ionising radiation received by a living organism.

Which graph shows the correct trend between the variables?




0625/21/O/N/25 Q37

**34** A uranium nucleus $^{238}_{92}U$ decays to form thorium by emitting an $\alpha$-particle. Thorium decays to protactinium by emitting a $\beta$-particle.

Which row gives the atomic number and mass number of this isotope of protactinium?

| | atomic number | mass number |
|---|---|---|
| **A** | 89 | 234 |
| **B** | 90 | 235 |
| **C** | 91 | 234 |
| **D** | 95 | 235 |

0625/22/F/M/25 Q34

**35** Which row shows what happens when a nucleus decays by emitting a $\beta$-particle?

| | change in the nucleus | particle emitted |
|---|---|---|
| **A** | a neutron changes to a proton and an electron | electron |
| **B** | a neutron changes to a proton and an electron | proton |
| **C** | a proton changes to a neutron and an electron | electron |
| **D** | a proton changes to a neutron and an electron | neutron |

0625/22/F/M/25 Q35

**36** Which precaution does **not** always reduce a scientist's exposure when working with sources of ionising radiation?

**A** ensuring the scientist only works with sources of radiation that have long half-lives

**B** increasing the distance between the scientist and the source of radiation

**C** limiting the time for which the scientist handles the source of radiation

**D** placing a lead shield between the scientist and the source of radiation

0625/22/F/M/25 Q36

**37** A detector is placed near a radioactive isotope and records a count rate of 700 counts / min.

The half-life of the isotope is 8 min.

The average background count rate is 60 counts / min.

What is the count rate measured by the detector after 16 min?

**A** 100 counts / min

**B** 160 counts / min

**C** 175 counts / min

**D** 220 counts / min

0625/22/F/M/25 Q37

Date: _____ Score: _____

**38** The nucleus of an isotope of nitrogen, N, absorbs a neutron. It then decays into an isotope of carbon, C, and emits x.

$$^{1}_{0}n + {}^{14}_{7}N \rightarrow {}^{14}_{6}C + x$$

What is x?

**A** $\alpha$-particle

**B** $\beta$-particle

**C** $\gamma$-radiation

**D** proton

0625/22/F/M/25 Q38

**35** A detector near a radioactive source shows a reading of 2000 counts / minute.

How is the corrected count rate determined?

**A** by moving the detector closer to the source

**B** by placing an absorber between the detector and the source

**C** by repeating and averaging the measurements

**D** by taking away the background radiation count rate

0625/22/M/J/25 Q35

**36** Which statement about alpha, beta and gamma radiation is **not** correct?

**A** Alpha particles are deflected most in an electric field.

**B** Beta particles are deflected most in a magnetic field.

**C** Gamma radiation is **not** deflected by a magnetic field.

**D** Gamma radiation is least ionising.

0625/22/M/J/25 Q36

**37** A sample of a radioactive material has a half-life of 20 minutes.

Which statement is correct?

**A** After 30 minutes, less than half of the material has decayed.

**B** After 40 minutes, all of the radioactive material has decayed.

**C** After 60 minutes, an eighth of the radioactive material remains.

**D** After 120 minutes, a sixth of the radioactive material remains.

0625/22/M/J/25 Q37

**35** Radon gas is a source of radiation that contributes to background radiation.

What does **not** contribute to background radiation?

**A** electromagnetic rays

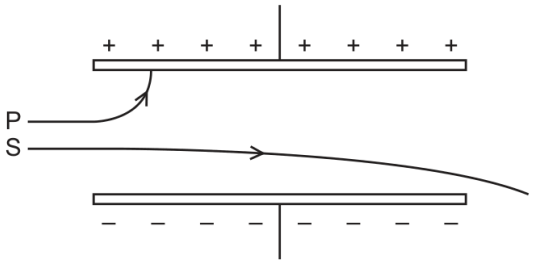
**B** food and drink

**C** rocks

**D** seismic waves

0625/22/O/N/25 Q35

**36** Two beams of radiation, P and S, enter an electric field, as shown.




Which types of radiation are P and S?

| | P | S |
|---|---|---|
| **A** | beta ($\beta$) | alpha ($\alpha$) |
| **B** | beta ($\beta$) | gamma ($\gamma$) |
| **C** | gamma ($\gamma$) | alpha ($\alpha$) |
| **D** | gamma ($\gamma$) | gamma ($\gamma$) |

0625/22/O/N/25 Q36

**37** Which change takes place in the nucleus when it decays by beta ($\beta$) emission?

**A** neutron + electron $\rightarrow$ proton

**B** neutron $\rightarrow$ proton + electron

**C** proton + electron $\rightarrow$ neutron

**D** proton $\rightarrow$ neutron + electron

0625/22/O/N/25 Q37

**33** A beta particle has less mass than an alpha particle, but the speed of a beta particle emitted during radioactive decay is approximately 20 times greater than the speed of an alpha particle.

Which statement correctly compares properties of these two particles?

**A** Alpha particles are less penetrating because they have less kinetic energy than beta particles.

**B** Alpha particles are more penetrating because they have less kinetic energy than beta particles.

**C** Alpha particles are less ionising because they have more kinetic energy than beta particles.

**D** Alpha particles are more ionising because they have more kinetic energy than beta particles.

0625/23/M/J/25 Q33

**35** Which effect on living things is **not** caused by ionising radiation?

**A** cancer

**B** cell death

**C** infection

**D** mutation

0625/23/M/J/25 Q35

**36** Isotope X is radioactive.

Which statement about X **must** be correct?

**A** A nucleus of X is unstable.

**B** X emits $\alpha$-particles.

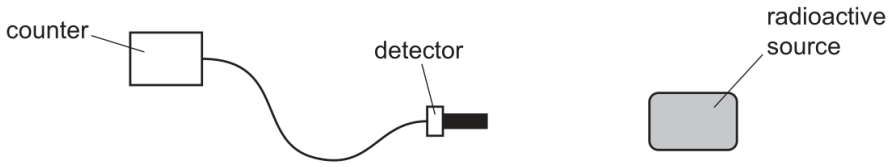
**C** X emits $\beta$-particles.

**D** X emits $\gamma$-radiation.

0625/23/M/J/25 Q36

**37** An experiment is done to measure the radiation from a radioactive source. The source has a half-life of 10 minutes.

The source is placed close to a detector that is connected to a counter, as shown.




The average background count rate is 20 counts / minute.

At the start of the experiment, the count rate recorded by the counter is 1000 counts / minute.

What is the count rate 10 minutes later?

**A** 490 counts / minute

**B** 500 counts / minute

**C** 510 counts / minute

**D** 530 counts / minute

0625/23/M/J/25 Q37

**33** Some students set up an experiment to measure the count rate for a radioactive source.

First, they measure the radioactive background count rate in the laboratory by taking three measurements. Each measurement is taken over a 10-minute period.

The table shows their results.

| number of counts in 10 minutes | | |
|---|---|---|
| first measurement | second measurement | third measurement |
| 170 | 164 | 185 |

Next, the students measure the counts near the radioactive source. This count rate is measured as 949 counts per minute.

What is the corrected count rate for the source?

**A** 932 counts per minute

**B** 776 counts per minute

**C** 430 counts per minute

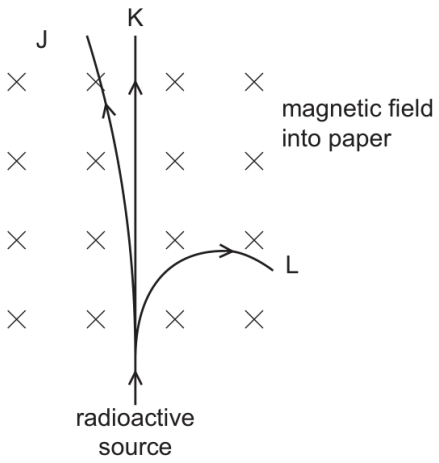
**D** 78 counts per minute

0625/23/O/N/25 Q33

**34** A radioactive source emits $\alpha$-particles, $\beta$-particles and $\gamma$-rays into a vacuum where there is a magnetic field.

The magnetic field acts perpendicularly into the plane of the paper.

The paths J, K and L of the three types of radiation through the magnetic field are shown.




Which radiation follows path J, path K and path L?

| | J | K | L |
|---|---|---|---|
| **A** | $\alpha$-particles | $\beta$-particles | $\gamma$-rays |
| **B** | $\alpha$-particles | $\gamma$-rays | $\beta$-particles |
| **C** | $\beta$-particles | $\alpha$-particles | $\gamma$-rays |
| **D** | $\beta$-particles | $\gamma$-rays | $\alpha$-particles |

0625/23/O/N/25 Q34

**35** Radioactive decay is a change in an unstable nucleus which may result in emission of alpha-particles or beta-particles.

Which statement describes how these emissions happen?

**A** Alpha emission is a spontaneous and random process but beta emission is **not**.

**B** Alpha emission and beta emission are both spontaneous and random processes.

**C** Beta emission is a spontaneous and random process but alpha emission is **not**.

**D** Neither alpha emission nor beta emission are spontaneous and random processes.

0625/23/O/N/25 Q35

**36** Which change occurs in the nucleus of a radioactive atom when a beta-particle is emitted?

**A** neutron $\rightarrow$ alpha-particle

**B** neutron $\rightarrow$ proton + electron

**C** proton $\rightarrow$ neutron

**D** proton $\rightarrow$ neutron + electron

0625/23/O/N/25 Q36

**37** A radioactive source that emits $\alpha$-particles, $\beta$-particles and $\gamma$-radiation is stored safely in a container.

From which material should the container be made?

**A** alumimium

**B** copper

**C** paper

**D** lead

0625/23/O/N/25 Q37

**9** Strontium-90 ($^{90}_{38}Sr$) is a radioactive isotope that contains 38 protons and 52 neutrons. Strontium-90 decays to form an isotope of yttrium (Y) by emitting beta ($\beta$) particles.

**(a) (i)** Suggest how the nucleus of a stable isotope of strontium differs from a nucleus of strontium-90. Explain your answer.

suggestion .....

explanation .....

.....

[2]

**(ii)** Complete the nuclide equation for the decay of strontium-90 to yttrium.

$$^{90}_{38}Sr \longrightarrow {}^{.....}_{.....}Y + {}^{.....}_{.....}\beta$$

[2]

**(iii)** Explain why scientists limit the amount of time they are exposed to radioactive strontium.

.....

..... [2]

**(b)** Yttrium is also unstable. A scientist places a sample of yttrium near a radiation detector. Table 9.1 shows the count rate recorded by the detector as the sample decays.

**Table 9.1**

| time / h | recorded count rate / counts / min |
|---|---|
| 0 | 68 |
| 50 | 49 |
| 100 | 38 |
| 150 | 32 |
| 200 | 26 |
| 250 | 24 |
| 300 | 20 |
| 350 | 21 |
| 400 | 20 |

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Fig. 9.1 shows a graph of the count rate **due to yttrium** against time.

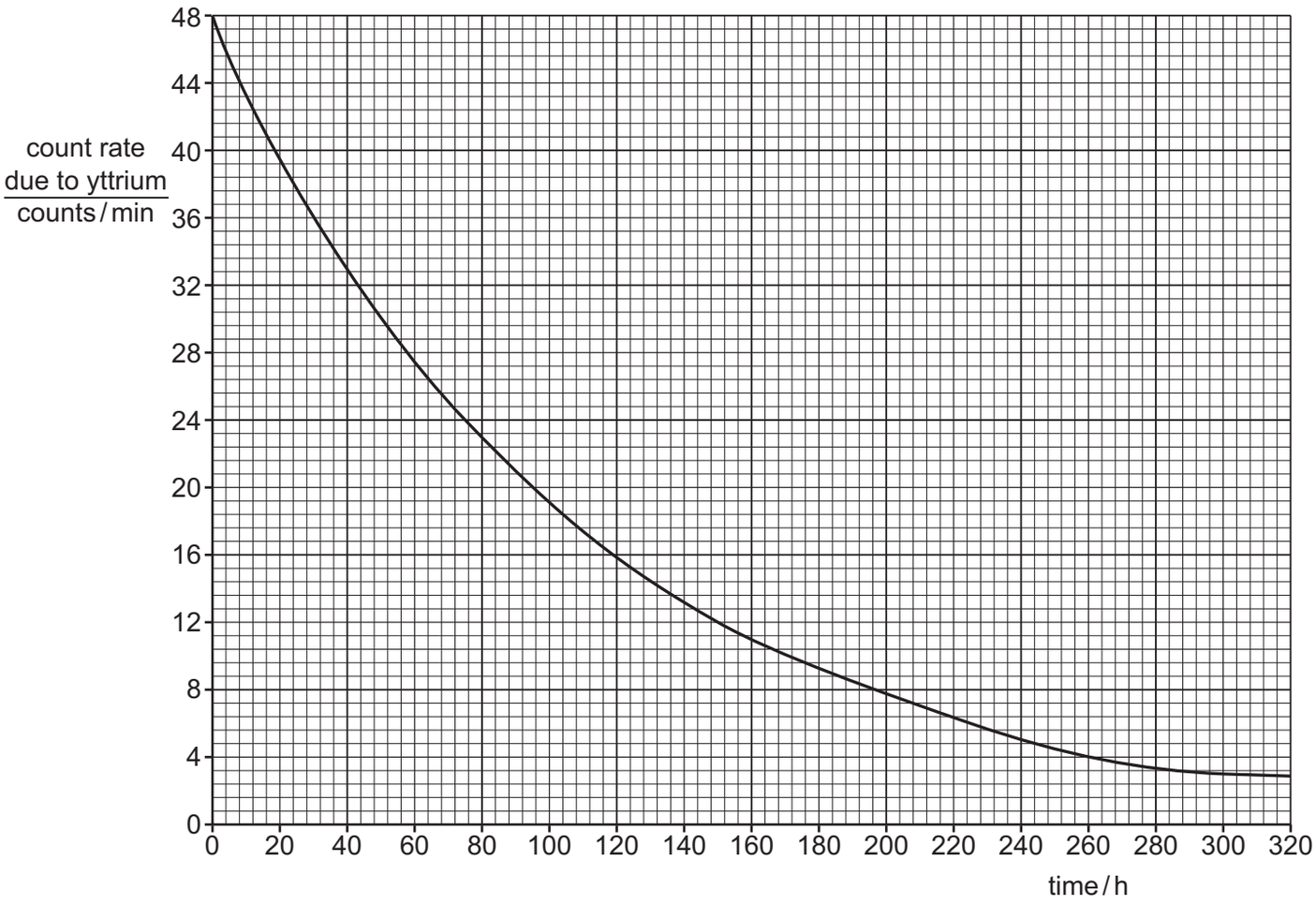


**Fig. 9.1**

**(i)** Use Fig. 9.1 to determine the half-life of yttrium. Show your working.

half-life = ..... h [3]

**(ii)** Explain the difference between the count rate in Table 9.1 and the count rate due to yttrium plotted on the graph in Fig. 9.1.

.....

.....

..... [1]

[Total: 10]

* 0000800000016 *



**9** Strontium-90 is a radioactive isotope of strontium. The nuclide notation for strontium-90 is:

$^{90}_{38}Sr$

**(a) (i)** Explain what isotopes are.

.....

..... [1]

**(ii)** Complete Table 9.1 for strontium-90.

**Table 9.1**

| particle | number in one atom | location |
|---|---|---|
| | 38 | outside nucleus |
| neutron | | |
| | 38 | inside nucleus |

[2]

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**(b)** Strontium-90 is used to measure the thickness of metal sheets in industry.
Strontium-90 decays by emitting beta ($\beta$) particles which pass through a metal sheet to a detector.

**(i)** One metal sheet is 0.75 mm thick. Suggest why strontium-90 is a suitable radioactive source to measure the thickness of the metal sheets.

.....

.....

..... [2]

**(ii)** The half-life of strontium-90 is approximately 27 years. Fig. 9.1 shows the shape of a decay curve.

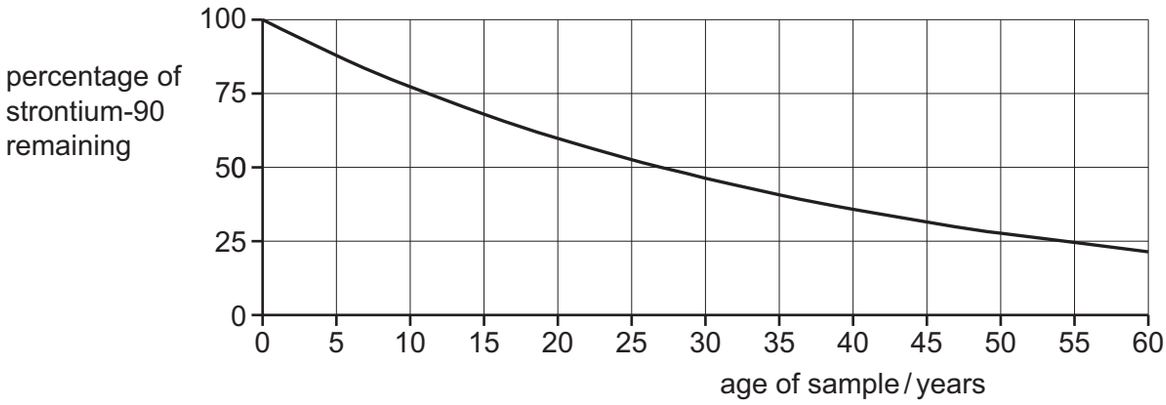


**Fig. 9.1**

The strontium-90 source is replaced with a new source after 15 years. Using Fig. 9.1, suggest why a strontium-90 source that is more than 15 years old needs to be replaced with a new source.

.....

..... [2]

[Total: 7]

**9** **(a)** **(i)** State what is meant by background radiation.

.....

..... [1]

**(ii)** State **one** source that makes a significant contribution to background radiation.

..... [1]

**(b)** Radioactive isotopes are used in some medical treatments.

Radium-223 ($^{223}_{88}$Ra) is an isotope of radium. Radium-223 decays by the emission of alpha ($\alpha$) particles to an isotope of radon (Rn).

**(i)** Complete the nuclide equation for this decay.

$$^{223}_{88}\text{Ra} \longrightarrow \overset{.....}{\underset{.....}{\quad}}\text{Rn} + \overset{.....}{\underset{.....}{\quad}}\alpha$$

[2]

**(ii)** Radium-223 is injected into the body to treat a specific organ.

Explain why the source must be inside the body.

.....

..... [1]

**(c)** An isotope of technetium is injected into the body to detect cancer in one of the organs. The half-life of this isotope is 6 hours and it decays by emitting gamma ($\gamma$) radiation. The radiation is detected outside the body.

**(i)** Explain why a source of gamma radiation is used.

.....

..... [1]

**(ii)** Explain why a source with a short half-life must be used.

.....

..... [1]

[Total: 7]

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**9** **(a)** Fig. 9.1 shows a beam of radiation in a vacuum. The beam contains $\alpha$-particles, $\beta$-particles and $\gamma$-radiation.

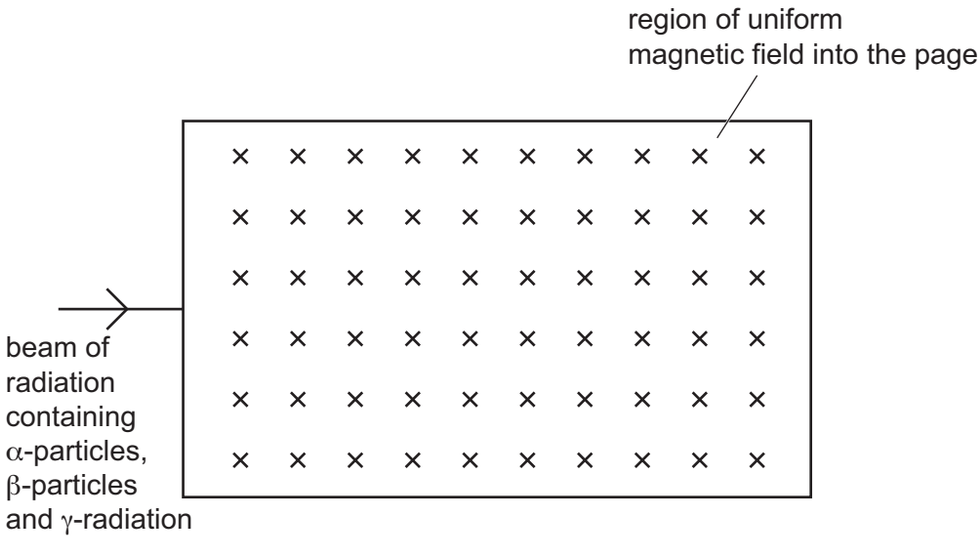


**Fig. 9.1**

The beam enters a region where there is a strong, uniform magnetic field. The direction of the magnetic field is into the page.

On Fig. 9.1 draw and label the paths within the magnetic field of:

**(i)** $\alpha$-particles (label this path $\alpha$) [1]

**(ii)** $\beta$-particles (label this path $\beta$) [2]

**(iii)** $\gamma$-radiation (label this path $\gamma$) [1]

**(b)** Table 9.1 shows five radioactive sources, the main type of radiation emitted by each source and the half-life of each source.

**Table 9.1**

| radioactive source | type of radiation emitted | half-life |
|---|---|---|
| P | alpha | 460 years |
| Q | alpha | 10 days |
| R | beta | 29 years |
| S | beta | 14 days |
| T | gamma | 30 years |

**(i)** Define half-life of a radioactive isotope.

.....

..... [1]

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**(ii)** Fig. 9.2 shows a simplified diagram of a machine that produces thin sheets of aluminium of constant thickness.

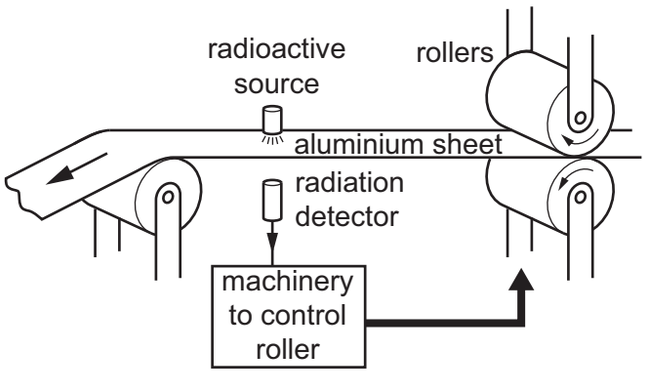


**Fig. 9.2**

The radiation detector is used to measure the thickness of the aluminium sheets and control the gap between the rollers.

State the most suitable radioactive source in Table 9.1 for the machine in Fig. 9.2.

Explain why this radioactive source is the most suitable and why the other sources are unsuitable.

most suitable source .....

explanation .....

.....

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[4]

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**11** **(a)** Define the half-life of a radioactive source.

.....

..... [1]

**(b)** A protactinium (Pa) nucleus decays into a uranium (U) nucleus by the emission of a beta particle (β-particle).

**(i)** Complete the nuclear equation for the decay.

$$^{234}_{91}\text{Pa} \rightarrow$$

[3]

**(ii)** State the change that occurs in the nucleus during the decay.

..... [1]

**(c)** A different element decays by the emission of an alpha particle (α-particle).

Give **two** reasons why α-particles are more strongly ionising than β-particles.

1 .....

.....

2 .....

.....

[2]

[Total: 7]

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**8** **(a)** Alpha ($\alpha$) particles are directed at a very thin sheet of gold foil.

The following observations provide evidence for the nuclear model of the atom.

State what conclusions may be made about the model of the atom from these observations.

**(i)** Observation: most of the particles pass straight through the foil.

conclusion .....

.....

..... [1]

**(ii)** Observation: a few particles bounce back or are deflected through large angles.

conclusion .....

.....

..... [1]

**(b)** Fig. 8.1 shows three beams of ionising radiation, alpha, beta and gamma, passing between two parallel plates. The plates are charged and there is a uniform electric field between them.

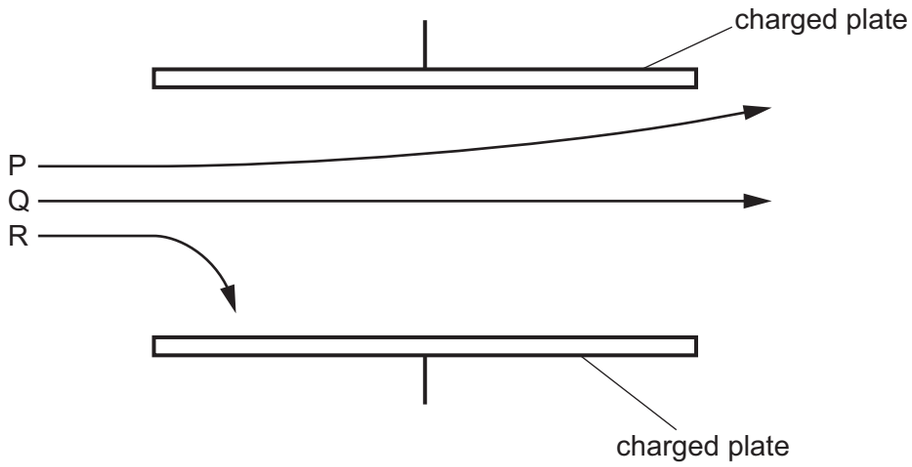


**Fig. 8.1**

**(i)** State and explain which path P, Q or R shows a beam of alpha particles.

path .....

explanation .....

.....

.....

[2]

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**(ii)** State and explain the direction of the electric field between the plates.

direction .....

explanation .....

.....

.....

[1]

**(c)** Curium-242 is a radioactive isotope. It emits alpha particles. It has a half-life of 160 days.

After 480 days, the mass of curium-242 in a sample is $2.4 \times 10^{-3}$ g.

Calculate the initial mass of curium-242 in the sample.

initial mass of curium-242 = ..... [2]

[Total: 7]

# 6 SPACE PHYSICS

27 questions · Paper 2 & Paper 4

# Vocabulary · Topic 6 SPACE PHYSICS

词汇表（中英对照）

| | | | |
|---|---|---|---|
| **solar system** | 太阳系 | **redshift** | 红移 |
| **planet** | 行星 | **Hubble constant** | 哈勃常数 |
| **moon** | 月球 | **Big Bang** | 大爆炸 |
| **comet** | 彗星 | **CMBR** | 宇宙微波背景辐射 |
| **asteroid** | 小行星 | **light-year** | 光年 |
| **galaxy** | 星系 | **protostar** | 原恒星 |
| **Milky Way** | 银河系 | **red giant** | 红巨星 |
| **star** | 恒星 | **white dwarf** | 白矮星 |
| **Sun** | 太阳 | **supernova** | 超新星 |
| **orbit** | 轨道 | **neutron star** | 中子星 |
| **orbital speed** | 轨道速度 | **black hole** | 黑洞 |
| **gravitational field** | 引力场 | | |

# 6.1 The Earth and the Solar System

7 MCQ + 7 theory questions · Paper 2 & Paper 4

· Know that the Earth is a planet that rotates on its axis, which is tilted, once in approximately 24 hours, and use
· Know that the Earth orbits the Sun once in approximately 365 days and use this to explain the periodic nature of th
· Know that it takes approximately one month for the Moon to orbit the Earth and use this to explain the periodic nat
· Define average orbital speed from the equation $v = 2\pi r/T$ where r is the average radius of the orbit and T is the or
· Describe the Solar System as containing: (a) one star, the Sun (b) the eight named planets and know their order fro
· Know that, in comparison to each other, the four planets nearest the Sun are rocky and small and the four planets f
· Know that planets, minor planets and comets have elliptical orbits, and recall that the Sun is not at the centre of

**38** A student defines three periods of time:

- the time $T_S$ taken for the Earth to orbit the Sun
- the time $T_R$ taken for the Earth to rotate once on its axis
- the time $T_M$ taken for the Moon to orbit the Earth.

Which row shows the shortest period of time and the longest period of time?

| | shortest period of time | longest period of time |
|---|---|---|
| **A** | $T_S$ | $T_R$ |
| **B** | $T_R$ | $T_S$ |
| **C** | $T_R$ | $T_M$ |
| **D** | $T_M$ | $T_S$ |

0625/21/M/J/25 Q38

**38** A satellite is in orbit around the Earth. The orbit has radius $r$ and period $T$.

Which expression gives the orbital speed of the satellite?

**A** $\frac{r}{2\pi T}$ **B** $\frac{T}{2\pi r}$ **C** $\frac{2\pi r}{T}$ **D** $\frac{2\pi T}{r}$

0625/21/O/N/25 Q38

**39** Comets move round the Sun.

Which statement describes the orbit of a comet?

**A** The orbit is circular with the Sun at the centre of the circle.

**B** The orbit is circular with the Sun **not** at the centre of the circle.

**C** The orbit is elliptical with the Sun at the centre of the ellipse.

**D** The orbit is elliptical with the Sun **not** at the centre of the ellipse.

0625/21/O/N/25 Q39

**38** Which statement is correct?

**A** The Earth orbits the Sun once in approximately 24 hours.

**B** The Earth orbits the Moon once in approximately 1 week.

**C** The Moon orbits the Earth once in approximately 1 month.

**D** The Sun orbits the Earth once in approximately 365 days.

0625/22/M/J/25 Q38

**38** In which direction does the Sun appear to move every day, when viewed from the Earth?

**A** east to west

**B** north to south

**C** south to north

**D** west to east

0625/23/M/J/25 Q38

**40** The orbit of Mercury is closer to the Sun than the orbit of the Earth is to the Sun.

How do the values of the gravitational field strength of the Sun near Mercury and the orbital speed of Mercury compare to those values for the Earth?

| | gravitational field strength of the Sun near Mercury | orbital speed of Mercury |
|---|---|---|
| **A** | greater | faster |
| **B** | greater | slower |
| **C** | smaller | faster |
| **D** | smaller | slower |

0625/23/M/J/25 Q40

Date: _____ Score: _____

**38** An artificial satellite orbits planet X at a distance $r = 7.0 \times 10^6$ m from the centre of planet X.

The average orbital speed $v$ of the satellite is 3500 m / s.

What is the orbital period?

**A** 21 minutes

**B** 52 minutes

**C** 1 hour and 45 minutes

**D** 3 hours and 29 minutes

0625/23/O/N/25 Q38

**10** Jupiter and the Earth are planets in our Solar System.

**(a)** Describe the composition of Jupiter and the Earth.

Jupiter .....

the Earth .....
[2]

**(b)** The gravitational field strength at the surface of the Earth is approximately 9.8 N / kg. The gravitational field strength at the surface of Jupiter is approximately 23 N / kg.

**(i)** Define gravitational field strength.

.....

..... [2]

**(ii)** State **one** factor which causes the difference between the gravitational field strength at the surface of Jupiter and the gravitational field strength at the surface of the Earth.

.....

..... [1]

**(c)** State and explain the difference between the orbital speed of Jupiter and the orbital speed of the Earth.

statement .....

explanation .....

.....
[3]

[Total: 8]

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**10** Fig. 10.1 shows the orbit of the Earth and the orbit of a comet around the Sun.

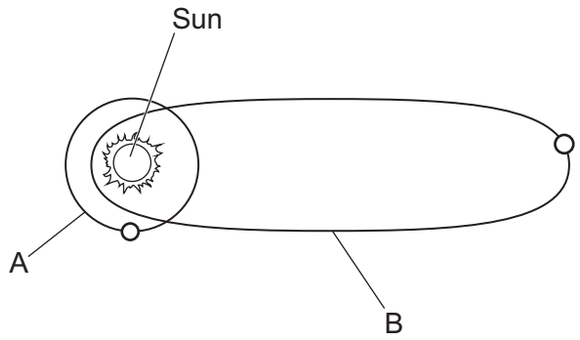


**Fig. 10.1**

**(a)** State which orbit, A or B, is the orbit of the comet.
Explain your answer.

orbit of comet is .....

explanation .....

.....
[1]

**(b)** Describe and explain how the motion of the comet changes as it orbits the Sun.

.....

.....

..... [2]

**(c)** At one position in its orbit, the comet is $6.6 \times 10^{-6}$ light-years away from the Earth.

**(i)** State the meaning of light-year.

.....

..... [1]

**(ii)** Determine the distance in metres between the comet and the Earth.

distance = ..... m [2]

[Total: 6]

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**10** Fig. 10.1 shows the path of the Earth as it orbits the Sun. X is a position on the Earth where scientists observe the apparent motion of the Sun throughout the year.

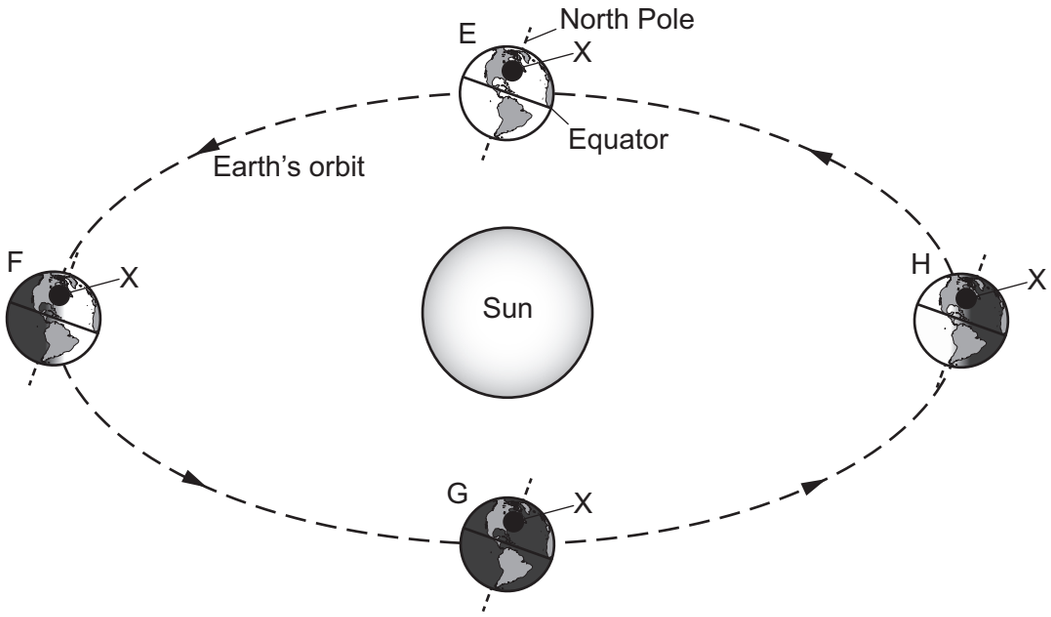


**Fig. 10.1**

**(a)** Determine how many days it takes the Earth to move around its orbit from F to G. Explain your answer.

number of days = .....

explanation .....

.....

[2]

**(b)** Fig. 10.1 shows four positions E, F, G and H of the Earth in its orbit of the Sun.

**(i)** Identify the position of the Earth when it is summer at X. ..... [1]

**(ii)** Identify the position of the Earth when it is winter at X. ..... [1]

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**(c)** The orbital speed of the Earth around the Sun is approximately $3.0 \times 10^4$ m/s.

Calculate the average radius of the Earth's orbit.

radius = ..... [3]

**(d)** Earth is a planet in the Solar System. State **one other** type of naturally occurring object that is present in the Solar System.

..... [1]

[Total: 8]

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**10 (a)** Table 10.1 shows data for planets A, B, C and D.

**Table 10.1**

| planet | surface temperature / K | diameter / km | gravitational field strength / N / kg |
|---|---|---|---|
| A | 153 | 143 000 | 23.0 |
| B | 623 | 4 800 | 3.7 |
| C | 53 | 50 000 | 11.0 |
| D | 93 | 120 000 | 9.0 |

**(i)** State and explain which one of these planets is closest to the Sun.

.....

..... [2]

**(ii)** Calculate the surface temperature of planet D in °C.

surface temperature = ..... °C [1]

**(iii)** An object falls through a height of 2 m on each of the planets in Table 10.1.

State and explain on which planet the object falls 2 m in the shortest time. Ignore any effect due to the atmosphere.

.....

..... [2]

**(b)** Fig. 10.1 shows some of the stages in the life cycle of a massive star.

massive star expands

↓

.....

↓

supernova

↓

nebula which contains .....

↓

leaves behind

..... or .....

**Fig. 10.1**

**(i)** Complete Fig. 10.1 by adding the correct terms. [4]

**(ii)** State the quantity that the brightness of a supernova in a galaxy can be used to determine.

..... [1]

[Total: 10]

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**10** **(a)** Fill in the gaps in the description of how an accretion model explains the formation of the Solar System.

The planets nearest the Sun are small and .....

The planets furthest from the Sun are large and .....

All the planets were formed when a cloud of gas and dust collapsed due to

.....

The rotation of material in the cloud forms an .....

[4]

**(b)** The time taken for light to travel from the Moon to the Earth is 1.3 s.

Calculate the distance of the Moon from the Earth. Give your answer in m.

distance = ..... m [3]

**(c)** The Whirlpool galaxy is $2.3 \times 10^{7}$ light-years from the Earth. The current value of the Hubble constant is $2.2 \times 10^{-18}$ per second.

Calculate the speed at which the Whirlpool galaxy is moving away from the Earth.

speed = ..... [3]

[Total: 10]

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**9** **(a)** Cosmic microwave background radiation (CMBR) can be observed in every direction in space around us.

**(i)** State when CMBR was produced.

..... [1]

**(ii)** State and explain how the wavelength of this radiation has changed since it was formed.

.....

.....

..... [2]

**(b)** Table 9.1 shows data about planets in our solar system.

**Table 9.1**

| planet | average radius of orbit around the Sun / $10^6$ km | time for one orbit / Earth years | average surface temperature / °C |
|---|---|---|---|
| Venus | 108 | 0.62 | 460 |
| Earth | 150 | 1.00 | 15 |
| Mars | 228 | 1.90 | –65 |
| Jupiter | 778 | 12.00 | –110 |
| Saturn | 1430 | 29.00 | –140 |

**(i)** The orbit of Venus is approximately circular.

Calculate the average orbital speed $v$ of Venus. Give your answer in km / h.

orbital speed = ..... km / h [3]

**(ii)** State how the average surface temperature of a planet changes as its distance from the Sun increases.

..... [1]

[Total: 7]

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**10** **(a)** Sketch a diagram to show the orbital path of a comet around the Sun.

Mark the position of the Sun on your diagram using an S.

[2]

**(b)** Draw a cross on your diagram to show where the comet is travelling with the slowest speed. Label the cross **L**. [1]

**(c)** Explain why the speed of a comet changes as it travels around its orbit.

Use the conservation of energy in your explanation.

.....

.....

.....

..... [3]

[Total: 6]

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# 6.2 Stars and the Universe

12 MCQ + 1 theory questions · Paper 2 & Paper 4

· Know that the Sun is a star of medium size, consisting mostly of hydrogen and helium, and that it radiates most of
· Know that stars are powered by nuclear reactions that release energy and that in stable stars the nuclear reactions
· State that: (a) galaxies are each made up of many billions of stars (b) the Sun is a star in the galaxy known as th
· Know that one light-year is equal to 9.5 × 1015 m
· Describe the life cycle of a star: (a) a star is formed from interstellar clouds of gas and dust that contain hydro
· Know that the Milky Way is one of many billions of galaxies making up the Universe and that the diameter of the Mil
· Describe redshift as an increase in the observed wavelength of electromagnetic radiation emitted from receding star

**39** Which statement about the Universe is correct?

**A** The Universe consists of 100 000 galaxies.

**B** Redshift is a decrease in the observed wavelength of electromagnetic radiation emitted from receding galaxies.

**C** Light emitted from distant galaxies appears redshifted.

**D** Redshift of the light from distant galaxies gives evidence that the Universe is contracting.

0625/21/M/J/25 Q39

**40** Which statement about the cosmic microwave background radiation (CMBR) is correct?

**A** CMBR now has a frequency higher than that of infrared radiation.

**B** CMBR has always had exactly the same frequency.

**C** CMBR had a shorter wavelength in the past.

**D** CMBR was produced a long time after the Universe was formed.

0625/21/M/J/25 Q40

**40** A galaxy is at a distance of 60 million light-years from the Earth. The galaxy is moving away at a speed of 1300 km / s.

One light-year is about $9.5 \times 10^{12}$ km.

There are about $3.2 \times 10^{7}$ seconds in one year.

Which expression can be used to estimate the age of the Universe in years?

**A** $\dfrac{60 \times 10^{6} \times 3.2 \times 10^{7}}{1300 \times 9.5 \times 10^{12}}$

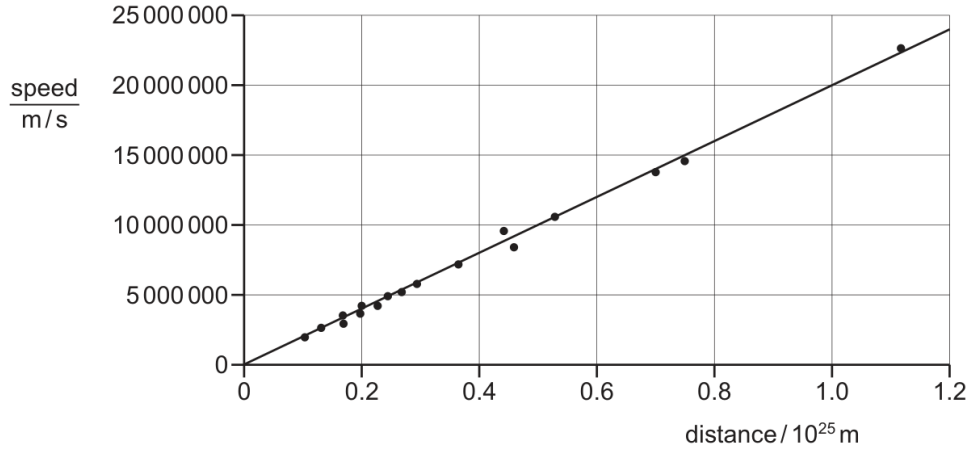
**B** $\dfrac{60 \times 10^{6} \times 9.5 \times 10^{12}}{1300 \times 3.2 \times 10^{7}}$

**C** $\dfrac{1300 \times 9.5 \times 10^{12}}{60 \times 10^{6} \times 3.2 \times 10^{7}}$

**D** $\dfrac{1300 \times 3.2 \times 10^{7}}{60 \times 10^{6} \times 9.5 \times 10^{12}}$

0625/21/O/N/25 Q40

**40** A student plots a graph of the speed of different galaxies moving away from the Earth against their distance from the Earth.




Using the best straight line shown on the graph, what is the value of the Hubble constant $H_0$?

**A** $2.0 \times 10^{-18}\,s^{-1}$

**B** $2.2 \times 10^{-18}\,s^{-1}$

**C** $4.5 \times 10^{17}\,s$

**D** $5.0 \times 10^{17}\,s$

**39** Stars are powered by nuclear reactions that release energy.

Which statement describes the nuclear reaction in stable stars?

**A** Nuclear fission converts helium into hydrogen.

**B** Nuclear fission converts hydrogen into helium.

**C** Nuclear fusion converts helium into hydrogen.

**D** Nuclear fusion converts hydrogen into helium.

0625/22/M/J/25 Q39

**40** A galaxy is moving away from the Earth at a velocity of 3000 km / s.

What is the distance between the Earth and the galaxy?

**A** $7.3 \times 10^{-25}$ m

**B** $7.3 \times 10^{-22}$ m

**C** $1.4 \times 10^{21}$ m

**D** $1.4 \times 10^{24}$ m

0625/22/M/J/25 Q40

**39** The galaxy Draco II is a distance of $6.6 \times 10^{20}$ m from Earth.

What is this distance in light-years?

**A** $1.4 \times 10^{4}$ light-years

**B** $6.9 \times 10^{4}$ light-years

**C** $1.4 \times 10^{6}$ light-years

**D** $6.9 \times 10^{6}$ light-years

0625/22/O/N/25 Q39

**40** What is a possible result of a supernova explosion?

**A** a neutron star

**B** a red giant star

**C** a red supergiant star

**D** a white dwarf star

0625/22/O/N/25 Q40

**39** A distant galaxy is moving away from the Milky Way at 4 400 000 m / s.

What is the approximate distance between the Milky Way and this distant galaxy?

**A** $2 \times 10^{12}$ m **B** $5 \times 10^{11}$ m **C** $2 \times 10^{24}$ m **D** $5 \times 10^{23}$ m

0625/23/M/J/25 Q39

**17** Star X has a larger diameter than star Y but they both emit radiation with the same power.

Which statement is correct?

**A** The surface temperature of X is less than the surface temperature of Y.

**B** The surface temperatures of X and Y are the same.

**C** The surface temperature of X is greater than the surface temperature of Y.

**D** The surface areas of X and Y are the same.

0625/23/O/N/25 Q17

**39** The table lists data about four different stars.

Which star is the Sun?

| | main elements in the star | regions of electromagnetic spectrum where most energy is radiated by the star |
|---|---|---|
| **A** | helium, hydrogen, carbon, nitrogen, oxygen | blue, ultraviolet, X-ray |
| **B** | helium, hydrogen, carbon, nitrogen, magnesium | red, infrared |
| **C** | helium, hydrogen | infrared, ultraviolet, visible |
| **D** | helium, hydrogen | red, infrared |

0625/23/O/N/25 Q39

**40** What can be formed directly following a supernova?

**A** a nebula and a neutron star

**B** a nebula and a white dwarf

**C** a red supergiant and a neutron star

**D** a red supergiant and a protostar

0625/23/O/N/25 Q40

**10 (a)** Define a light-year in words.

..... [1]

**(b)** It takes light 490 s to travel from the Sun to the Earth.

Calculate the distance from the Sun to the Earth.

distance = ..... [2]

**(c)** Fig. 10.1 is a scatter graph showing how the speed of galaxies moving away from the Earth varies with their distances from the Earth.

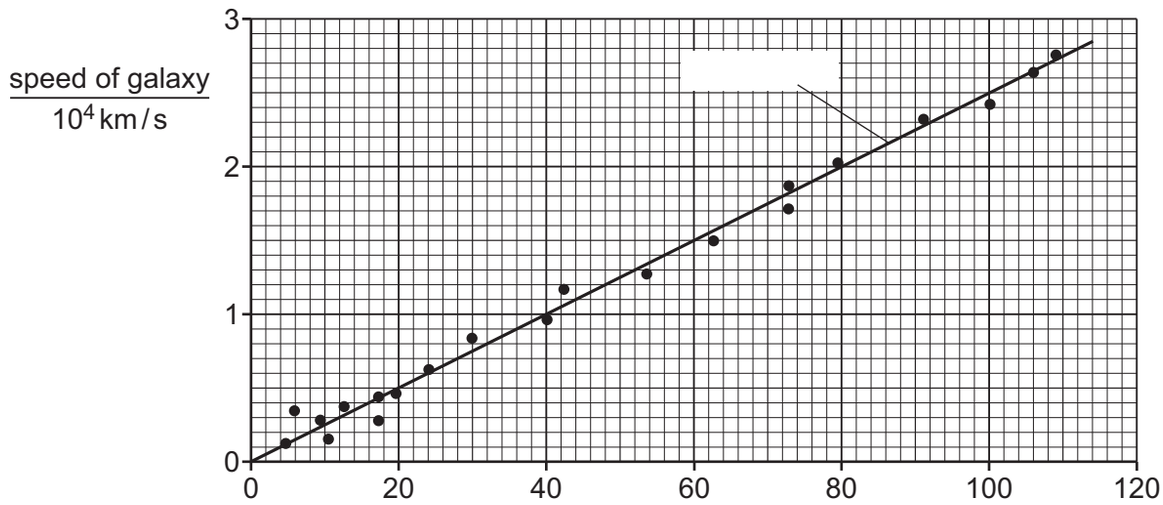



**Fig. 10.1**

A scientist draws a best-fit line on the scatter graph.

Use this best-fit line to determine a value for the Hubble constant.

Hubble constant = ..... [3]

**(d)** The speed of a receding galaxy can be estimated using redshift.

Describe what is meant by redshift.

.....

.....

.....

..... [2]

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[Total: 8]

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# Knowledge Recall

知识点默写

# Part 1 general physics

Define:

**Acceleration(a=Δv/t)**:_____unit:_____

**Force(F=Δp/t）**:_____unit:_____

**Impulse(=Ft):**_____unit:_____

**Density(=m/V)**:_____unit:_____

**Pressure(=F/A)** :_____unit:_____

**gravitational field strength(g=F/m)**:_____ _____unit:____

**Power(=W/t)**_____unit:_____

**Moment(=Fd)**:_____unit:__

**Momentum(=mv)**:_____unit:__

**Work(=FS)**_____unit:__

State what is meant by:

**Hooke’s law (F=kx):**_____

_____

**principle of the conservation of momentum**:_____

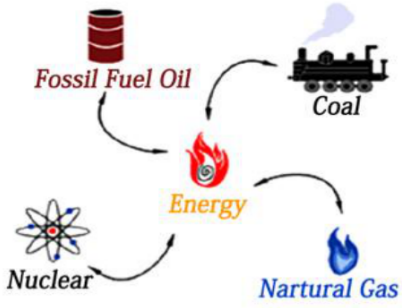
_____ $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

**note : v is “+”value in positive direction ,but “—”value in negative direction**

- Conditions for a body in _____平衡: i:_____ ii:_____
- _____再生能源&_____不可再生能源

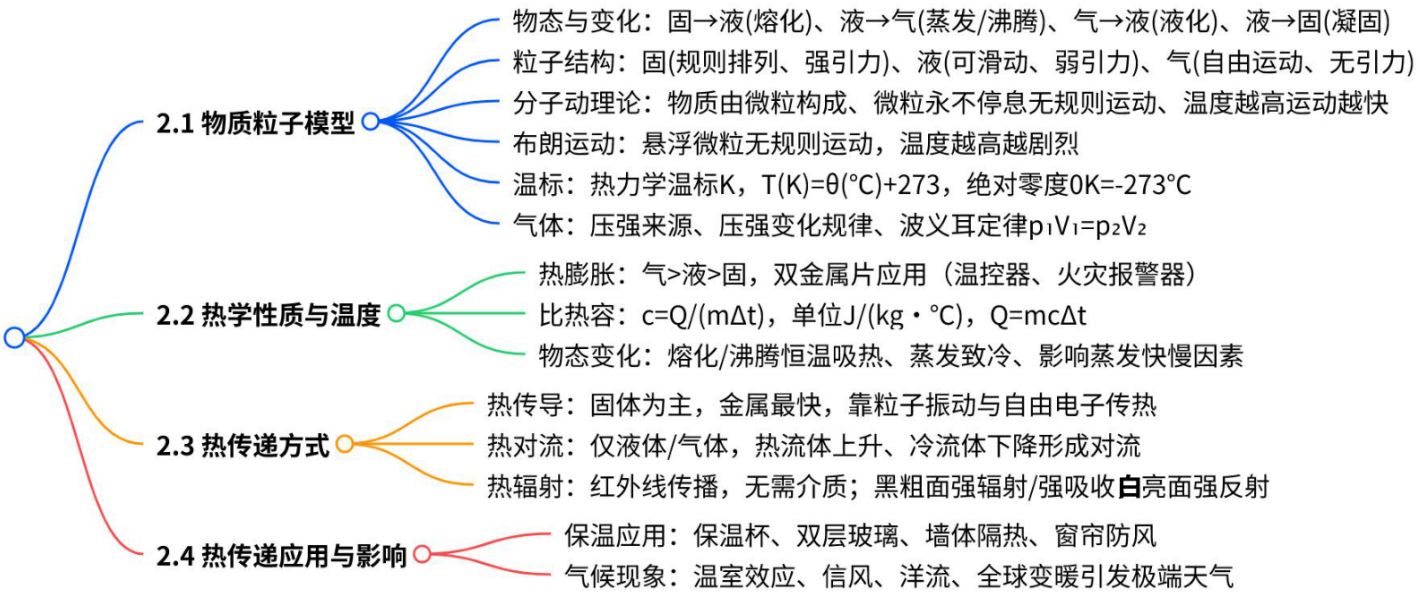
Non-Renewable Energy

不可再生能源都是燃料类型的，用完就没啦：




the Sun is the source of energy for **all** our energy resources

# IGCSE Physics----- Thermal Physics Key Summary




## 1 *Kinetic particle model of matter*

**(1)**the particle structure of solids, liquids and gases

| **Solid** Fixed shape and volume | **Liquid** Fixed shape changed volume | **Gas** no fixed shape and volume |
|---|---|---|
| •Strong attractive force hold Particles close together<br>•_____规则排列<br>•____振动 in fixed positions | •Weaker attractive forces than solids to hold Particles together.<br>•_____不规则排列<br>•Particles vibrate but can _____滑动over each other | • no attractive forces between particles<br>•Particles apart to each other<br>• Particles move and_____ |

(2) evidence for the kinetic molecular model of matter

_____布朗运动:_____

_____

(3) **simple kinetic molecular model of matter**物质的分子动理论模型

**i, Matter is made up of very tiny particles (** _____分子, _____原子, _____离子**)**

**ii, All the particles can move** _____随机的 **and** continuously持续的.

**iii, The higher the temperature, the** _____**the particles move.**

**temperature is a measure of** _____平均动能 **of molecules.**

- _____**内能--the sum of** _____**分子动能** and_____**分子势能.**

➢ why evaporation makes things cooler?为什么蒸发让物体变冷

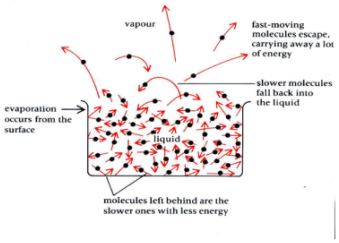
_____

_____

➢ **what is the difference between boiling and** _____**蒸发**:

(i) _____

(ii)_____




**（4）the** _____开尔文温度（热力学温标） **unit:**_____**(K)**

_____绝对零度:_____the point at which molecules have no kinetic energy

**absolute zero=0K=**_____**°C**

**T (K) = θ ( °C) +** _____

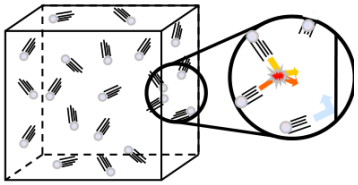
### (5) Gases

➢ **Explain gas pressure** in terms of the change of momentum of the particles striking the walls creating a force.

_____

_____

_____



➢ Pressure changes

•**If the volume is constant, then increasing the temperature will increase the pressure.**

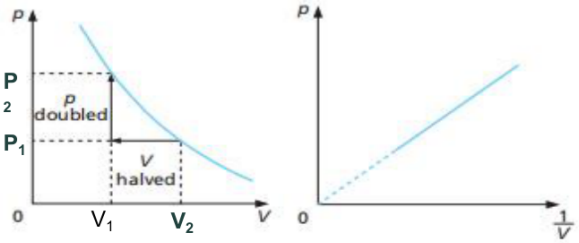
explain:_____.

**If the temperature is constant, then decreasing the volume will increase the pressure.**

explain:_____

➢ **For a fixed mass of gas, at constant temperature, its pressure is_____反比 to its volume.**

**$pV$ = constant or $P_1 \times V_1 = P_2 \times V_2$**




## 2 Thermal properties and temperature

(1) _____**热膨胀** of solids, liquids and gases _____加热后分子振动更剧烈，间距变大，物体体积变大

expansion is greatest in_____, less in_____ and lowest in_____

(2)_____比热容 $C = \dfrac{Q}{m\Delta t}$

- define :_____ **unit:** _____

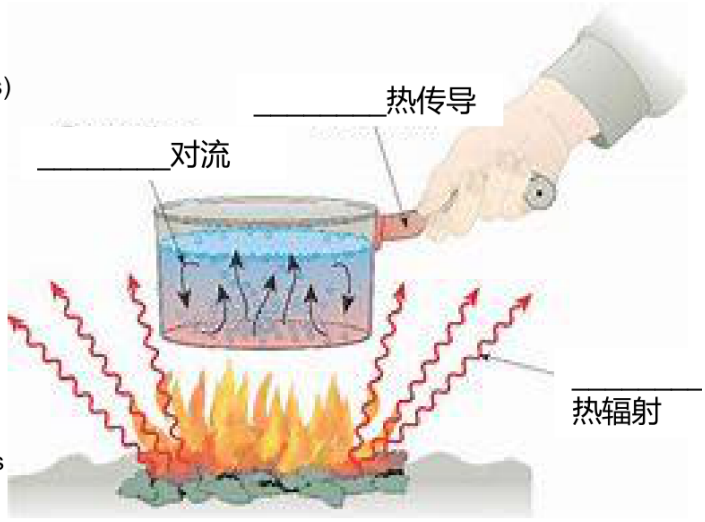
### 3._____热传递

(1)_____热传导is best in _____（solids，liquids or gas) energy passed by particle _____振动。

➢ metal is better _____热导体but worse____绝缘体 because energy passed by ion vibration and a better way ---electrons are free to move randomly and collide with atoms then pass on the energy.

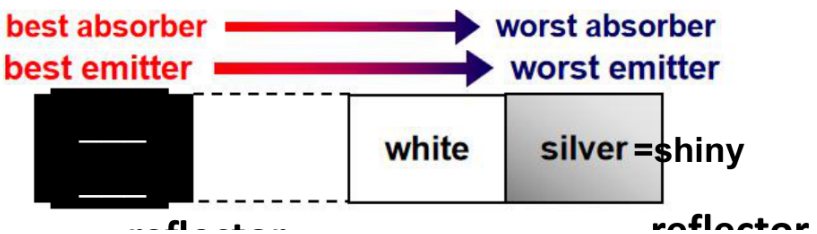
(2)_____对流 occurs in _____and _____ （solids，liquids or gas)

Warm up→ less dense → rises; cool down→ more dense→ sinks




•(3)_____热辐射： All objects **emit** (give out) thermal energy in the form of_____红外线 but very hot objects also give out light waves.

• An emitter will cool down quickly, an absorber will heat up more quickly and a reflector will not heat up quickly



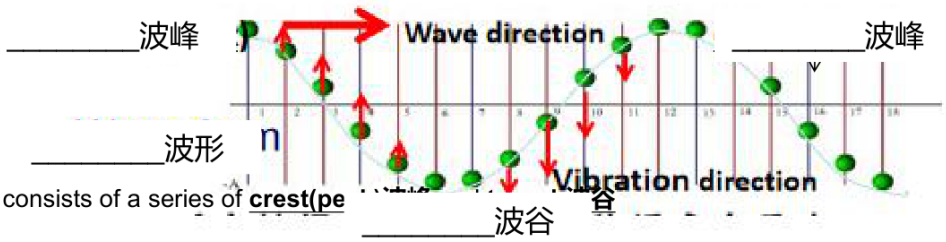
# 3 waves

## 1 General properties of waves

(1)two forms:根据传播方向分类

- _____横波:_____

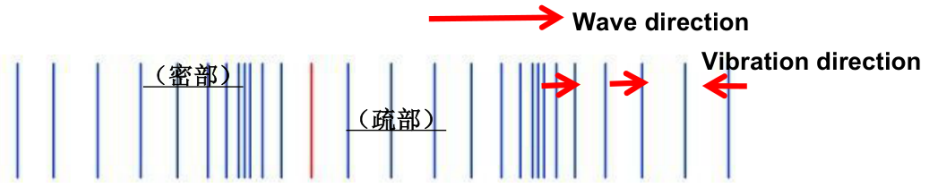
Examples：_____电磁波 ,_____**水波**




**wave form波形** consists of a series of **crest(pe** **谷**

- _____纵波:_____

Examples:_____**声波**



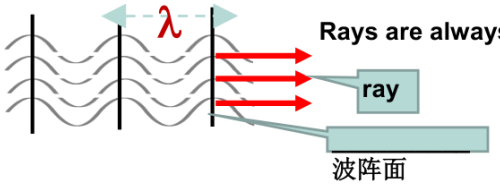

**wave form波形** consists of a series of _____密部**(partivle close together)** and _____**疏部(particles furthest away)**

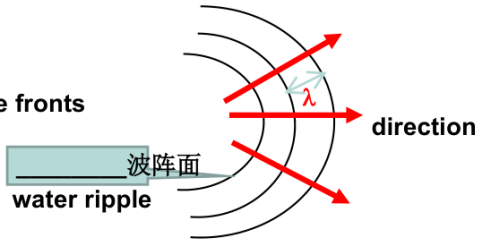
- In an earthquake there are two types of seismic waves:
  primary P-waves and secondary S-waves.

**P-waves are**_____These waves are called **primary waves** as they travel_____ and so are felt first.

**S- waves are**_____**called Secondary wave**

**(2) to describe the wave:**



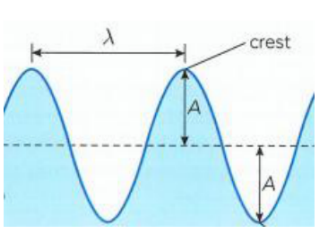

**(1)A---amplitude（振幅）**:_____

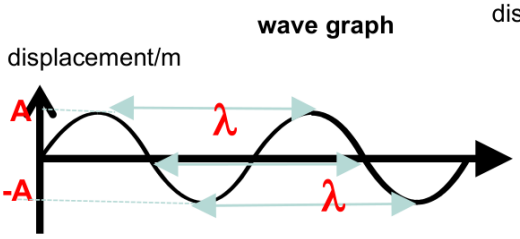
**(2)λ----wavelength（波长）**：_____unit :m

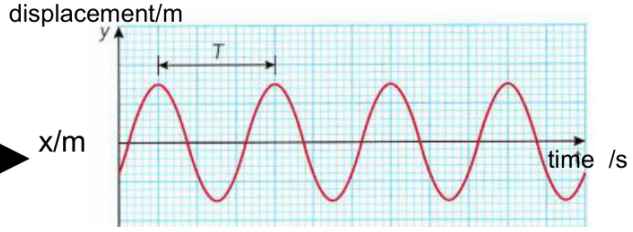
**(3)f---frequency（频率）**:_____. **f=1/T unit: hertz (Hz赫兹) = 1/second (1/s)**

**(4)T---period（周期）**：_____. **T=1/f unit:s**

**(5)v---wave speed（波速）v=**_____公式 **unit: m/s**

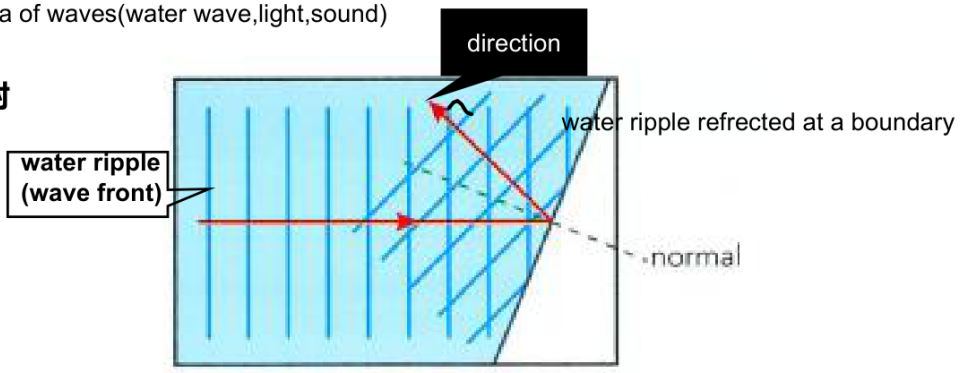


(4) the common phenomena of waves(water wave,light,sound)

➢ _____反射




- Waves bounce away from surface at same angle they strike it.
- **the law of reflection反射定律:** _____入射角）= _____反射角

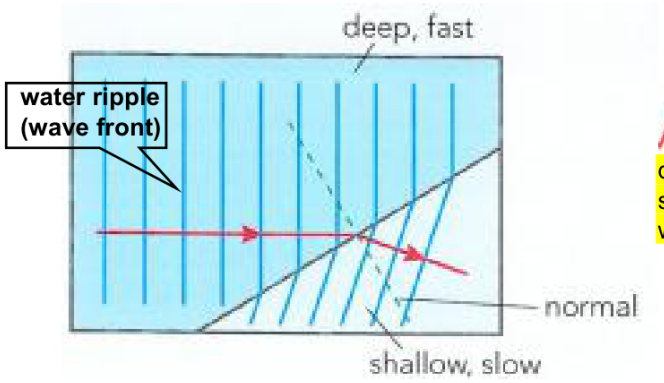
- The plane mirror image像 has these properties:
  - _____
  - _____
  - _____左右颠倒
  - Image is_____虚幻的:

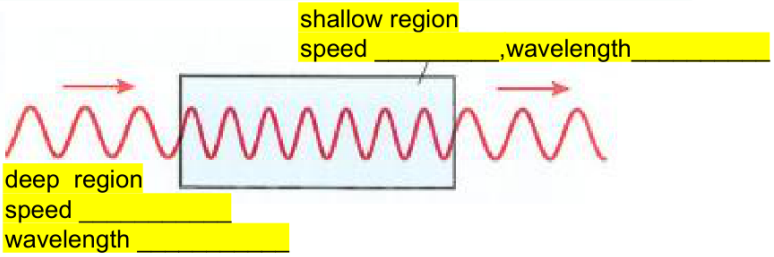
➢ _____折射

_____

_____由于不同介质中的波速不同导致的波的传播方向变化

e.g. water wave move _____慢 in shallow region, but _____快 in deep region,so water wave is refracted when it passing from the deep region to the shallower region.




The speed of light in vacumn(air) is _____, it will _____变慢in some medium(water, glass) .
The more the light is _____, the more it bends _____ the normal line.
the _____ the angle to the normal line.

_____折射率 n = _____ = _____

in medium: n>1
greater n-------> smaller speed---> smaller angle
(means "more refracted=more change in direction")

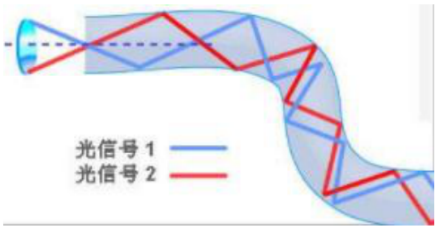
the medium with **greater n (smaller v) is** _____ **光密材料**

➢ _____(TIR)全反射

**When light passes from an _____optical dense to a _____dense medium, all the incident light is _____inside the denser medium**

**Critical angle C（临界角）:** _____

$$n = \frac{\sin 90^0}{\sin C} = \frac{1}{\sin C} \text{---} > \sin C = _____$$

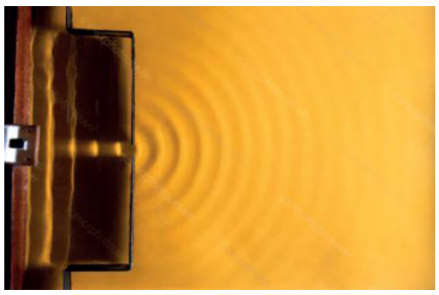


➢ _____衍射

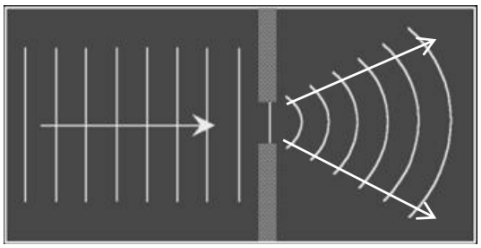
**the _____ of waves when they pass through an 狭缝 or round an _____**

波长越长衍射现象越容易

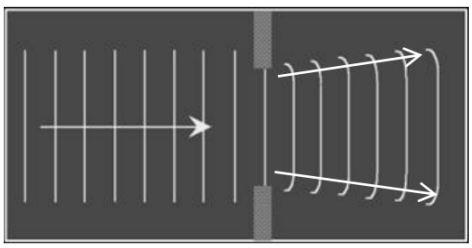
Diffraction water waves in a ripple tank



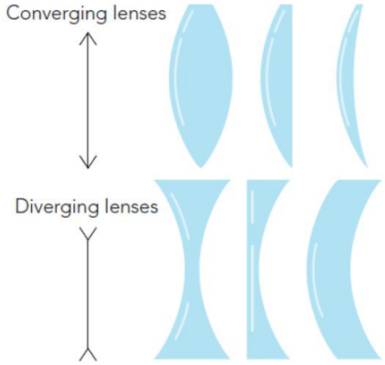
**more diffraction: _____**



**less diffraction: _____**



## 3.2.3 Thin lenses

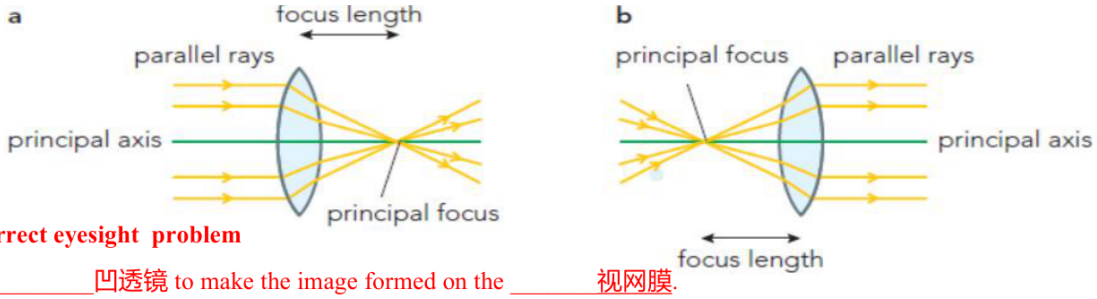



**_____（凸透镜/会聚透镜）**

**_____凹透镜/发散透镜）**

➢ **Physical terms about the lens**

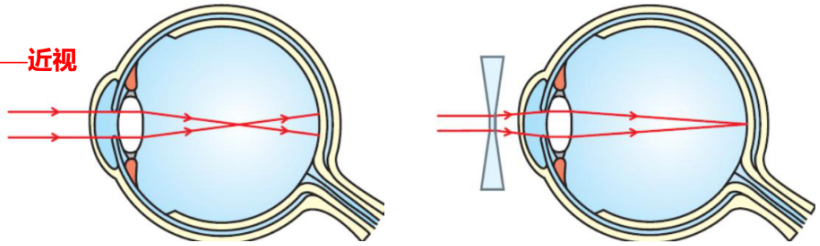
- _____焦点）：_____
- _____焦距）：_____
- _____主光轴）：line that goes through optical center.
- _____光心）：the center of the lens
- _____实像）：image can be caught on a screen;when object is further away from the optical centre than **one F**
- _____虚像）：image cannot be caught on a screen When the object is closer to the optical centre than **one F**.




➢ **Using lense to correct eyesight problem**

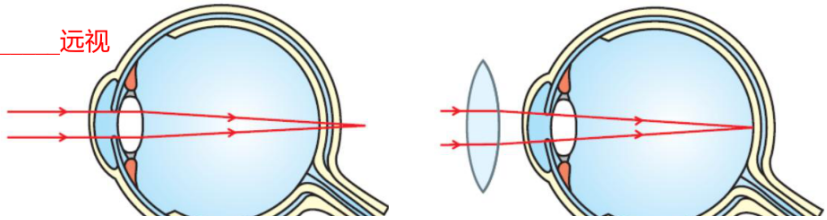
- ---Using _____凹透镜 to make the image formed on the _____视网膜.

_____近视



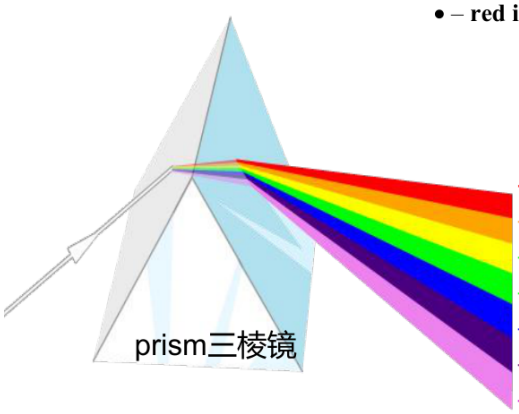
- ---Using_____凸透镜 to make the image formed on the _____视网膜.

_____远视



### 3.2.4 _____ of light光的色散

- **White light is a mixture of colors**, and the prism refracts each color by a different amount due to different speed of colourful light
- – **red is refracted least,violet most;**




– red-------- wavelength, speed , refracted
– orange
– yellow
– green
– blue
– indigo
– --------shortest , smallest , most

- _____单色光: _____单一频率的光 (used in laser激光)

## 3.3 _____电磁光谱

**fλ=c=_____ in vacumn（真空）;**
**the greater _____,the smaller _____,**

| Radiation | Wavelength range / m |
|---|---|
| 无线电波 _____ | > _____ |
| 微波 _____ | _____ |
| 红外线 _____ | _____ |
| 可见光 _____ | _____ |
| 紫外线 _____ | _____ |
| X-光波 _____ | _____ |
| γ-光波 _____ | _____ |

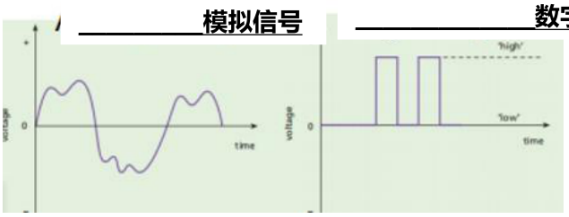
**Properties the E.M. waves（电磁波） have in common相同的:**
**i) they are all transverse wave**
**ii) they all can travel through vacuum**
**iii) they have same speed in vacuum**
**$c=3\times10^8$m/s**
**iv) they all can transfer energy and information**

（1）.typical uses of the different regions of the electromagnetic spectrum including:

- **Radio waves（无线电波）**:_____广播 (TV, AM/FM radio) _____无线电频率识别
- **Micro waves（微波）**: _____手机通信, _____卫星通信, _____无线网络, _____微波炉加热
- **Infrared**: _____遥控器,intruder alarm, _____电烤炉, _____热成像and _____安保系统, _____ 光纤通信
  **visible light:** vision, photography, illumination
- **UV(ultra violet) rays（紫外光）**: _____**晒黑** and _____**灭菌** , security marking, detecting fake ,bank notes,
- **X-rays（X-射线）** _____医学成像& CT scan,
- **Gamma rays（r-ray）**: _____医疗灭菌,_____癌症治疗

通信信号有两种:

- Analogue uses a whole range of continuous variations to transmit a signal.
- Digital signals use only 2 states, on(high) and off(low).



## 3.4 Sound

**(2) Speed of sound waves**:wave speed depends on medium.

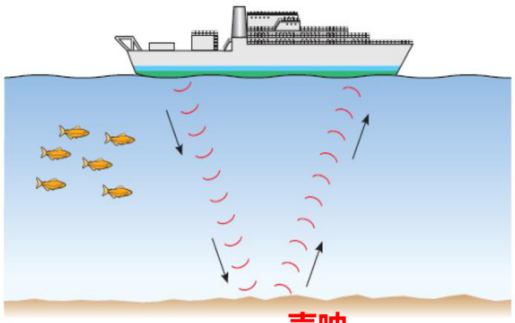
**eg:Air : m/s ; Water: m/s ; Concrete : ; Steel: m/s**
**sound wave move in solid ,then liqiud , in gas.**

(4) three properties of sound note:
**音调）** --increase with frequency
**响度）** -- incease with amplitude
**音色）**--determined by wave form
可听到频率）:Humans can hear frequencies between Hz.

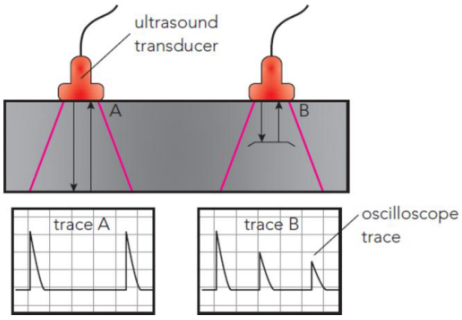
**(5) 回声**: The sound wave is reflected back from the surface.
use: 回声定位.

---Sound with frequency higher than 20kHz, is called 超声波）
---Sound with frequency lower than 20Hz, is called 次声波）

超声波应用



**声呐**




**material testing**

**医学成像（B超）**

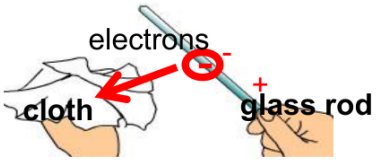
# 4 Electricity and Magnetism

## 4.2.1 Electric charge电荷

**amount of charge**: symble--**Q(q)** ;unit---**C( _____库伦)**

**_____(异种电荷吸引 and**
**_____(同种电荷排斥)**

(3) charging a body--the result of **loss or gain in electrons**(movement of electrons).
When you _____摩擦 a glass rod (or bar) with a cloth, electrons from the rod move onto the cloth. So **the cloth _____becomes negatively charged** overall, and the **rod _____ is positively charged**



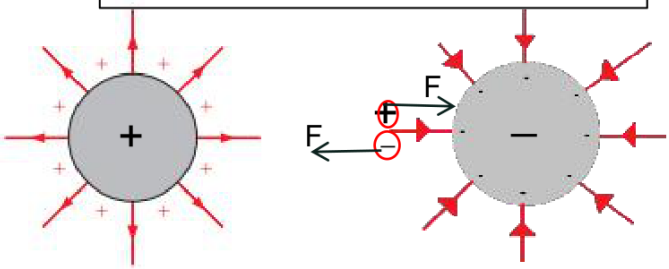

(4) and **conductors**（导体）:

_____绝缘体: A material through which **electrons** _____ , that ***can*** be charged by friction.

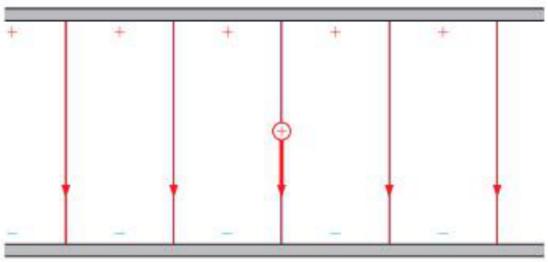
_____**conductors**导体: A material through which electrons_____.

(5)_____**电场):**_____
field lines indicate the **direction of**_____

**field lines pattern for _____点电荷**




**field lines pattern between two charged parallel plates**
**_____匀强电场**



charged conducting sphere=**point charge** concentrated at its centre
inside the conducting sphere,there is no electrical field.

**(6) define current:**_____ **I=Q/t** **unit:_____安培**

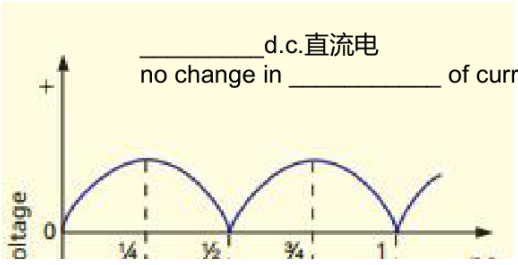
**(7) define E.M.F:**_____ **unit:_____伏特， 1V=_____**

$$\text{E.m.f(E)} = \frac{\text{W(work done)}}{\text{Q}} = \frac{\Delta\text{E(electrical energy supplied)}}{\text{Q}}$$

**(8) define voltage/Potential difference:**_____

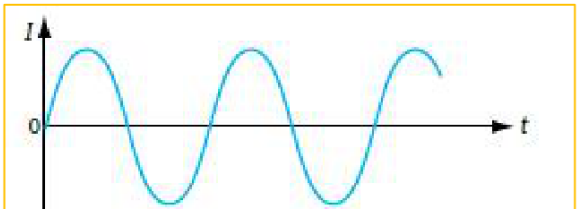
$$\text{V} = \frac{\text{W(work done)}}{\text{Q}} = \frac{\Delta\text{E(electrical energy dissipated )}}{\text{Q}}$$

**(9) Resistance** $R = \frac{V}{I}$ **unit:_____欧姆**




_____d.c.直流电
no change in _____ of current

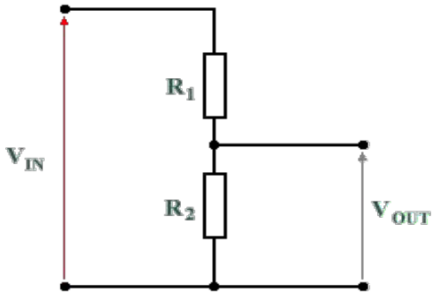
_____a.c.交流电
changed _____



**千瓦时1kWh=1000Wx3600s=3.6X10$^6$J**
**(1h=1hour=3600s)**

meaning:energy transferred in _____ at a rate of transfer of _____

**use of** potential divider circuit **串联分压电路的应用**




$V_1/V_2=R_1/R_2$

$V_{out}= V_{in}R_2/(R_1+R_2)$

**$V_{out}$ depends on the ratio of $R_1$ and $R_2$**

**基本关系就是**

**_____电压和电阻成正比**

**电压大小取决于电阻R1与R2之比**

## 4.4 Electrical safety

•**Damaged insulation（绝缘层破损）:** contact with the_____**火线** due to gap in the insulation causes _____**触电）** which can cause serious injury or shock.

precausion:

- _____**保险丝）:** A **fuse** protects a circuit. Thin piece of wire which_____It is placed on the _____火线 before the switch.

_____**跳闸开关:** a safety switch that automatically open(trip) electricity from flowing when a current exceeds a certain value.

- **Benefits of _____接地） a Metal Case:**
  - Many electrical appliances, have metal cases, the earth wire creates a safe route for current to flow through, if live wire touches casing Earth terminal connected to metal casing, so the current goes through _____地线instead of causing an _____**触电**.

# 4 magnetism

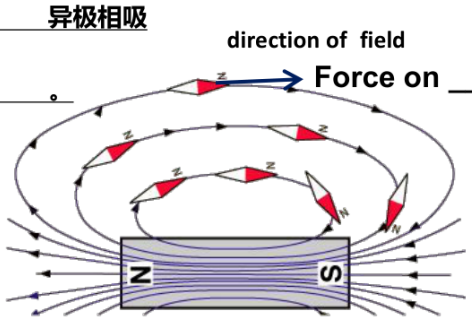
## 4.1 Simple phenomena of magnetism

### 4.1 .1 magnetism

(1)**forces between magnets磁铁**

**同极相斥 and 异极相吸**

(2) **磁场:** 。




(3) **磁性材料（ ）** ---such as i 铁, s 钢, n 镍 and c 钴

(5) Magnetic Properties of Iron and Steel

- **Iron铁** is **软磁铁--- temporary magnetism(**Gets magnetised faster but loses its magnetism as soon as inducing magnet is removed).
  **use：变压器铁芯 &电磁继电器 or 电磁铁**
- **steel钢** is **硬磁铁---permanent magnetism(**Slow to be magnetized but retains magnetism for a long time) **use:making magnet**

(6) three ways to increase the strength of an **electromagnets电磁铁 or solenoids螺线管**

- _____
- _____
- _____ .

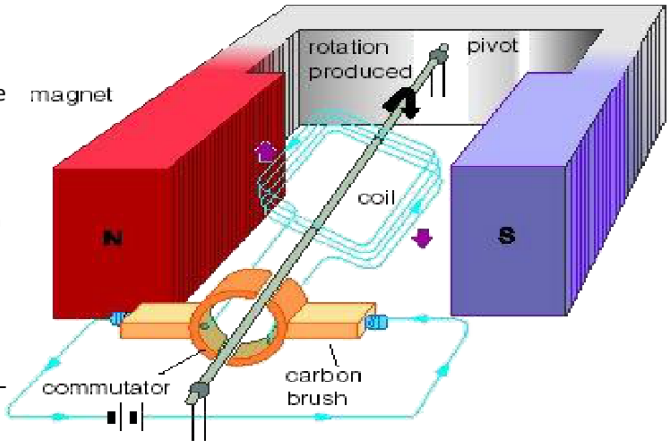
### 4.1.3 直流电动机

(1)how a DC motor runs on a direct current.

- When a current-carrying coil is in a magnetic field, magnet exerts force on current
- there are **two** _____ **with** _____**&**_____
  **to produce** _____

The coil is made of insulated copper wire and is free to _____between the poles of the magnet.

- The _____**开环换向器** is fixed to the coil and rotates with it.
- When the coil overshoots the vertical, **the commutator** ___ _____, **so the forces**_____ **and**_____ _____.




- The **brushes are two** _____ **which rub against the commutator and keep the coil** _____ **to battery.**

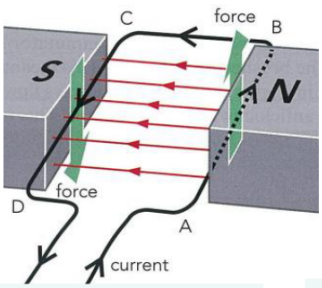
(2)

**TURNING EFFECT INCREASED BY:**

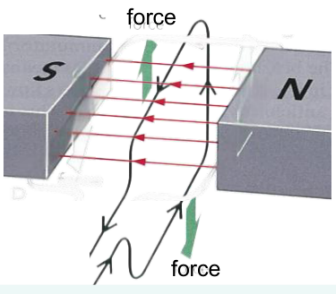
- _____
- _____
- _____

**REVERSING ROTATION CAN BE DONE BY:**

- _____
- _____




- **moment=**_____
- when the coil plane is_____to magnetic field.




- **moment=**_____

when the coil plane is at_____ to the magnetic field.

# 4.5 Electromagnetic effects

## 4.5.1 电磁感应 = 磁场变化

Show understanding that when a conductor moving across a magnetic field then _____变化的磁场linking with a conductor can_____**感应电动势** in the conductor

(1) Describe an experiment to demonstrate electromagnetic induction

(2)State the factors affecting the **magnitude大小** of an **induced e.m.f感应电动势**.

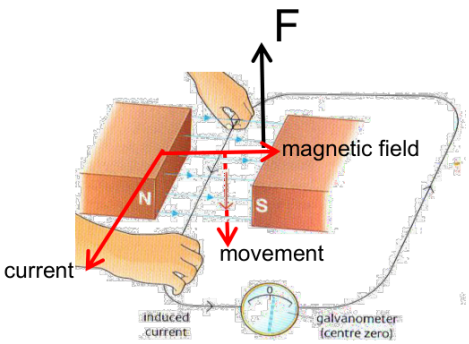
(i) _____

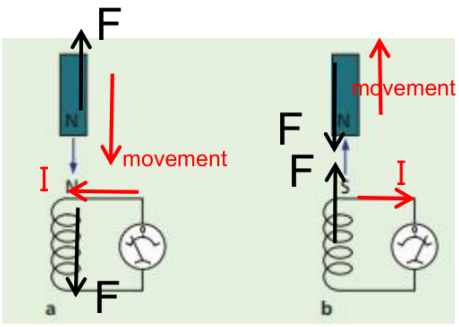
(ii)_____

(iii) _____

(3)Show understanding that the direction of an induced e.m.f (or current)_____

**"the force on induced current alway "_____" the _____of magnet or coil**




➢ example 1

wire"_____" the magnetic field downward

-------> _____

----------> _____ on induced current upward to _____ the motion downward

➢ example 2

- magnet push towards the coil

  ------> _____

  ------->force on induced current to_____ the magnet moving close

- magnet push away the coil

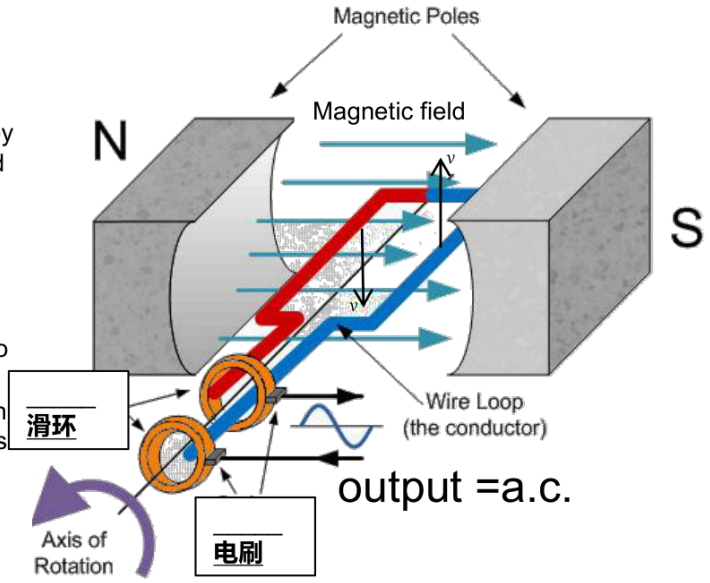
  ------> current induced

  ------->force on induced current to _____ the magnet moving away

## 4.5.2 The a.c. generator 交流发电机

(1) how to work

- The coil is made of insulated copper wire and is _____by turning the shaft; the _____**滑环** are fixed to the coil and rotate with it.
- The _____**电刷** are two_____which rub against the _____ and keep the coil _____ the outside part of the circuit, usually made of carbon.
- **slip rings used** _____
- When the coil is rotated, it_____magnetic field lines, so an EMF is generated, which makes a current flow.
- Each side of the coil travels upwards then downwards then upwards etc. so the current flows backwards then forwards then backwards etc. so it is an alternating current.




(2)current change with time

- **induced current I = _____**
  **when the coil plane is _____ to magnetic field,**

- **induced current I=_____**
  **when the coil plane is _____垂直 to the field, since it is cutting NO field lines.**

(3)**The induecd EMF can be increased by:**

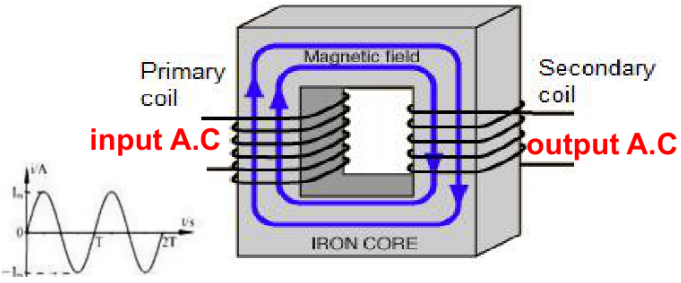
- o _____
- o _____
- o _____

## 4.5.3 _____变压器

(1) how to work:

.

- _____
  _____
- _____
  _____
- _____
  _____.




- _____**升压变压器）** increases the voltage
- _____**降压变压器）** decreases voltage. for ideal transformer (NO energy loss)

_____ $=I_S/I_P$

input power= output power

. $I_p V_p = I_s V_s$

the higherV ， the smaller I

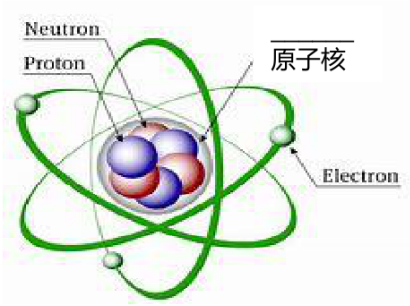
(2)the advantages of **high-voltage transmission高压输电**--- _____when the transmitted _____

explain: transmitted输送 current I=_____ **power lost in a resistor** $P$ = _____

having a lower current will the power loss, so voltage, current get

# IG physics Nuclear physics key Summary

## 1 .The nuclear model of the atom原子的核式结构




_____ 原子核

both_____质子 and_____中子are called _____**核子**

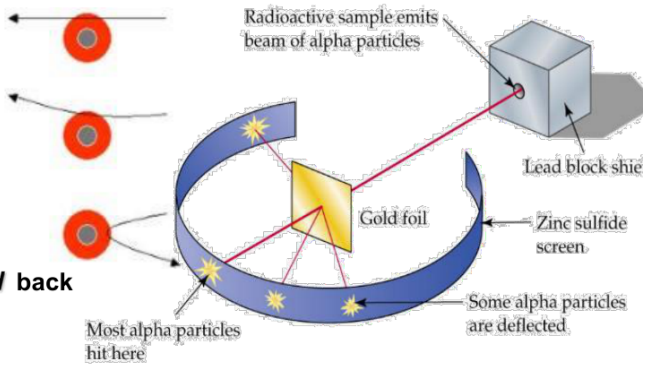
**2.Conclusion of _____阿尔法粒子散射实验**

- **Most *α particles are not deflected***

**Show that：_____**

**A few α *particles are deflected*偏转 and**
**Extremely few (less than one in million) α *particles are deflected* back by nearly 180°**

**Show that：_____**




3._____同位素: _____

**4 _____核裂变：_____**

- **_____核聚变：_____**

for all nuclear reaction, _____**and**_____ are conserved.

_____**核能** released in the form of **γ ray**
**due to there is total mass lost after reaction,**
**the total _____质能守恒.**

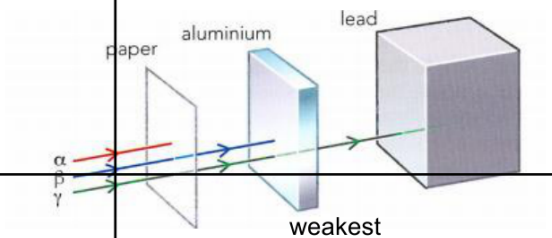
$^{1}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He$ + **nuclear energy（核能）γ**

total mass $^{1}_{1}H + {}^{2}_{1}H$ > total mass $^{3}_{2}He$

**5._____背景辐射: _____**

. It mainly comes from natural sources such (a) _____ (b) _____ (d)_____

- **characteristics（特点） of three kinds of radioactive放射性 emission射线**

| | **Alpha emission α 射线** | **Beta β射线** | **Gamma γ ray γ 射线** |
|---|---|---|---|
| **Nature** | **Helium nucleus $_2$ He** | **electron** | **Electromagnetic Radiation** |
| **Charge** | **+ 2e** | **- e** | **No charge** |
| **Mass** | **4u** 4 | **1/1840 u** | **Massless** |
| _____穿透性 | run a few centimetres in air stopped by a paper | run a few metres in air stopped by a thin aluminum | run untill stopped by a thick lead   |
| _____电离 | most stongly (due to most +charge) | weak | weakest |
| **deflection in magnetic and electric field电磁场中的偏转** | deflected | deflected more (opposite to α) | no deflection |
| **use** | _____烟雾报警器利用电离性_____ | **measure thickness due to the _____穿透性** | **(1)_____消毒;**<br>**r(2)adiotherapy –γ rays kill_____**<br>**(3)detect faults in manufactured goods** |

6._____放射性衰变

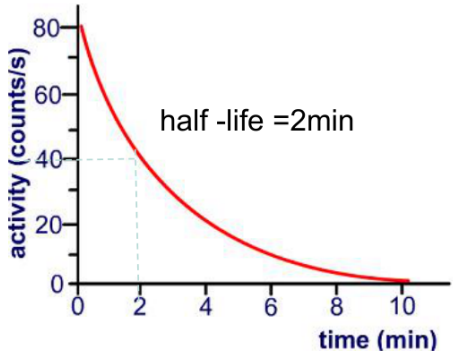
α-decay：equation: ${}^{A}_{Z}X \rightarrow$ ____ $+ {}^{4}_{2}\alpha$

β-decay：equation: ${}^{A}_{Z}X \rightarrow$ ____ $+ {}^{0}_{-1}e^{-}$

the electron come from inside the nucleus,one neutron has split into a proton and electron

${}^{1}_{0}n \rightarrow$ _____ + _____

7. _____半衰期:_____



# IG-Space physics key summary

## 1 Earth,Sun and Moon

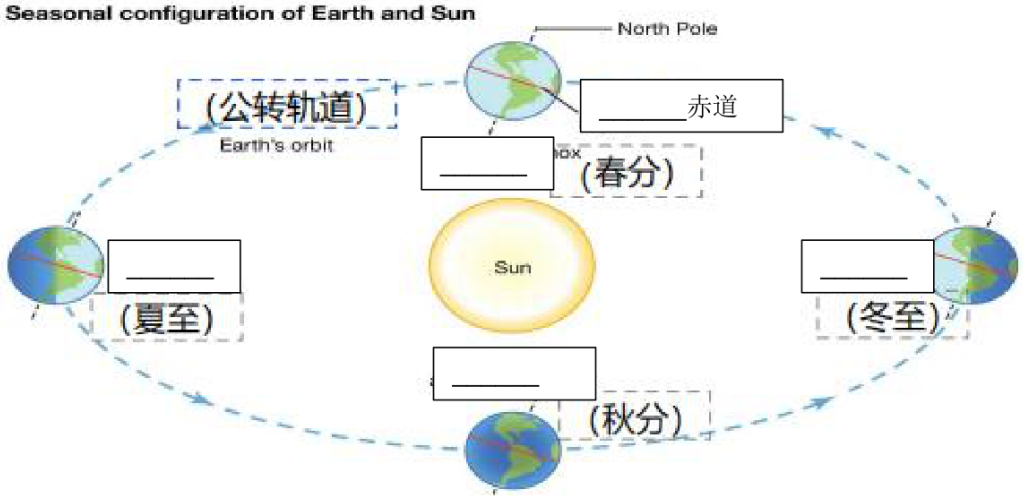
**(1）.Explanation of day and night:**

The Earth _____转动 on its _____地轴 ,It takes _____hours for Earth to complete one rotation.As the Earth spins on its axis,part of the Earth that faces the Sun has _____白天. And the part that faces a way from the Sun has ___黑夜

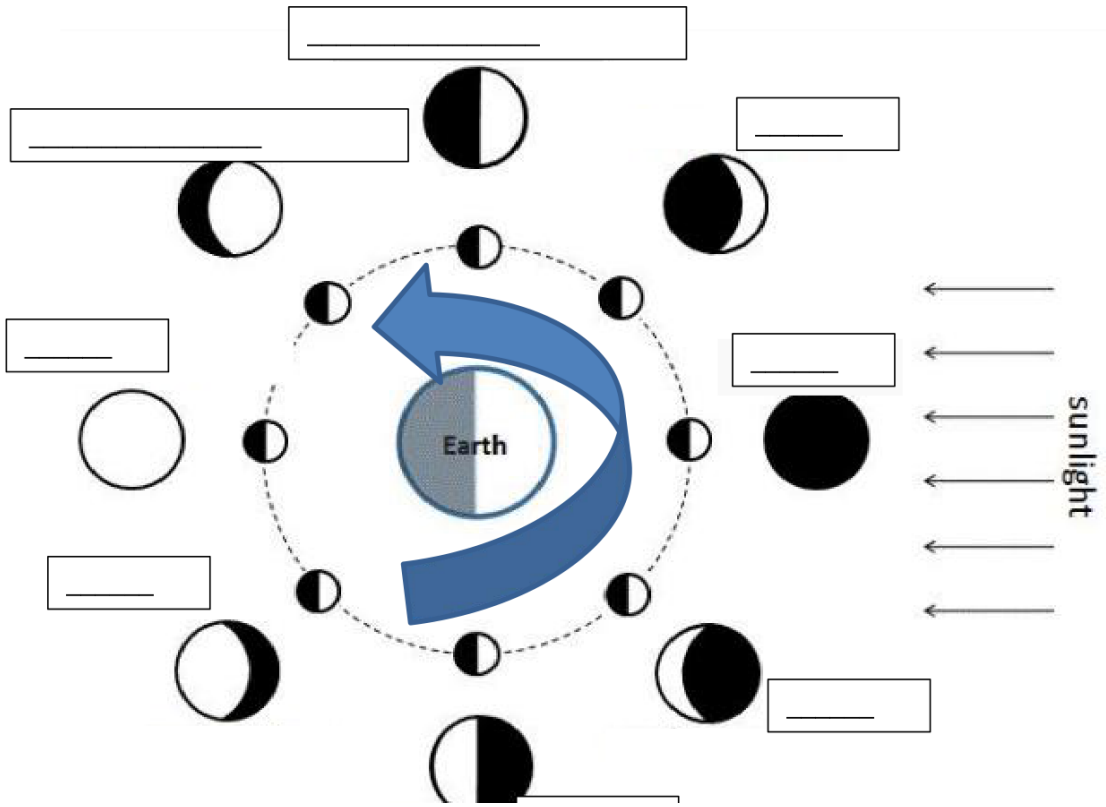
**（2）.Explanation of the periodic nature of the seasons.**

The Earth _____坏绕 the sun once_____一年. While it orbits the sun, it causes each _____半球 to either face the sun or face away from it.The_____半球 facing the sun has _____ and the _____ pointing _____it has winter.

• The _____地轴倾斜**causes seasons**.




（3)It takes approximately one month for the Moon to orbit the Earth and this is related to the periodic nature of the _____月相周期=27.5day



# IG-Space physics key summary

## (4). The Solar System :

- one _____恒星： the Sun.
- the eight planets : 轨道半径越大，公转周期越_____,转速越_____(判断变大或者变小）

four _____岩质内行星(体积小，密度大)：_____（水星）_____（金星体表温度最高）, Earth, _____（火星）

four _____气态外行星（体积大，密度小）:_____(木星）， _____（土星卫星最多）， _____天王星 and _____海王星.

- minor planets that orbit the Sun,an object which orbits the Sun but is not large enough or far enough from another object,that include the_____矮行星 such as _____冥王星.

●moons:that orbit the planets. is the_____天然卫星

●Smaller Solar System bodies, including

_____小行星 in the_____小行星带 and_____流星体: rocks that are smaller than the planet which orbit the Sun

_____彗星: aballofice, dust and gas which orbits the Sun in a highly _____ 椭圆轨道.

_____吸积盘: a rotating disc of matter formed by_____沉积物: ____ and _____pulled together by gravity

(5) **.Gravitational field(引力场)：**

. The force that keeps an object in orbit around the Sun is the _____万有引力of the Sun.

- the strength of the gravitational field depends on the _____ and _____

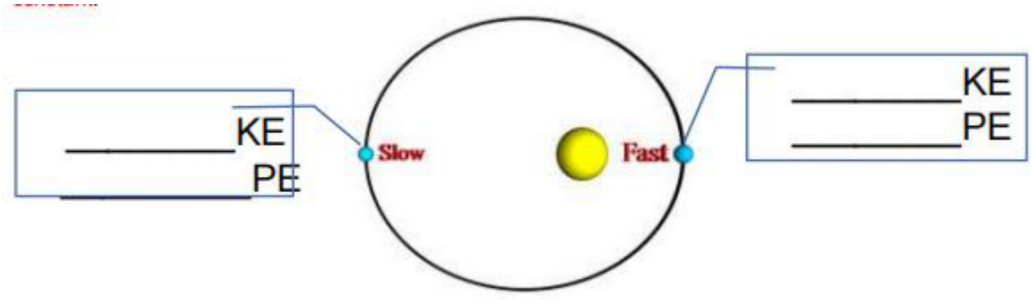
The strength of the Sun's gravitational field _____ and that the orbital speeds of the planets _____as the distance from the Sun _____(判断递增还是递减关系）.

理想圆轨道：orbital speed v = 2πr/T

r : _____环绕半径 T : _____环绕周期

实际_____椭圆轨道: moving fastest at closest to the Sun ,距离越大，速度越小

**总能量total energy =KE+PE** 不变



# IG-Space physics key summary

## Stars and the Universe

- The Sun is a star in the_____**星系** known as the _____**银河系.**
- TThe Sun is a star of medium size, consisting mostly of **hydrogen**(_____%) and **helium**(_____%).
  the sun radiates most of its energy in the_____(50%), _____(40%)and _____(10%)
- Stable stars like our sun are powered by the _____**核聚变**. This makes the Sun shine.
- _____光年：_____

### Describe the life cycle of a star:恒星的生命周期

**(a)** a star is formed from_____星际 clouds of _____ and _____ that contain **hydrogen**

**(b)** a _____原恒星is on interstellar cloud collapsing and increasing in temperature as a result of its internal _____万有引力。

**(c)** a _____原恒星 becomes a _____ when the _____ is balanced by an _____force due to the high temperature in the centre of the star

**(d)** all stars eventually run out of _____ as f_____ for the nuclear reaction

根据恒星质量不同演化两条路径

- For more massive stars

more massive stars expand to form _____红超巨星 when most of the hydrogen in the centre of the star has been corverted to helium，then explodes as a _____超新星, forming a _____星云 containing hydrogen and new heavier elements,leaving behind a **neutron star**（中子星）or a **black hole**（黑洞）

- For less massive star(such as SUN)

a ____红巨星from a less massive star forms a _____行星状星云 with a _____白矮星at its centre.

表格

| 英文术语 | 中文译名 | 关键考点 |
| --- | --- | --- |
| Nebula | 星云 | 恒星诞生的摇篮，主要成分氢和氦 |
| Protostar | 原恒星 | 引力坍缩阶段，未启动核聚变 |
| Main-sequence star | 主序星 | 氢聚变稳定进行，恒星最长的阶段 |
| Red giant/Red supergiant | 红巨星 / 红超巨星 | 氢耗尽后膨胀的阶段，由质量决定类型 |
| Planetary nebula | 行星状星云 | 中小质量恒星抛射的外层气体 |
| White dwarf | 白矮星 | 中小质量恒星的核心残骸，电子简并支撑 |
| Supernova | 超新星 | 大质量恒星的剧烈爆炸，抛射重元素 |
| Neutron star | 中子星 | 大质量恒星核心残骸，中子简并支撑 |
| Black hole | 黑洞 | 超大质量恒星核心坍缩的产物，引力极强 |

Stable star

# IG-Space physics key summary

## The universe

- _____红移 as an _____in the observed _____ of electromagnetic radiation emitted from _____远离的stars and galaxies.

- The _____红移 in the light from distant galaxies is evidence that the universe is _____膨胀and supports the_____（宇宙大爆炸理论）

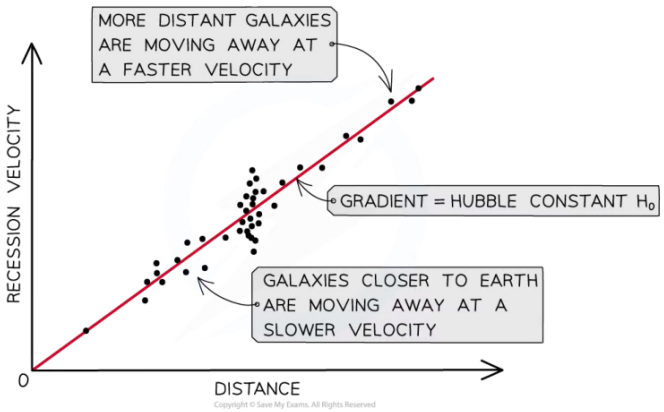
- The _____ of a specific frequeny is observed at all points in space Around us and is known as _____(宇宙微波背景辐射)(CMBR)

The CMBR was produced shortly after the _____ and that this radiation has been expanded into the_____ region of the electromognetic spectrum as the Universe _____. And that this is evidence for the idea that all the matter in the Universe was present at a _____.

- Hubble's law shows that _____

$$H_0 = \frac{v}{d}$$

v: _____
d: _____
$H_0$:_____




The equation $\frac{1}{H_0} = \frac{d}{v}$ represents an estimate for the _____ of the Universe.